

# **Seasonal and Annual Wind-Direction Regimes in Pakistan**

**A long-term, station-level wind-rose climatology based on Pakistan Meteorological Department (PMD) observations**

**Objective:** To characterize the seasonal evolution and annual distribution of prevailing wind directions across Pakistan using long-term meteorological observations from 61 stations.

**Spatial domain:** Pakistan

**Stations:** 61 PMD stations

**Temporal basis:** Full available record at each station

**Temporal resolution:** Monthly and annual climatologies

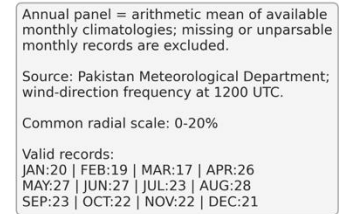
**Observation time:** 1200 UTC

**Directional resolution:** 16 compass sectors

**Author:** Muhammad Abid Khan, PhD, University of Tsukuba, Japan • Director's Award for Academic Excellence

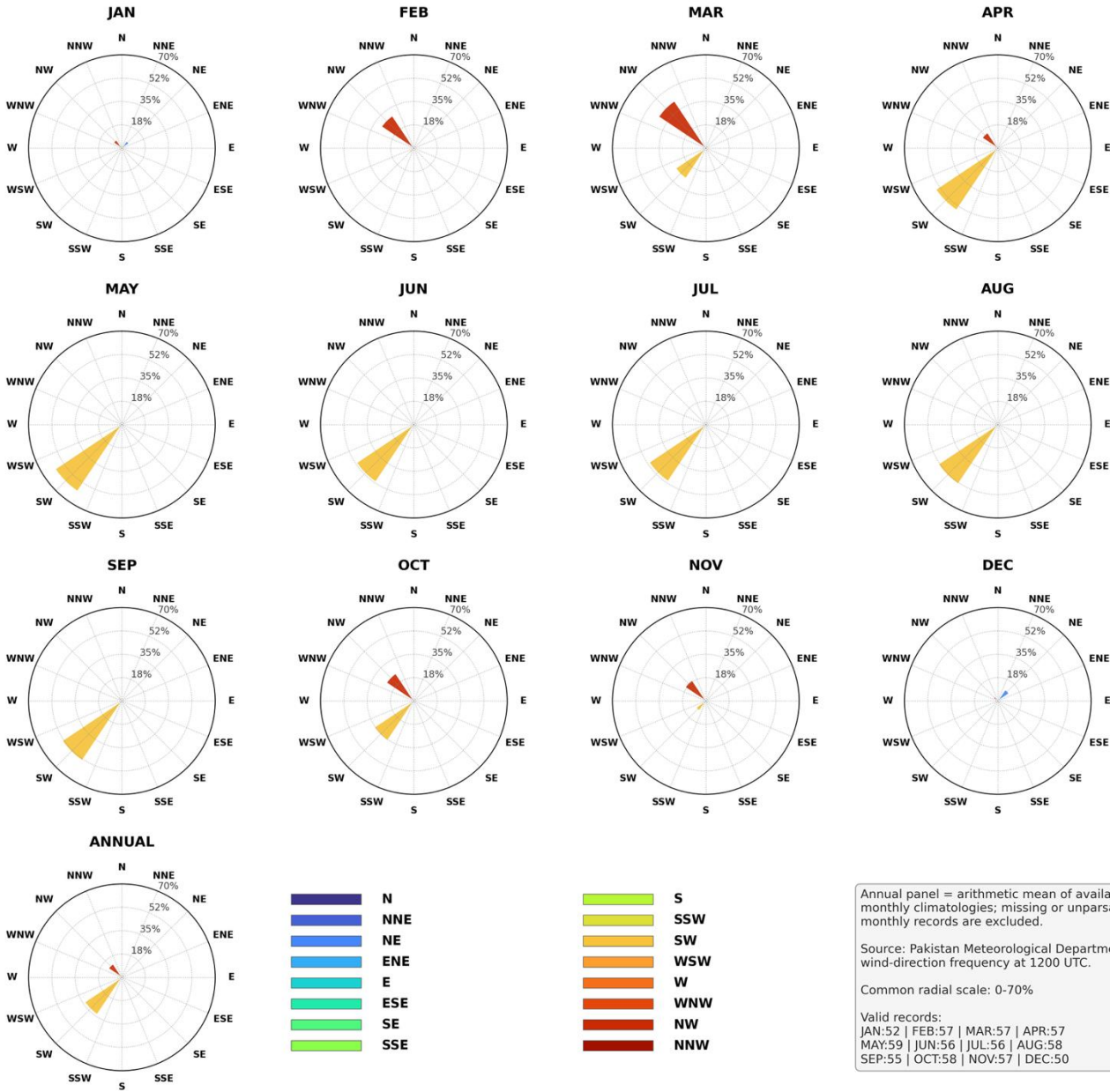
## 1959-2000 (42 years) | 1200 UTC | Long-term mean frequency (%)

**FEB**                      **MAR**



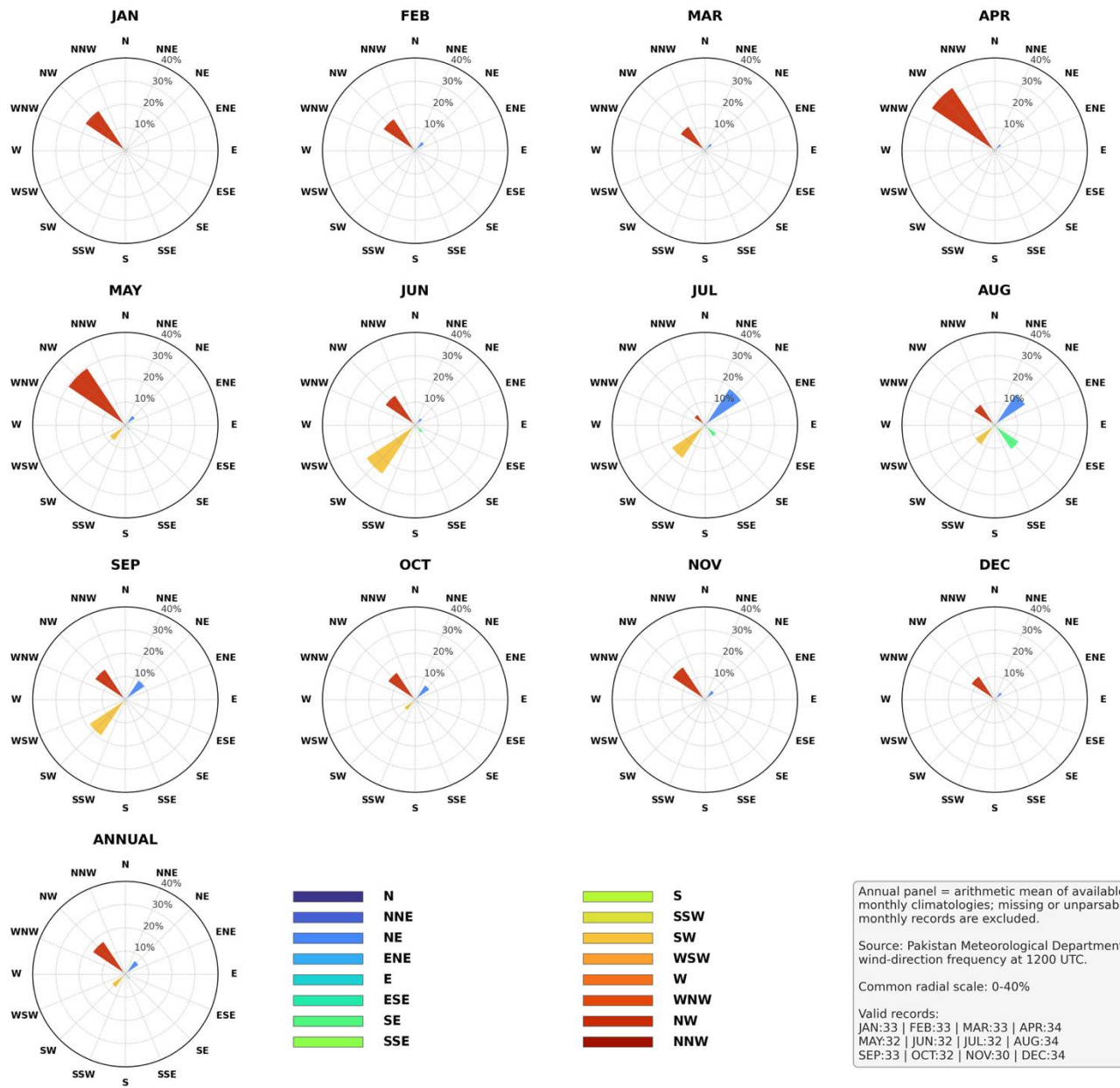
# Wind Rose Climatology: BADIN

1937-2000 (64 years) | 1200 UTC | Long-term mean frequency (%)



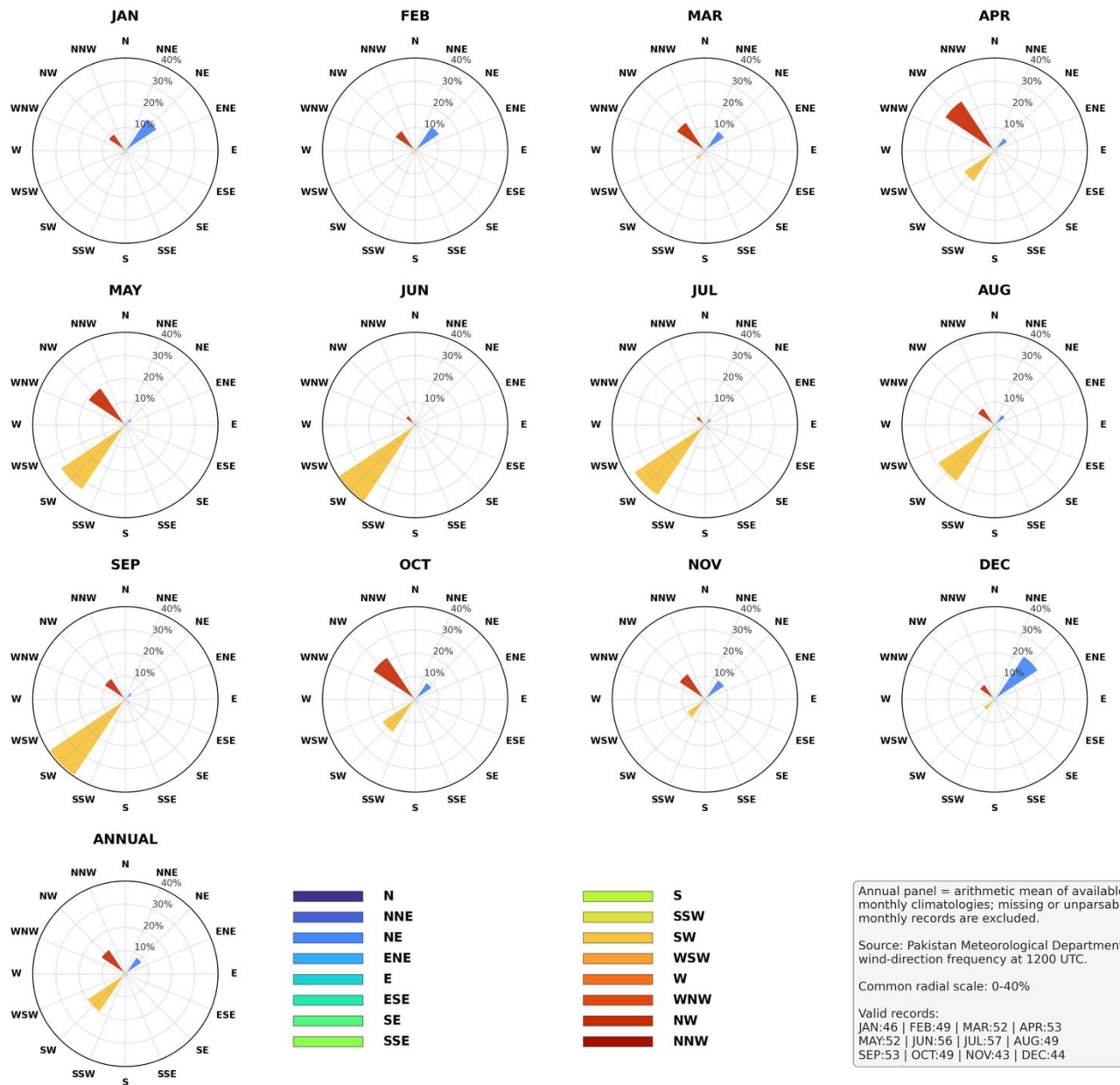
Wind Rose Climatology: BAHAWALNAGAR

1963-2000 (38 years) | 1200 UTC | Long-term mean frequency (%)



Wind Rose Climatology: BAHAWALPUR

1936-2000 (65 years) | 1200 UTC | Long-term mean frequency (%)



Annual panel = arithmetic mean of available monthly climatologies; missing or unparsable monthly records are excluded.

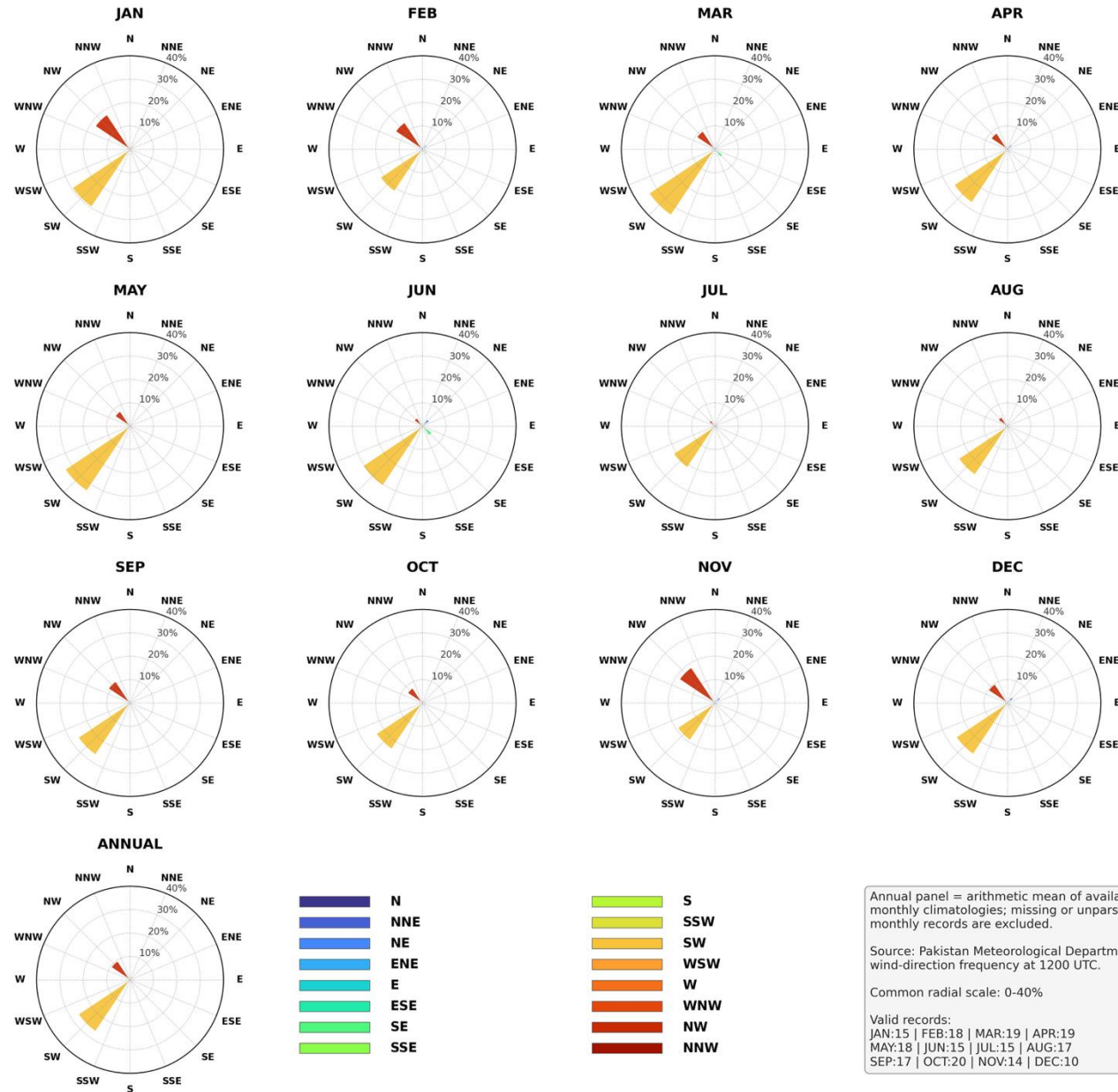
Source: Pakistan Meteorological Department; wind-direction frequency at 1200 UTC.

Common radial scale: 0-40%

Valid records:  
JAN:46 | FEB:49 | MAR:52 | APR:53  
MAY:52 | JUN:56 | JUL:57 | AUG:49  
SEP:53 | OCT:49 | NOV:43 | DEC:44

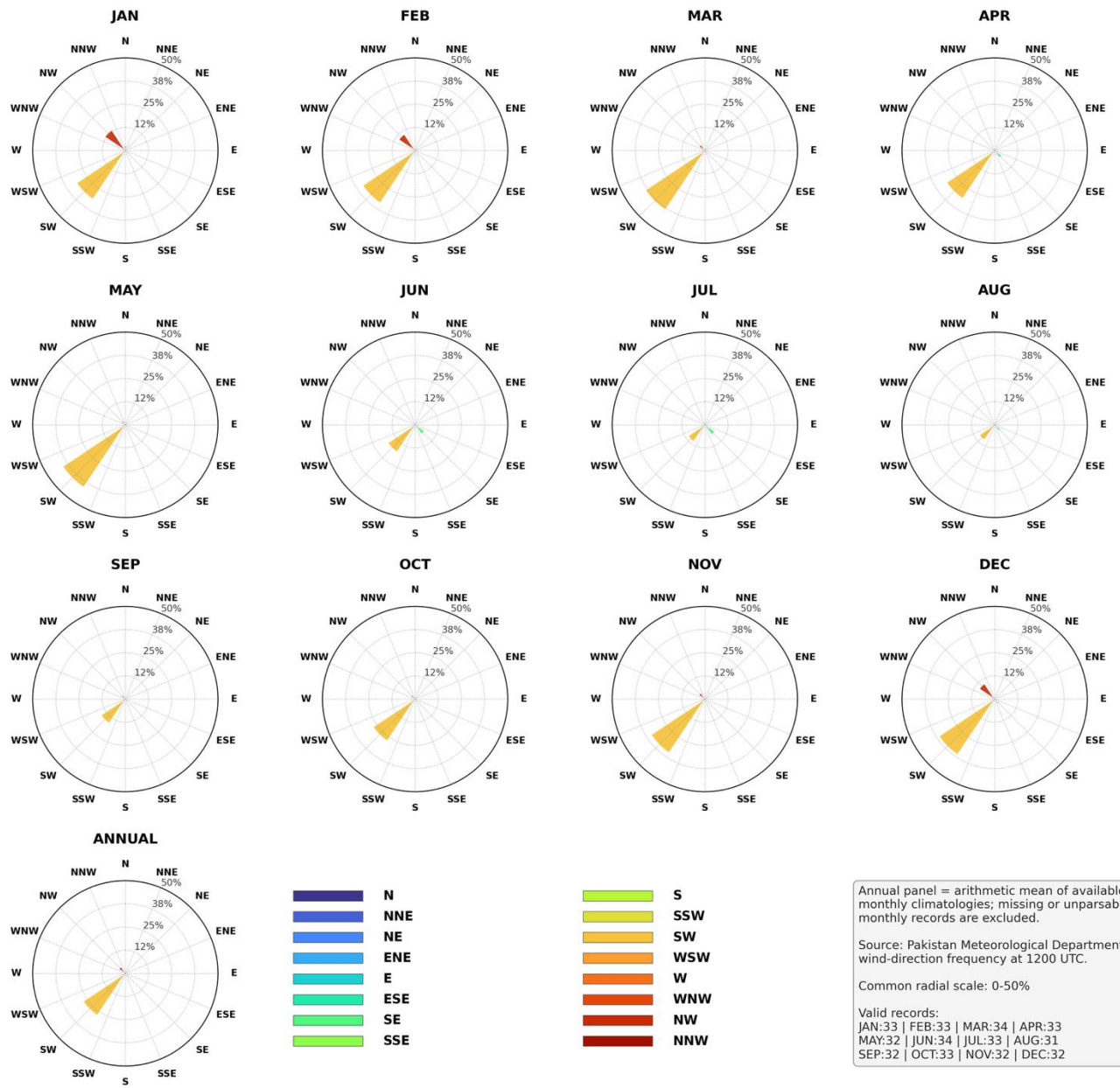
# Wind Rose Climatogy: BALAKOT

1960-2000 (41 years) | 1200 UTC | Long-term mean frequency (%)



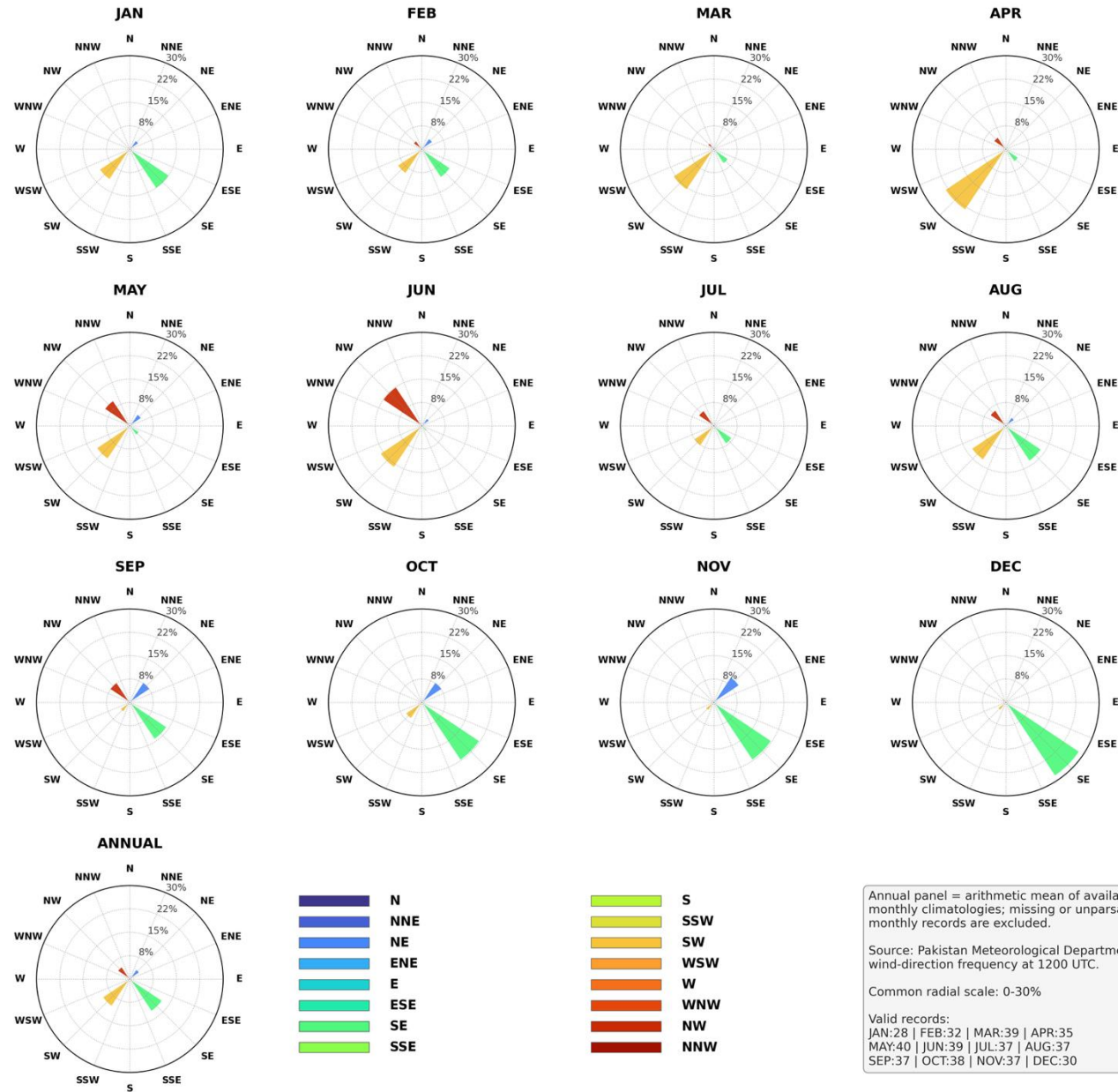
Wind Rose Climatology: BARKHAN

1967-2000 (34 years) | 1200 UTC | Long-term mean frequency (%)



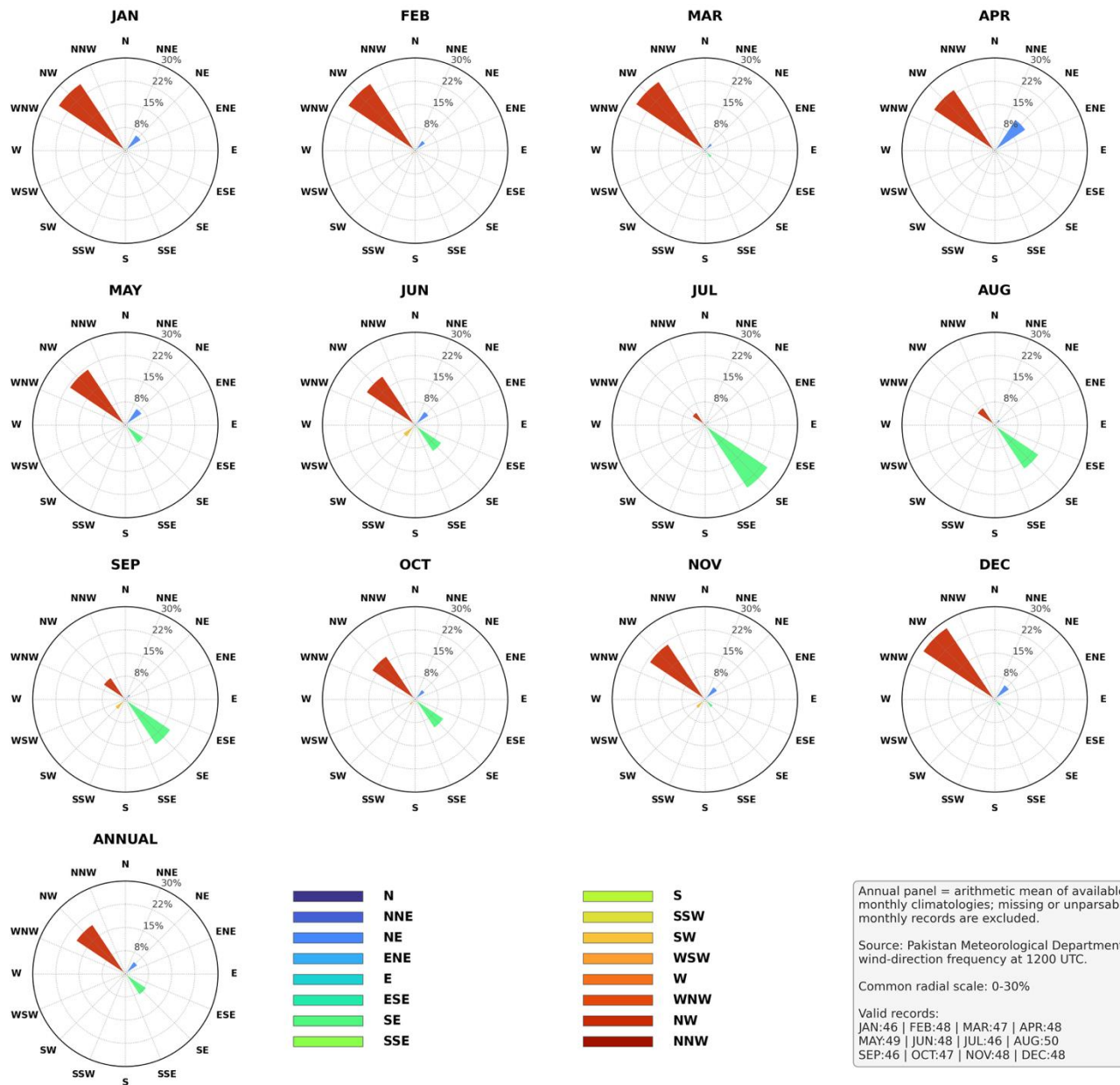
# Wind Rose Climatology: BUNJI

1953-2000 (48 years) | 1200 UTC | Long-term mean frequency (%)



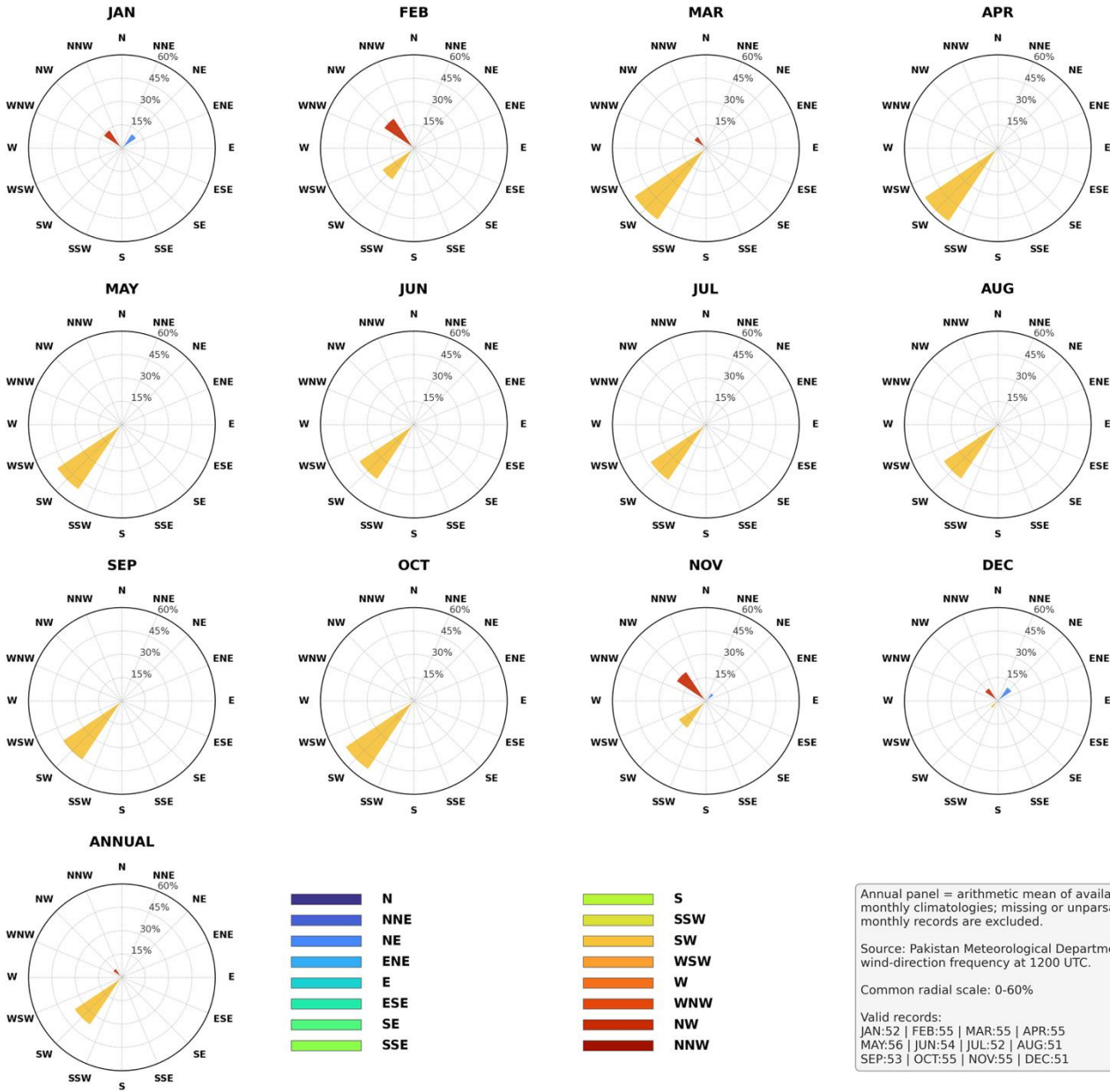
Wind Rose Climatology: CHERAT

1943-2000 (58 years) | 1200 UTC | Long-term mean frequency (%)



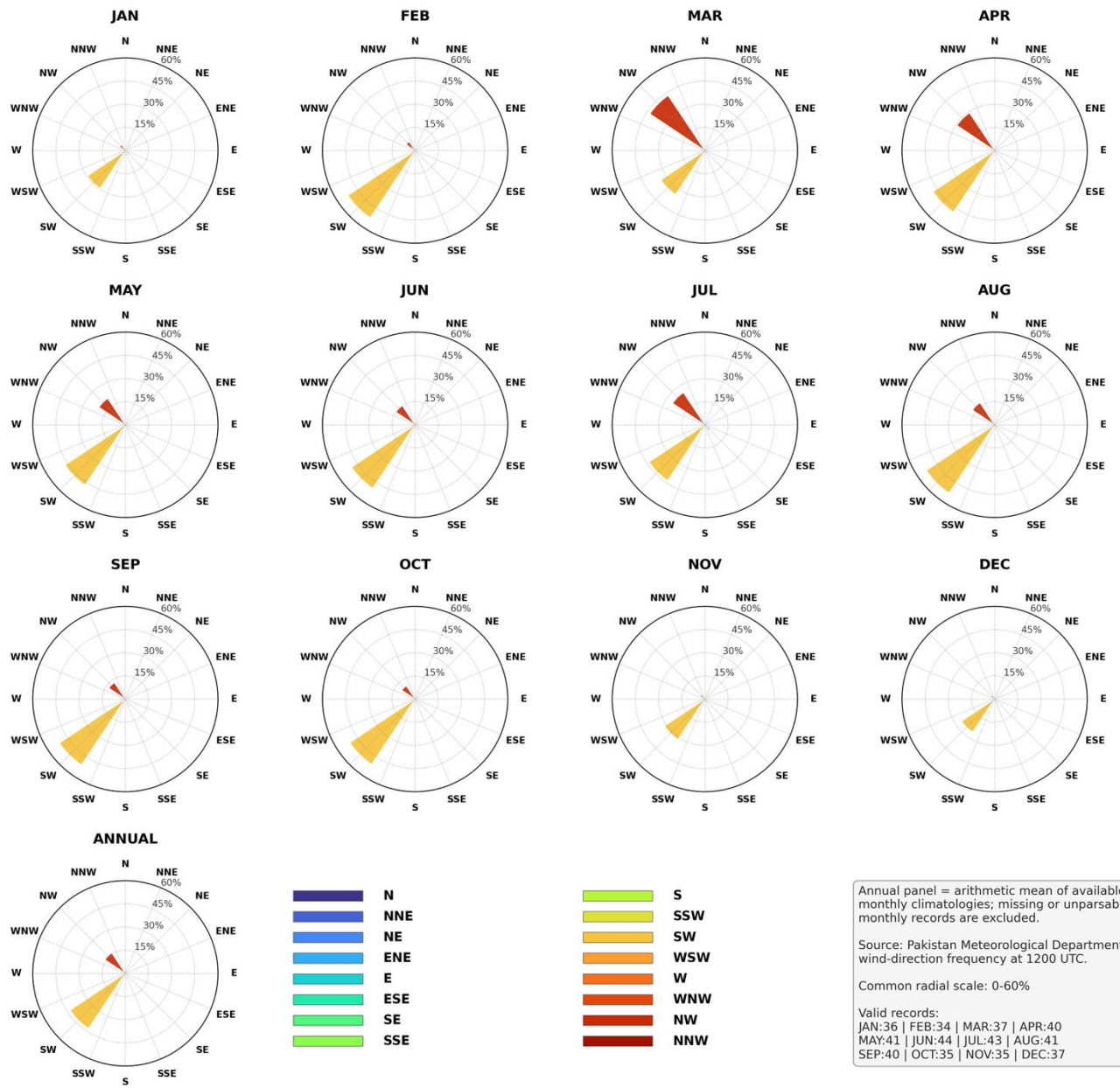
Wind Rose Climatology: CHHOR

1933-2000 (68 years) | 1200 UTC | Long-term mean frequency (%)



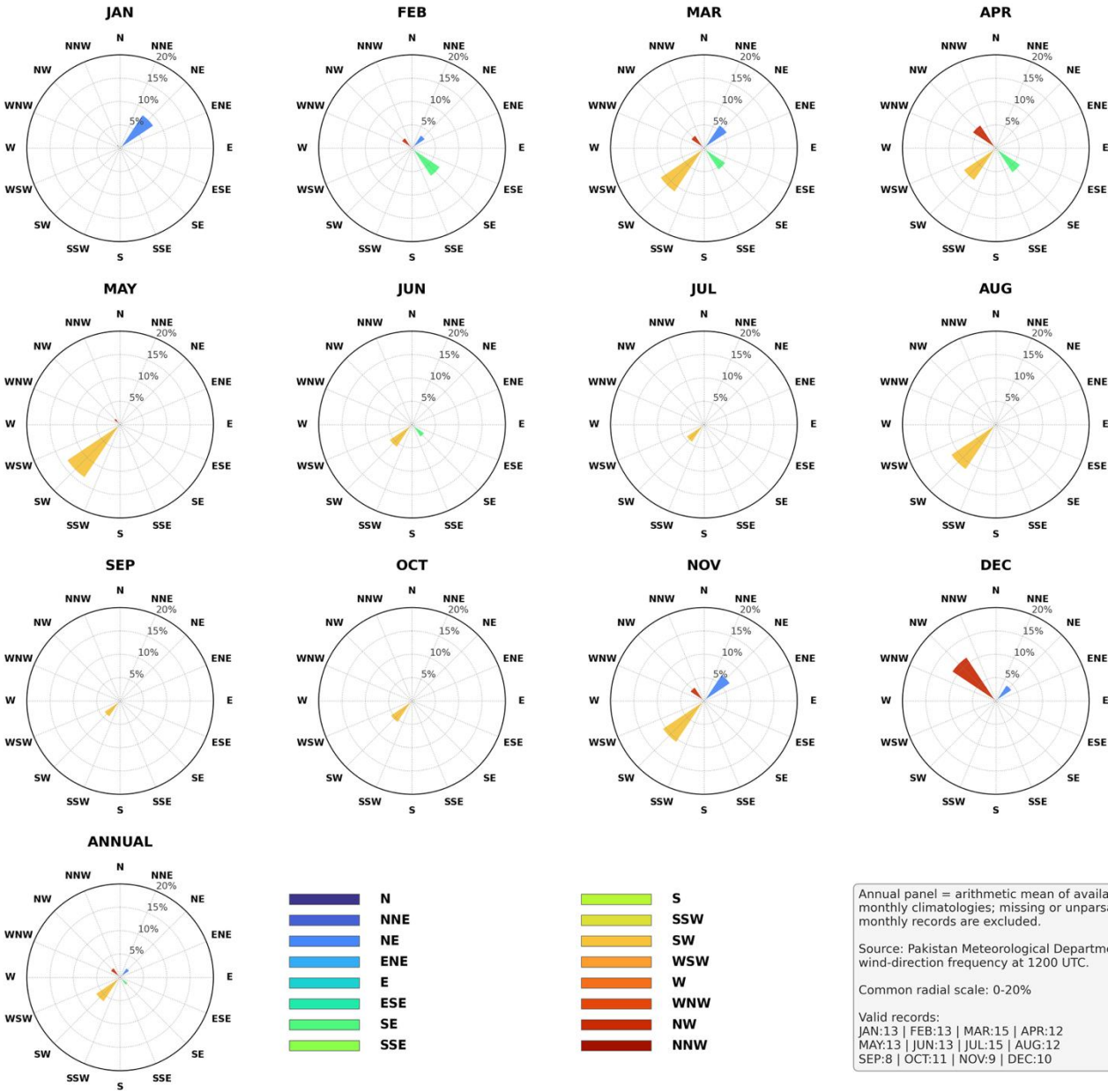
# Wind Rose Climatology: CHILAS

1953-2000 (48 years) | 1200 UTC | Long-term mean frequency (%)



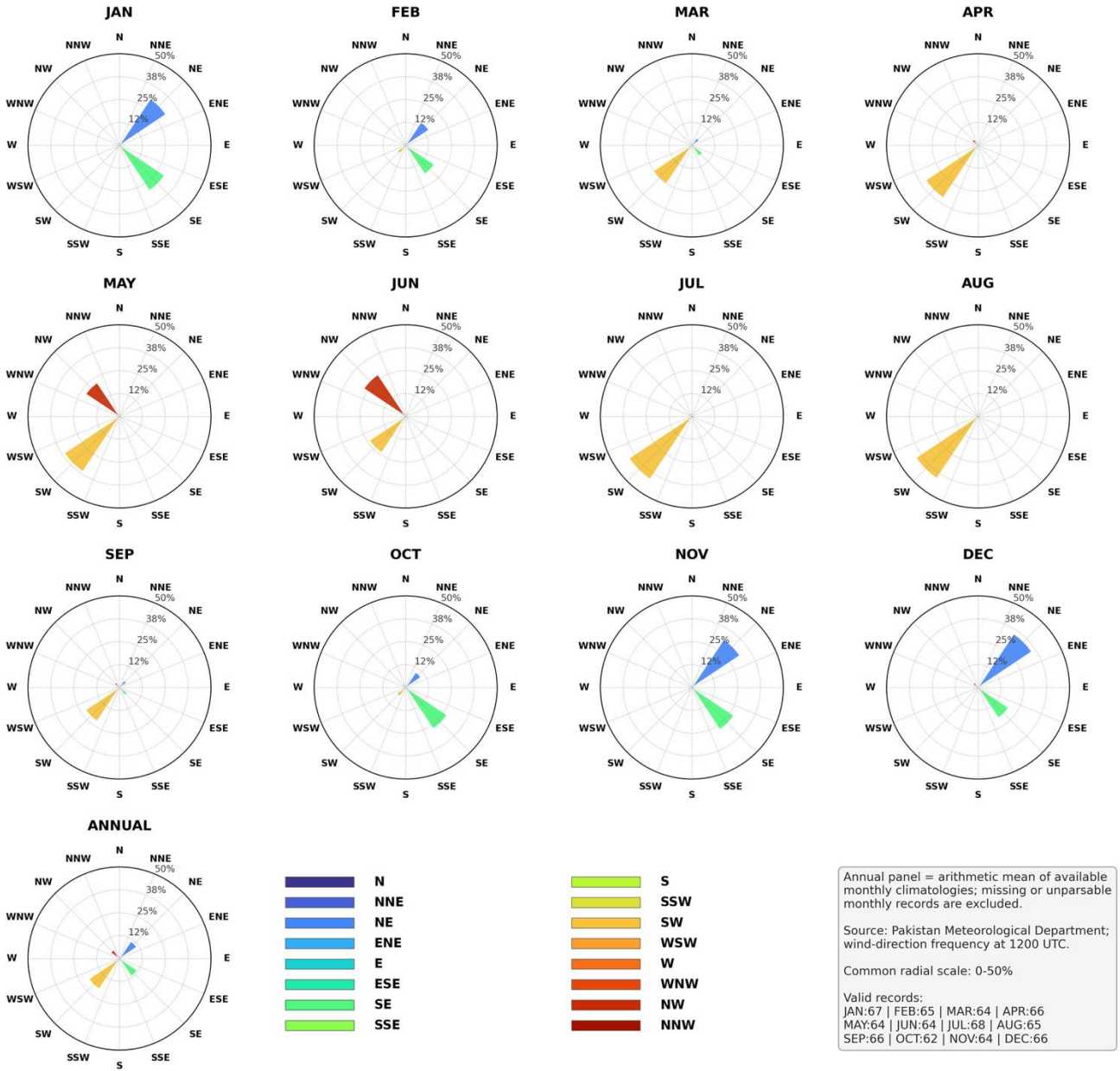
Wind Rose Climatology: CHITRAL

1964-2000 (37 years) | 1200 UTC | Long-term mean frequency (%)



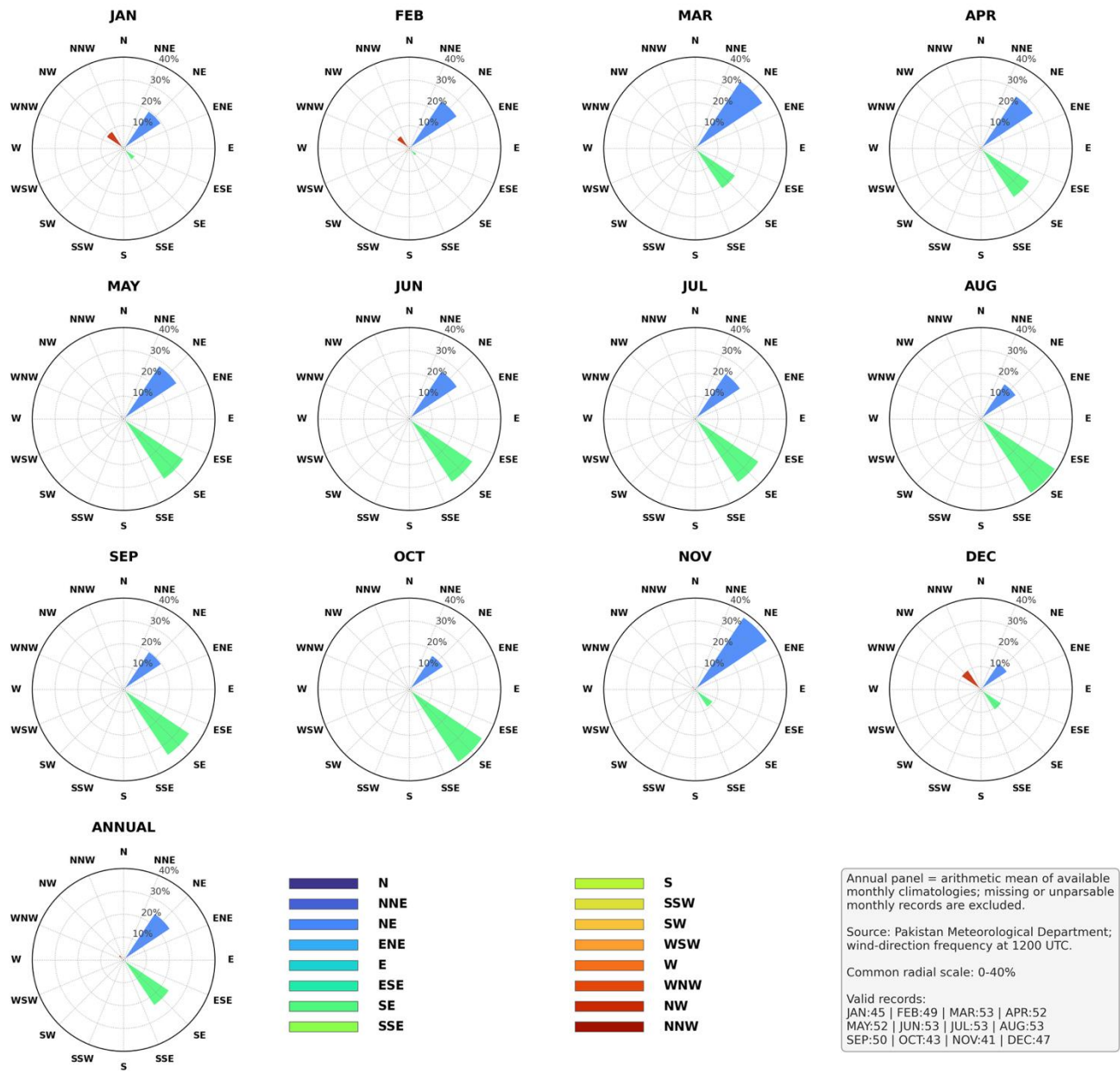
Wind Rose Climatolgy: DALBANDIN

1933-2000 (68 years) | 1200 UTC | Long-term mean frequency (%)



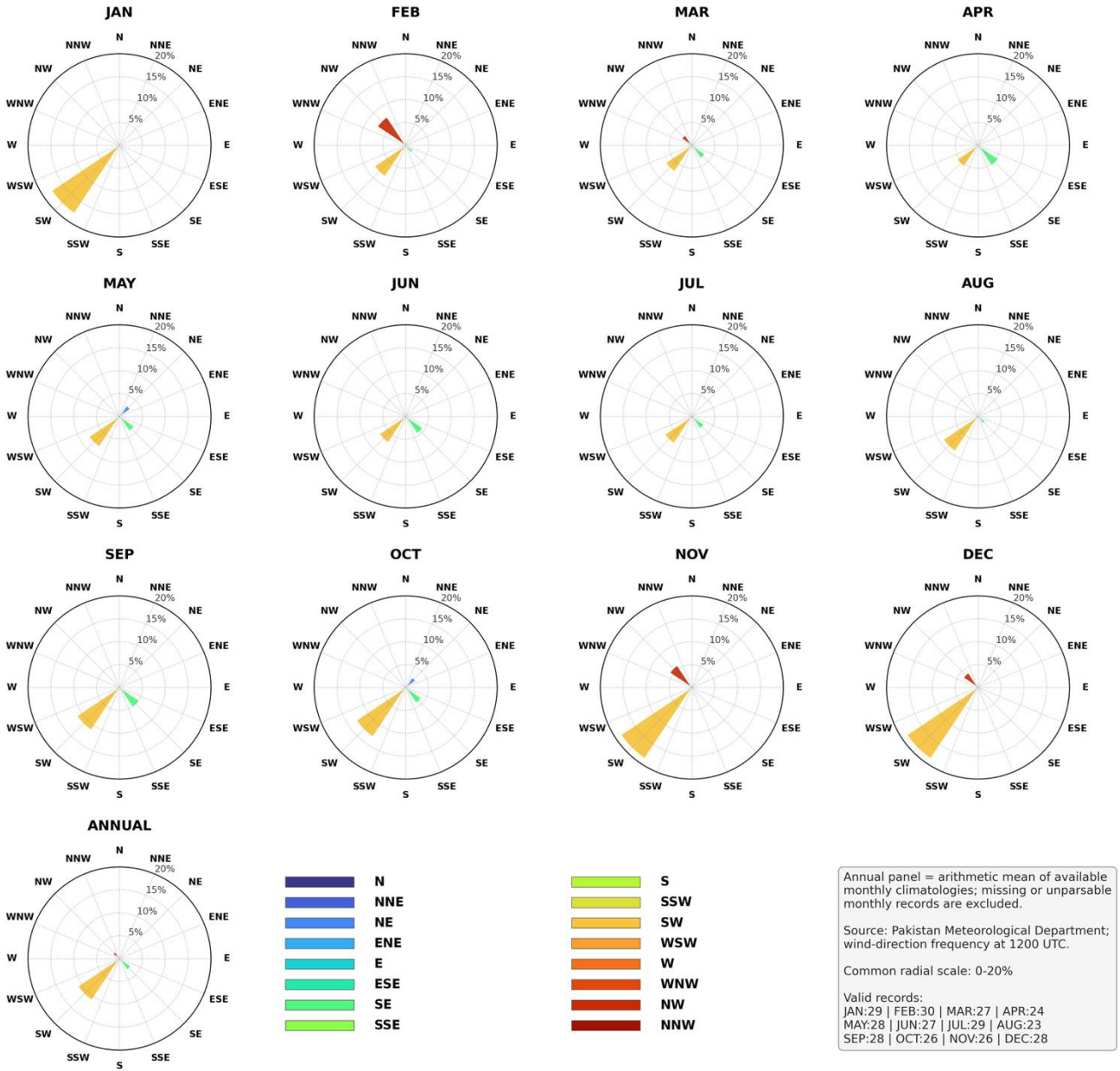
Wind Rose Climatology: DERA ISMAIL KHAN

1943-2000 (58 years) | 1200 UTC | Long-term mean frequency (%)



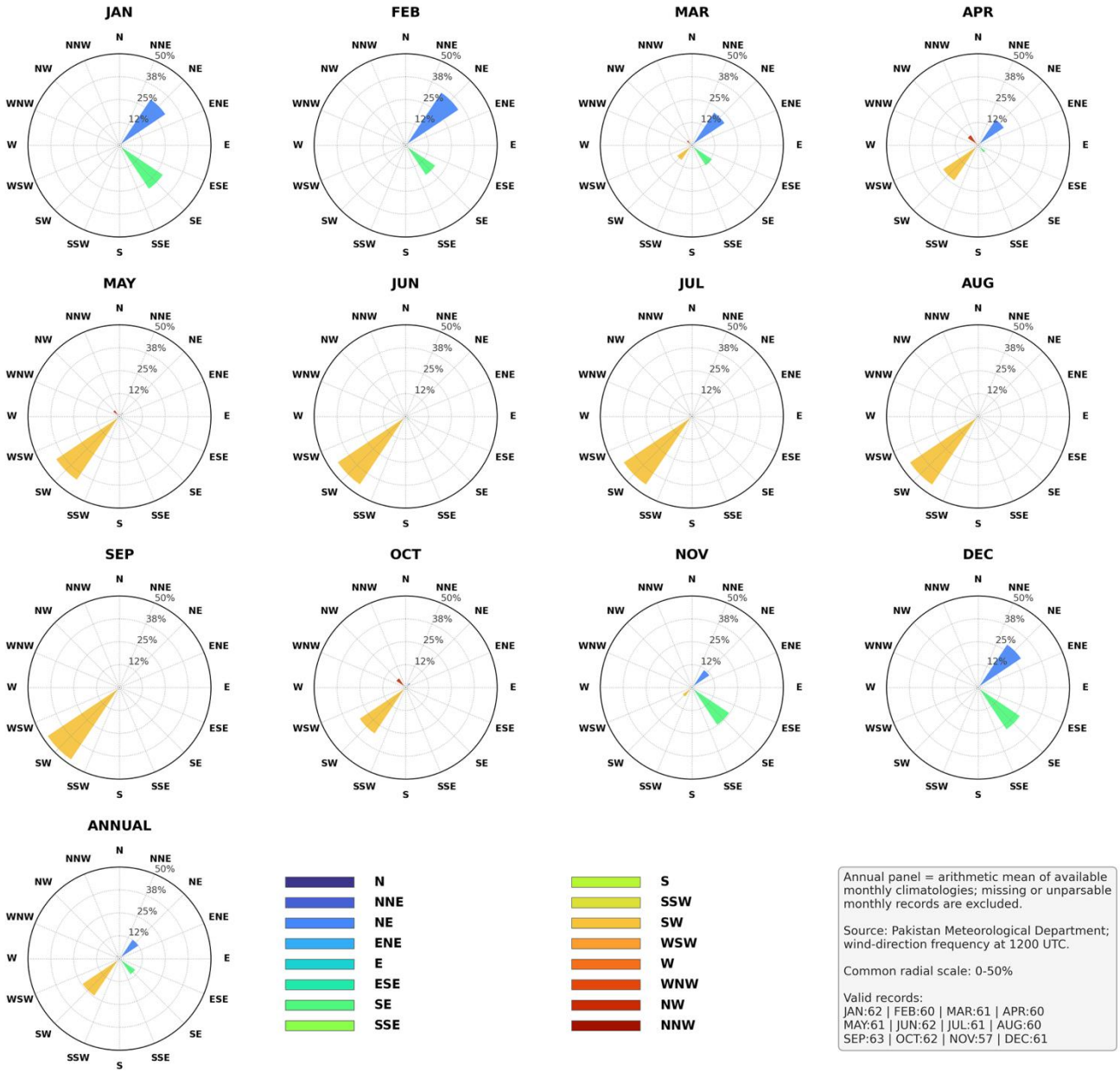
Wind Rose Climatology: DIR

1967-2000 (34 years) | 1200 UTC | Long-term mean frequency (%)



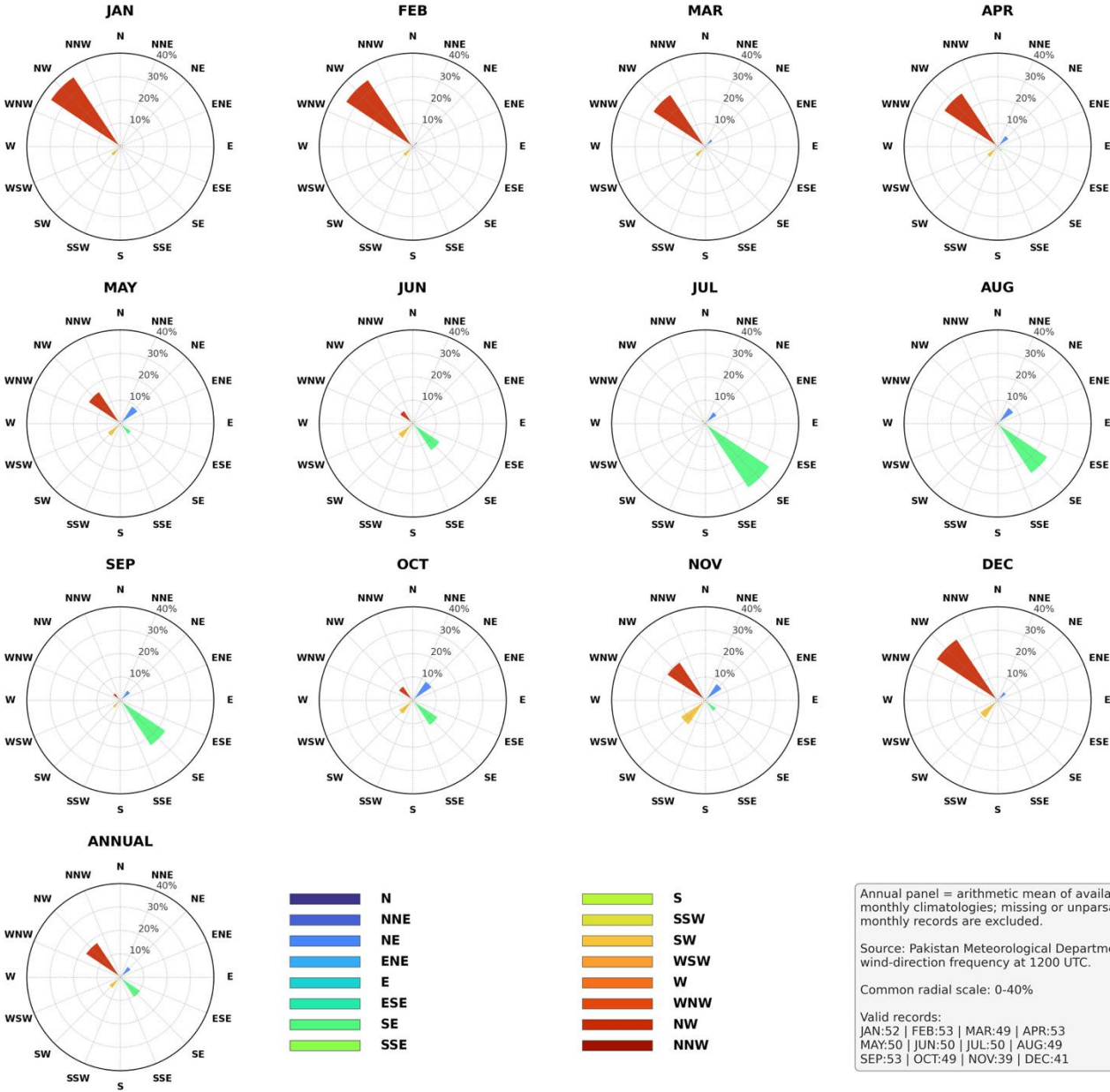
# Wind Rose Climatology: DROSH

1936-2000 (65 years) | 1200 UTC | Long-term mean frequency (%)



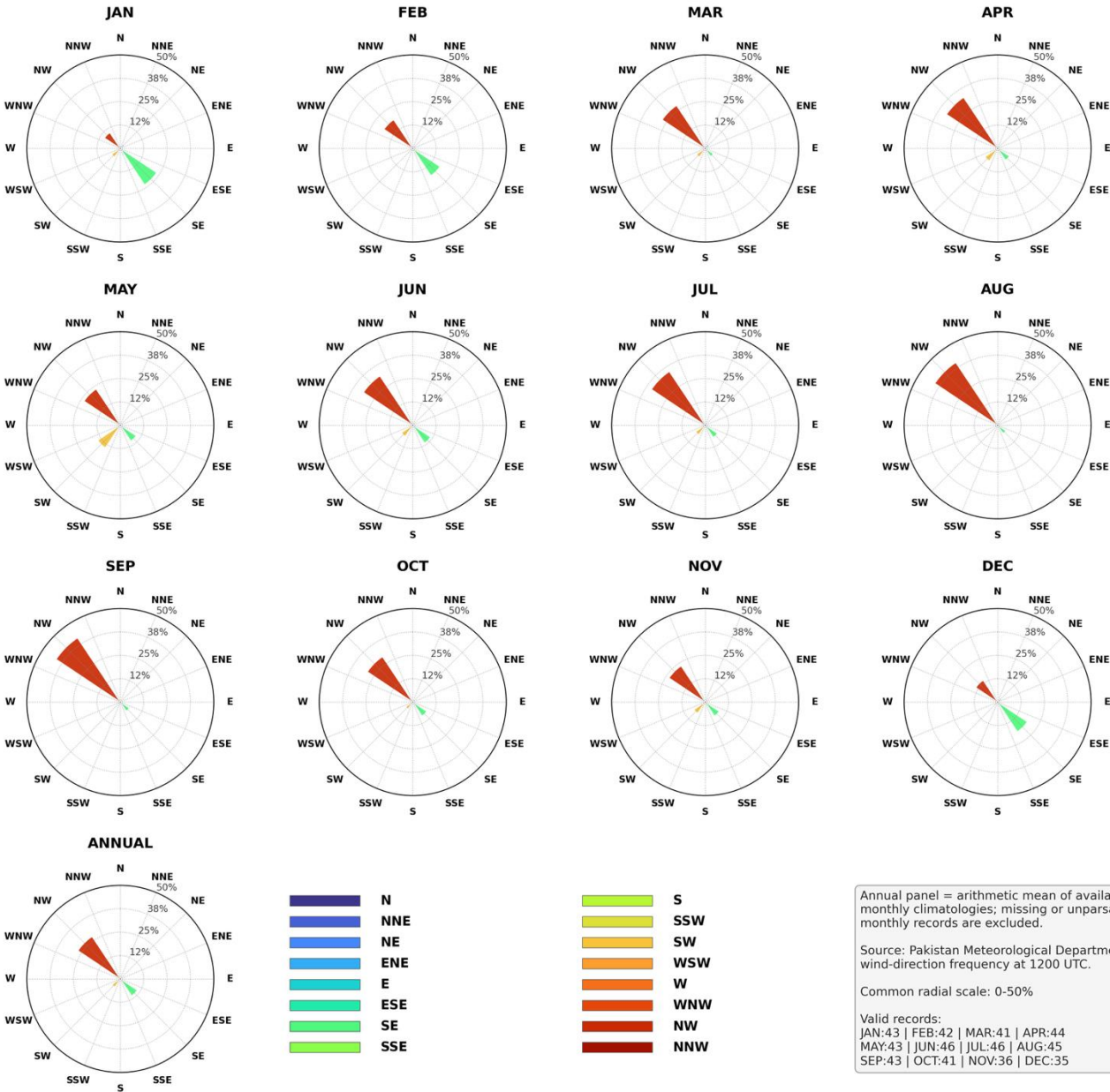
Wind Rose Climatology: FAISALABAD(LYALLPUR)

1936-2000 (65 years) | 1200 UTC | Long-term mean frequency (%)



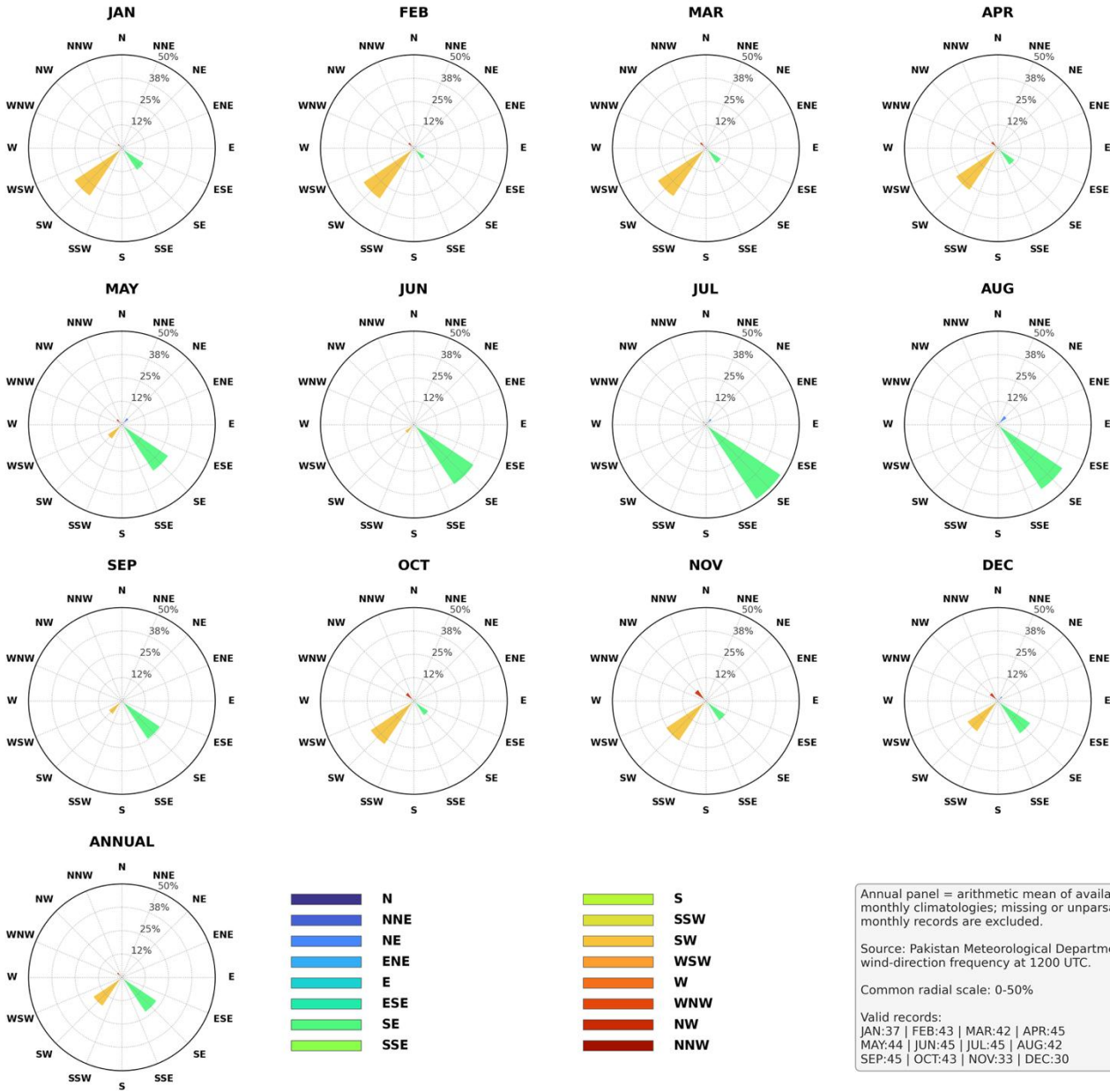
Wind Rose Climatology: GARHI DUPATTA

1955-2000 (46 years) | 1200 UTC | Long-term mean frequency (%)



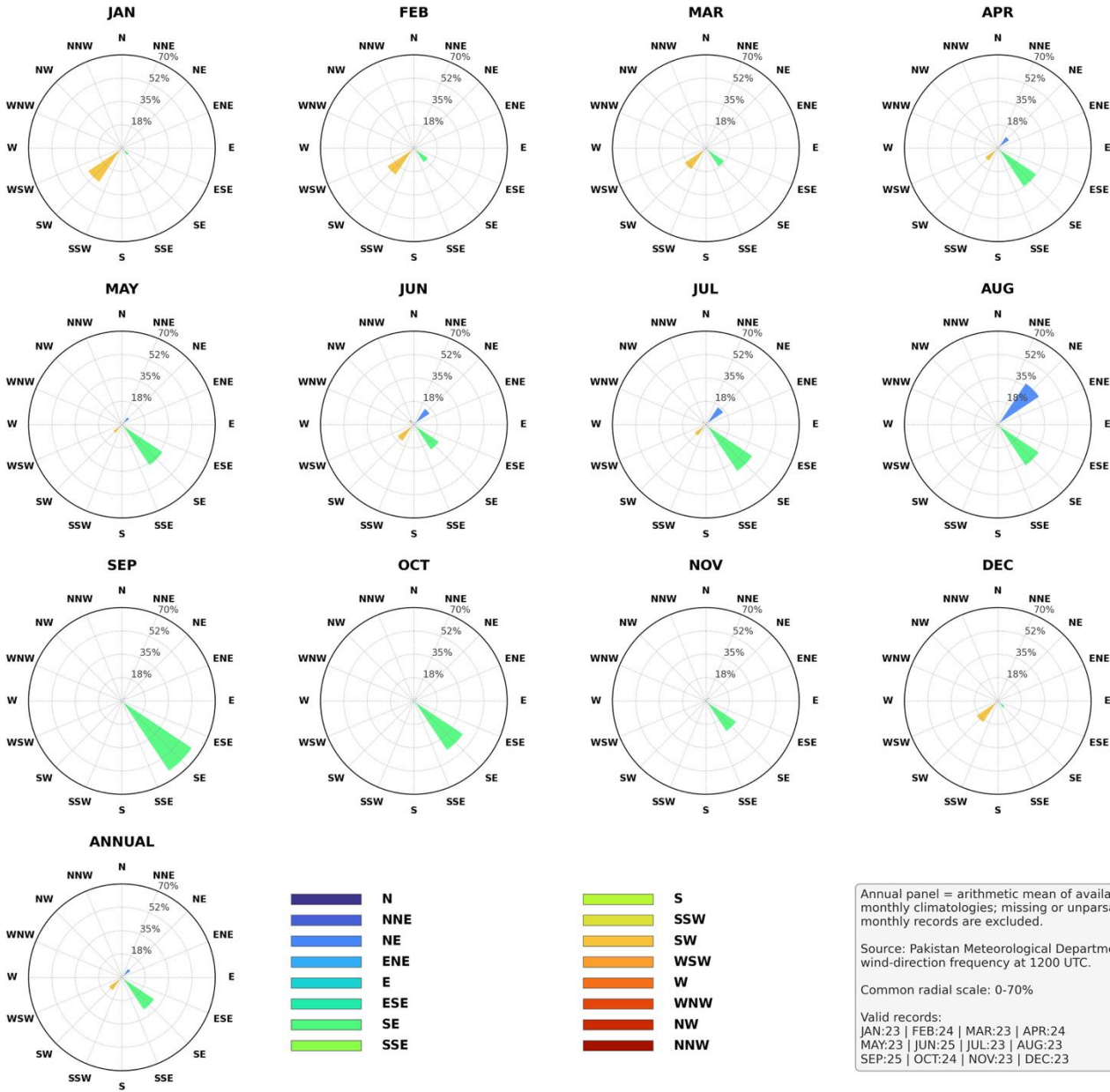
# Wind Rose Climatology: GILGIT

1952-2000 (49 years) | 1200 UTC | Long-term mean frequency (%)



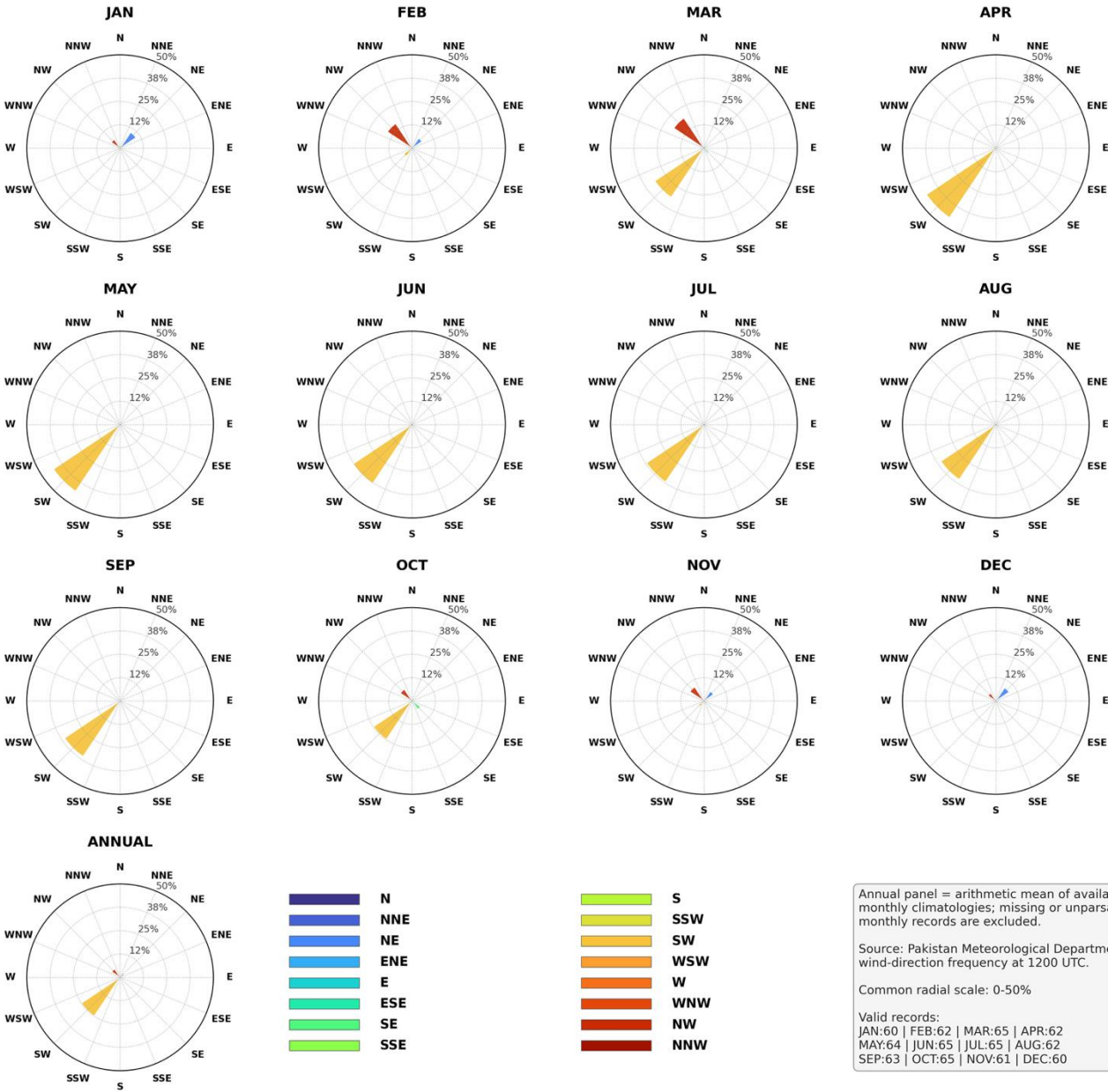
# Wind Rose Climatology: GUPIS

1955-2000 (46 years) | 1200 UTC | Long-term mean frequency (%)



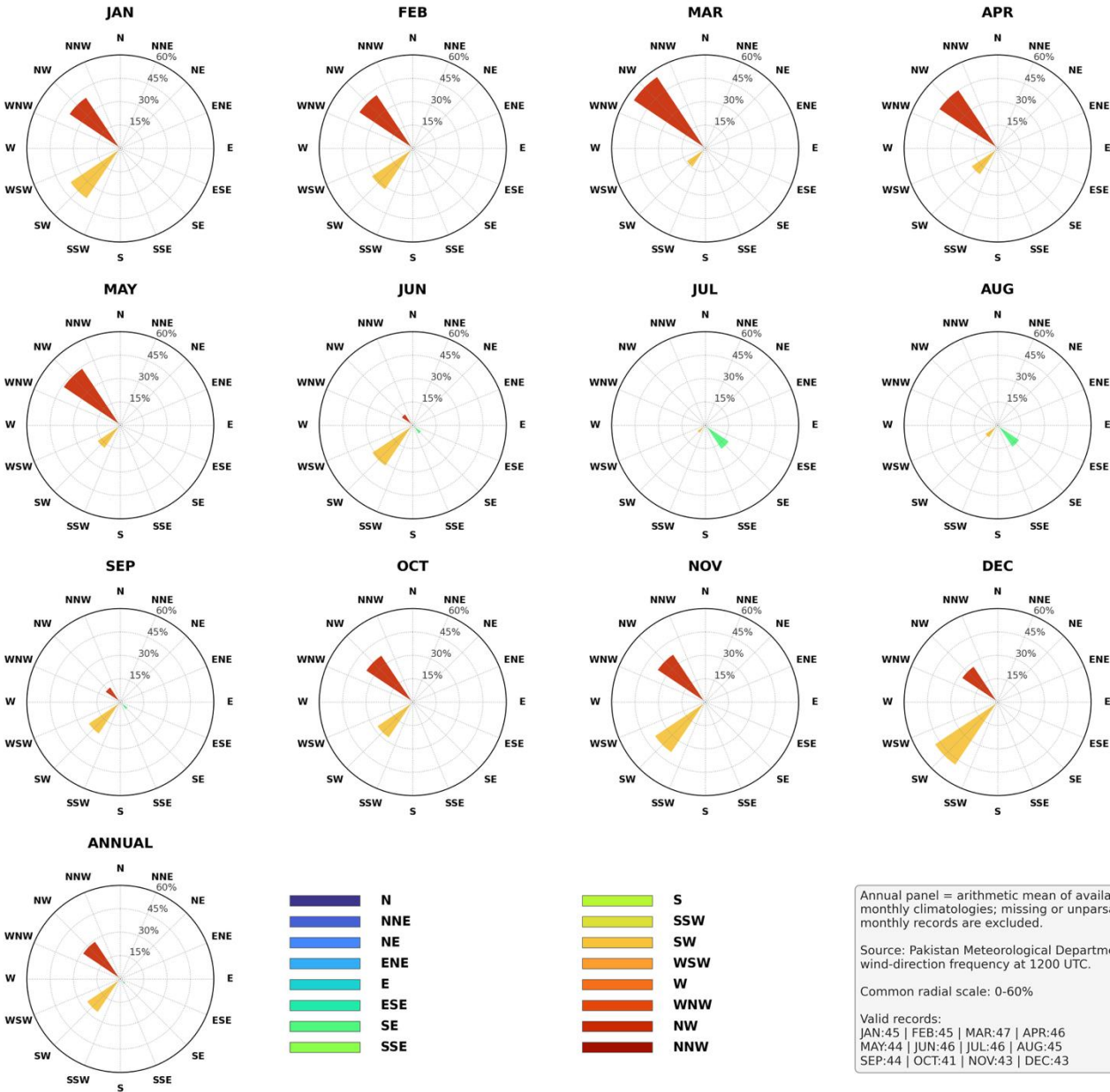
Wind Rose Climatology: HYDERABAD

1933-2000 (68 years) | 1200 UTC | Long-term mean frequency (%)



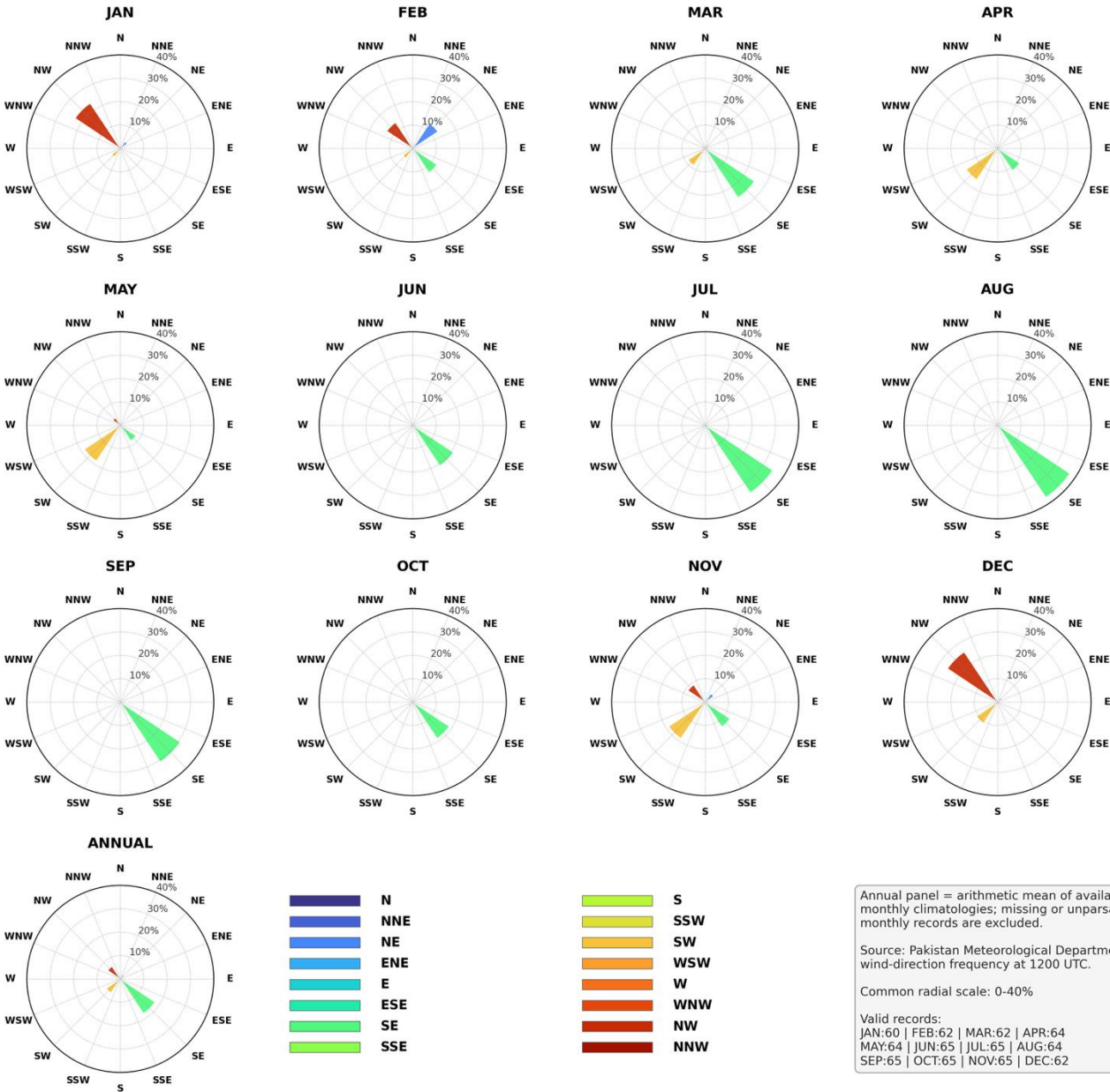
Wind Rose Climatolgy: ISLAMABAD

1954-2000 (47 years) | 1200 UTC | Long-term mean frequency (%)



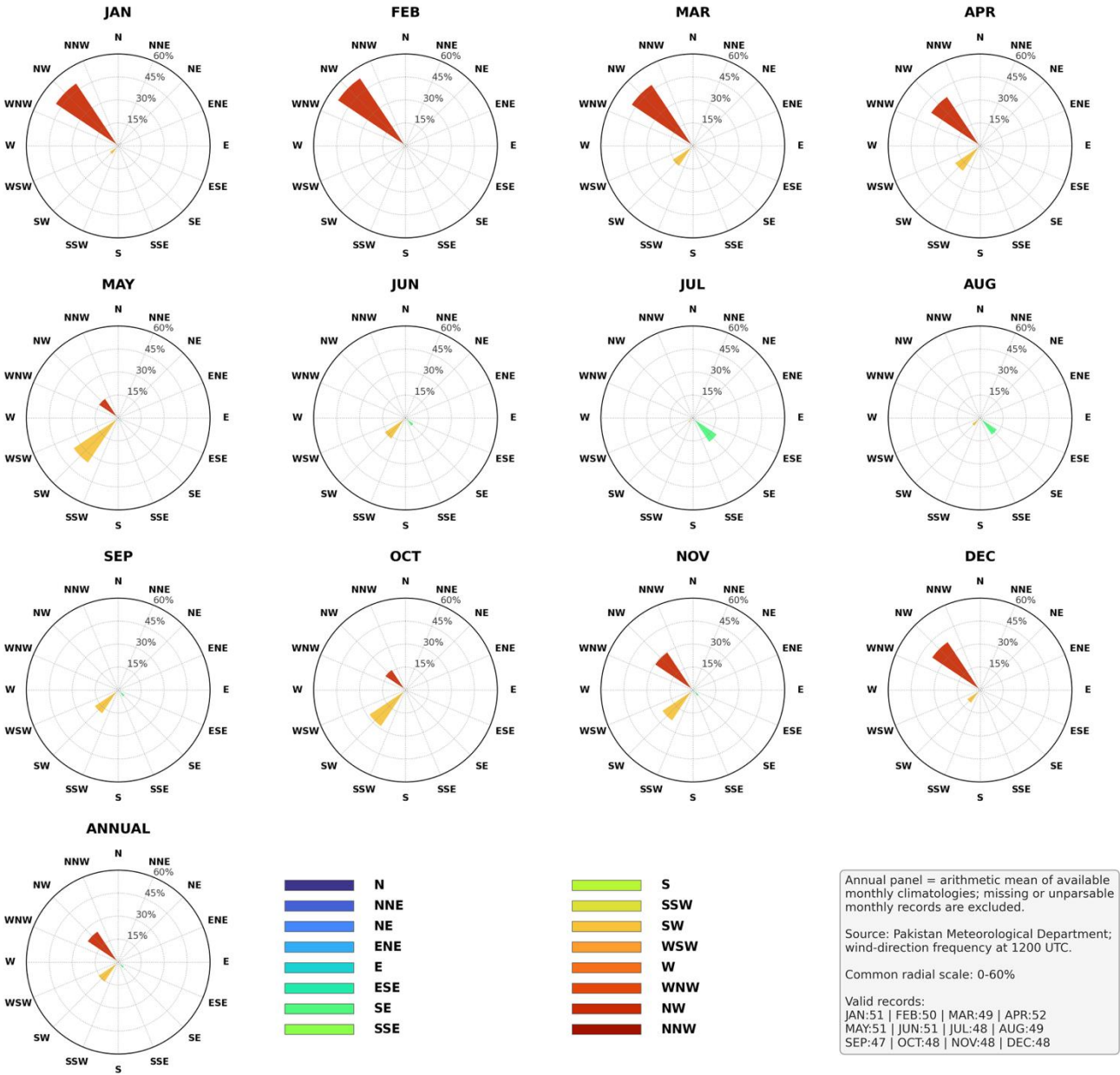
Wind Rose Climatolog: JACOBABAD

1934-2000 (67 years) | 1200 UTC | Long-term mean frequency (%)



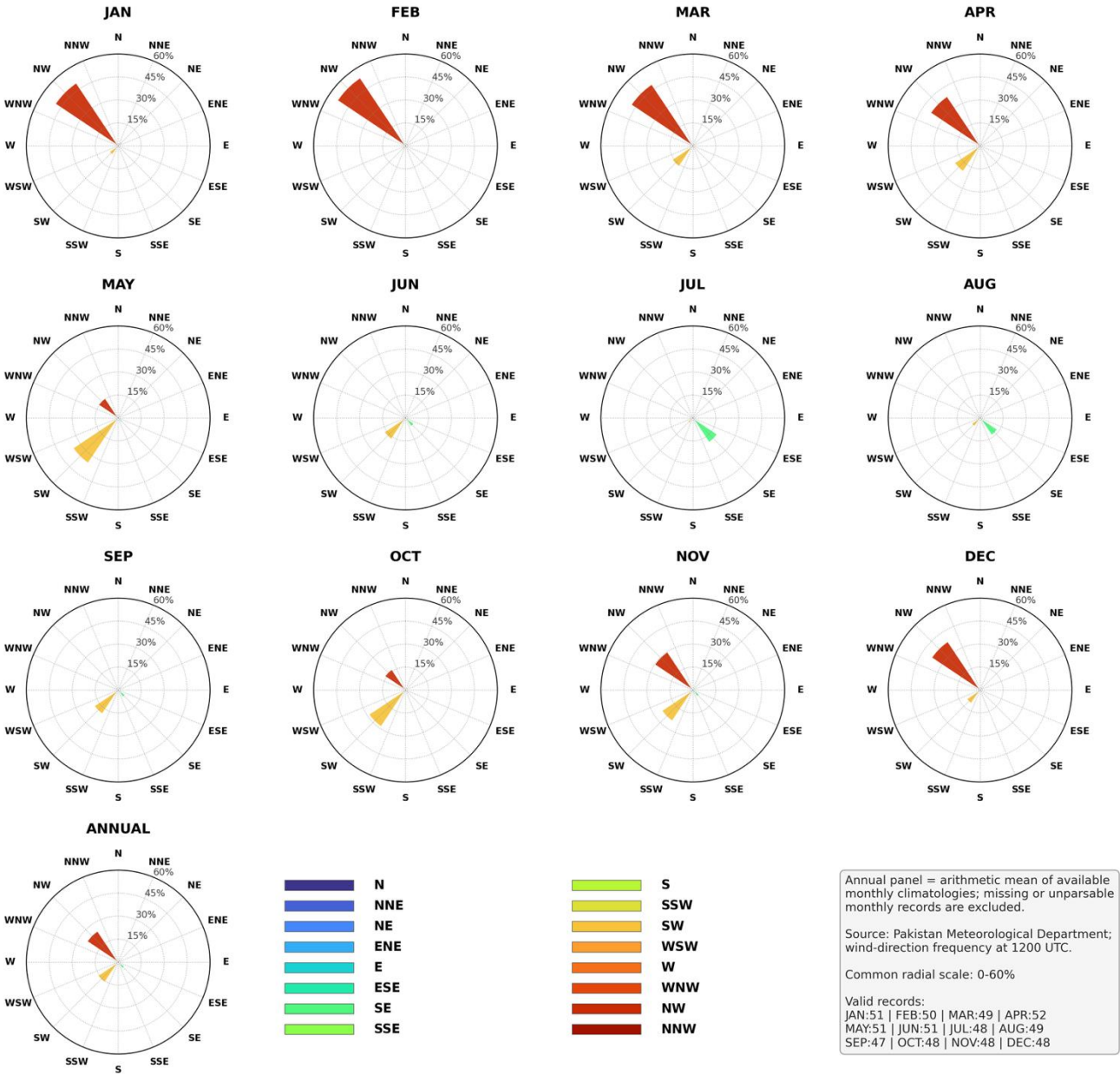
# Wind Rose Climatology: JHELUM

1948-2000 (53 years) | 1200 UTC | Long-term mean frequency (%)



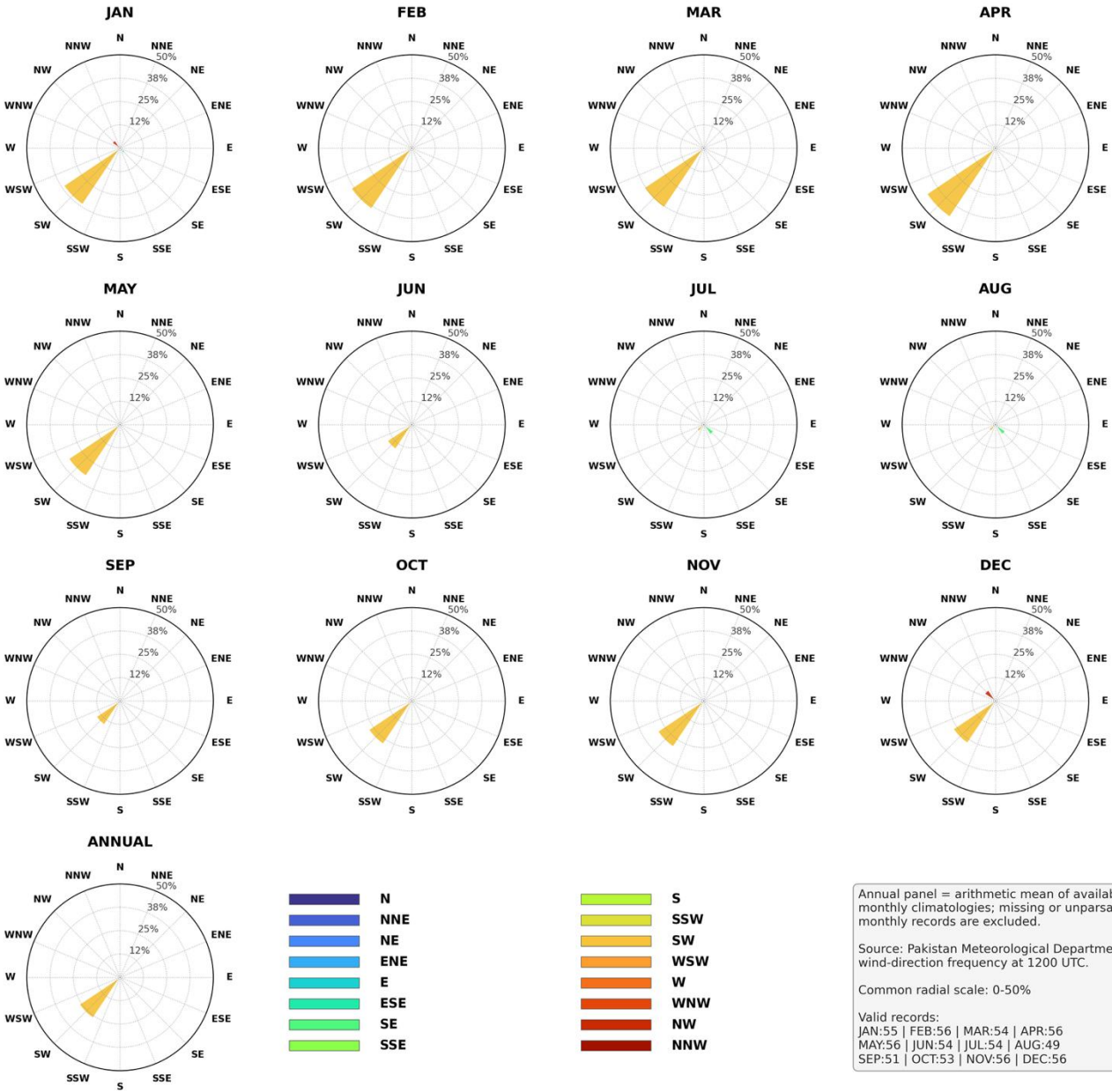
# Wind Rose Climatology: JHELUM

1948-2000 (53 years) | 1200 UTC | Long-term mean frequency (%)



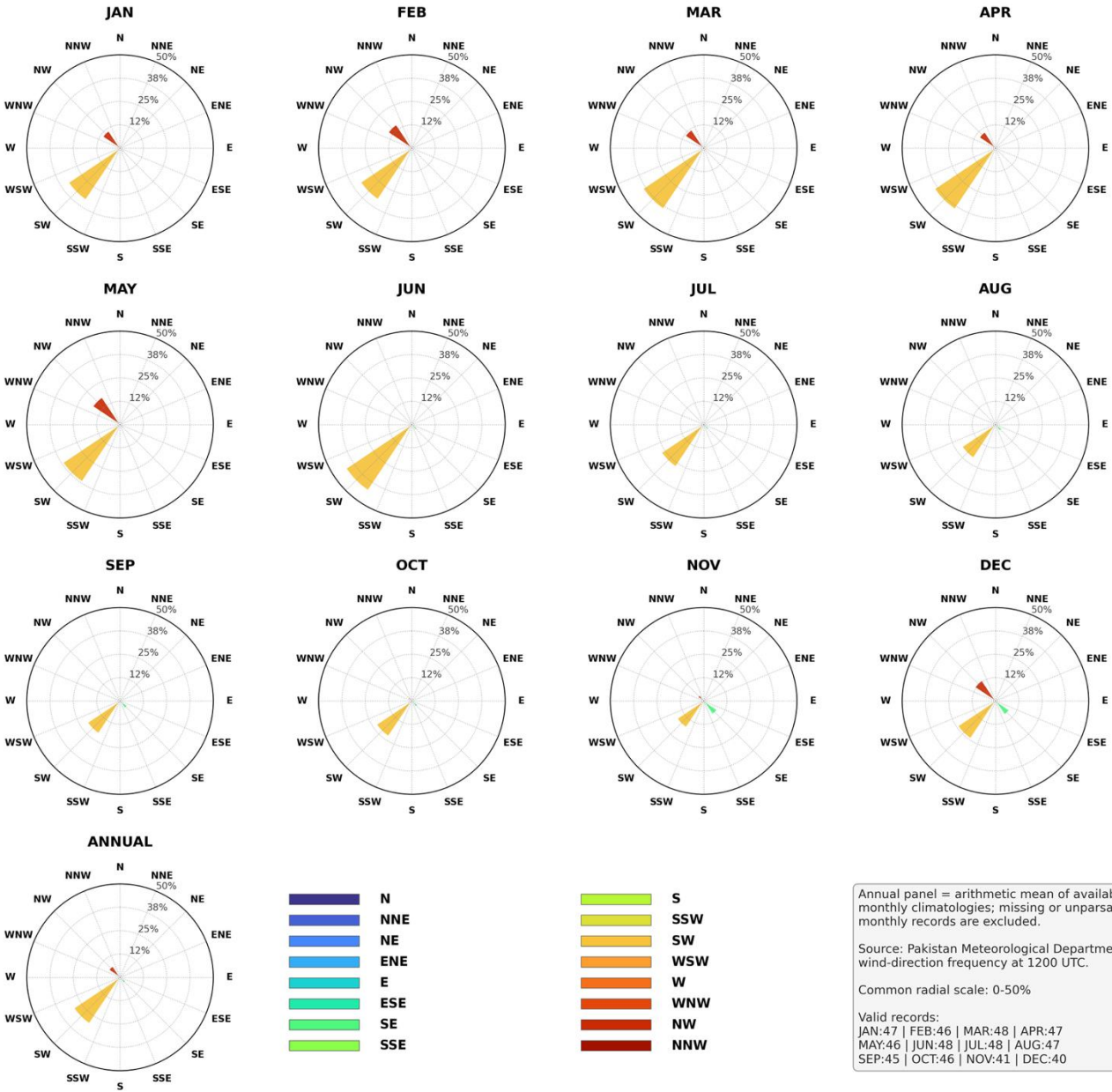
# Wind Rose Climatology: JIWANI

1938-2000 (63 years) | 1200 UTC | Long-term mean frequency (%)



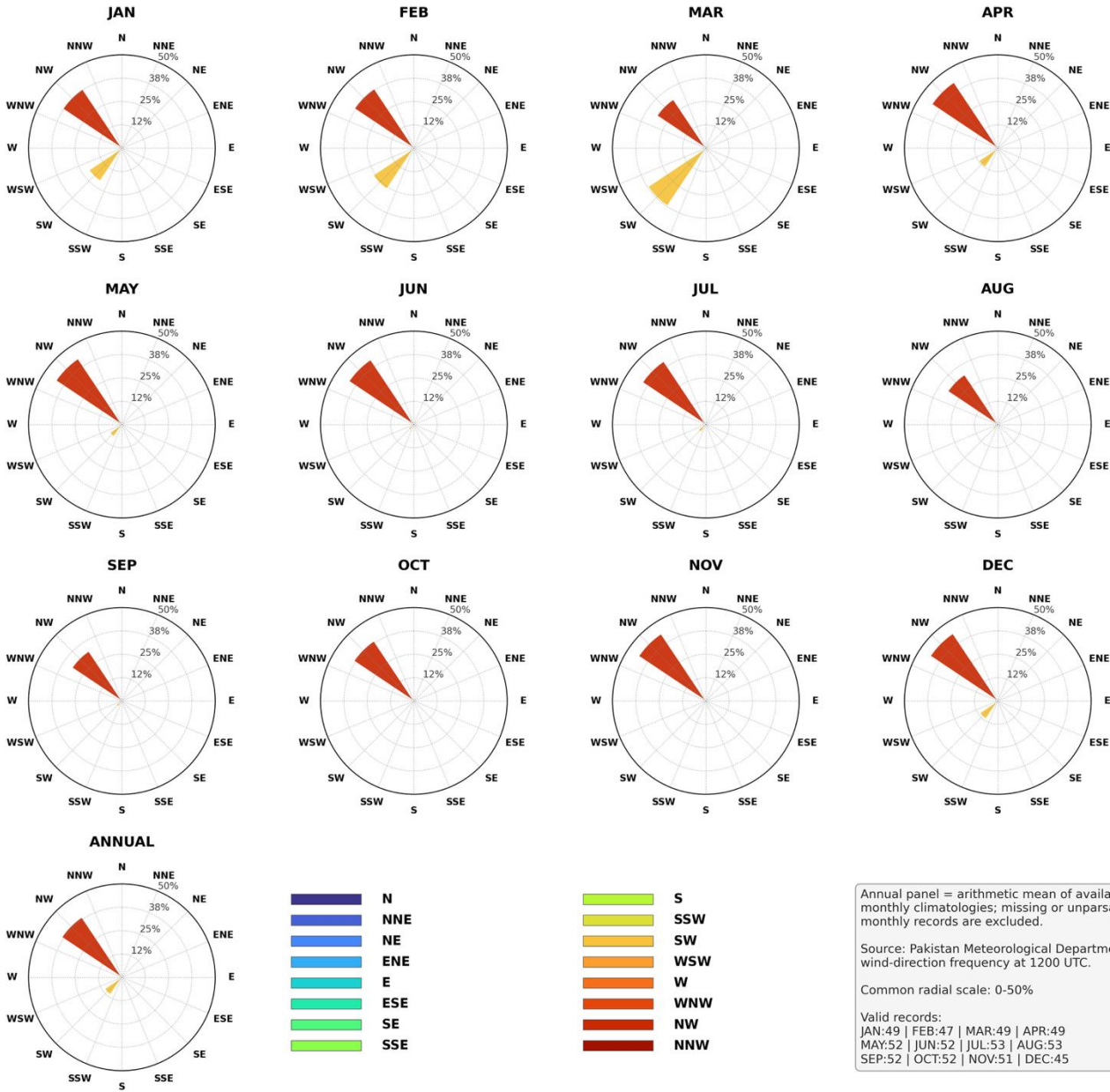
# Wind Rose Climatology: KAKUL

1953-2000 (48 years) | 1200 UTC | Long-term mean frequency (%)



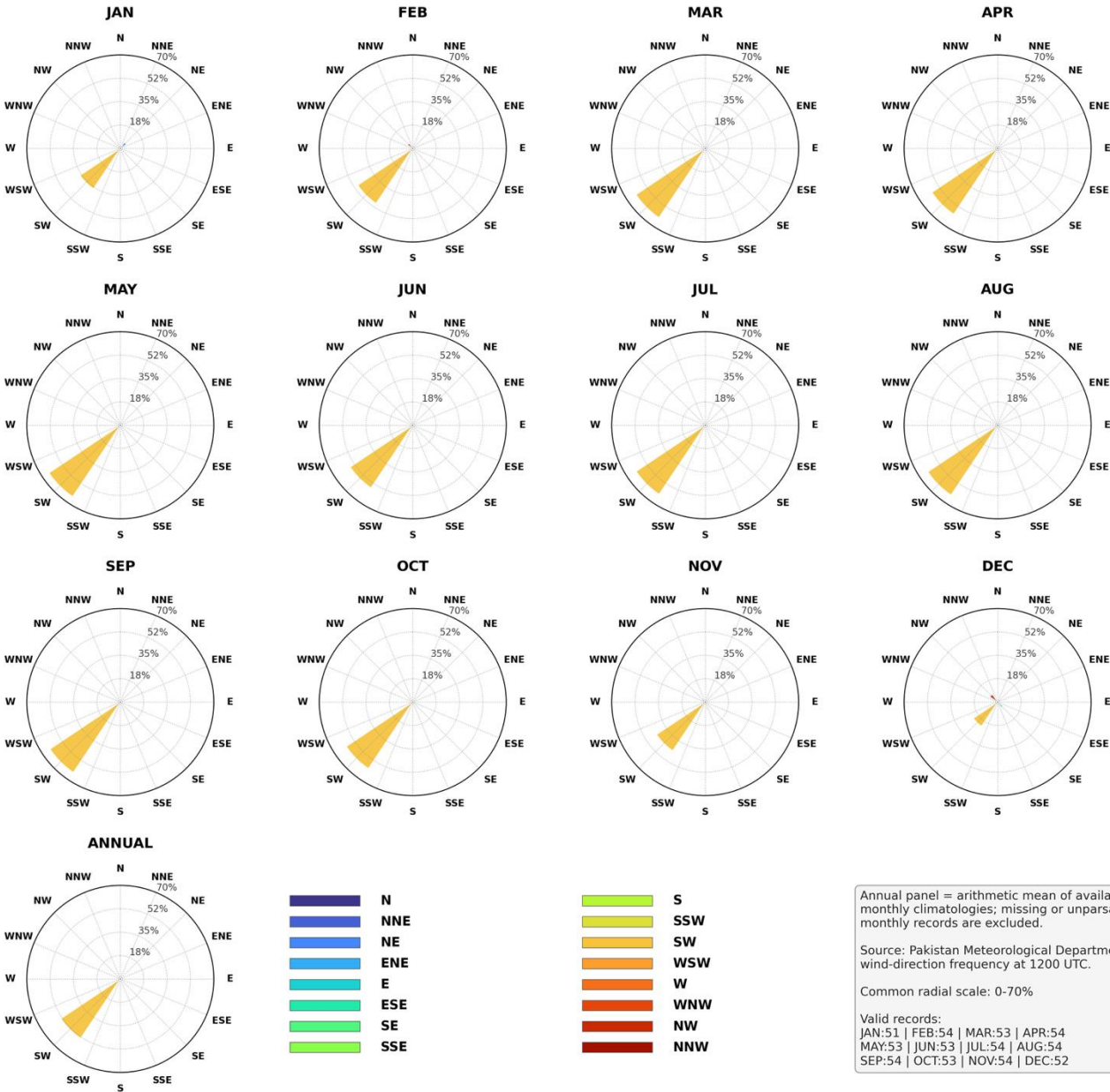
# Wind Rose Climatology: KALAT

1939-2000 (62 years) | 1200 UTC | Long-term mean frequency (%)



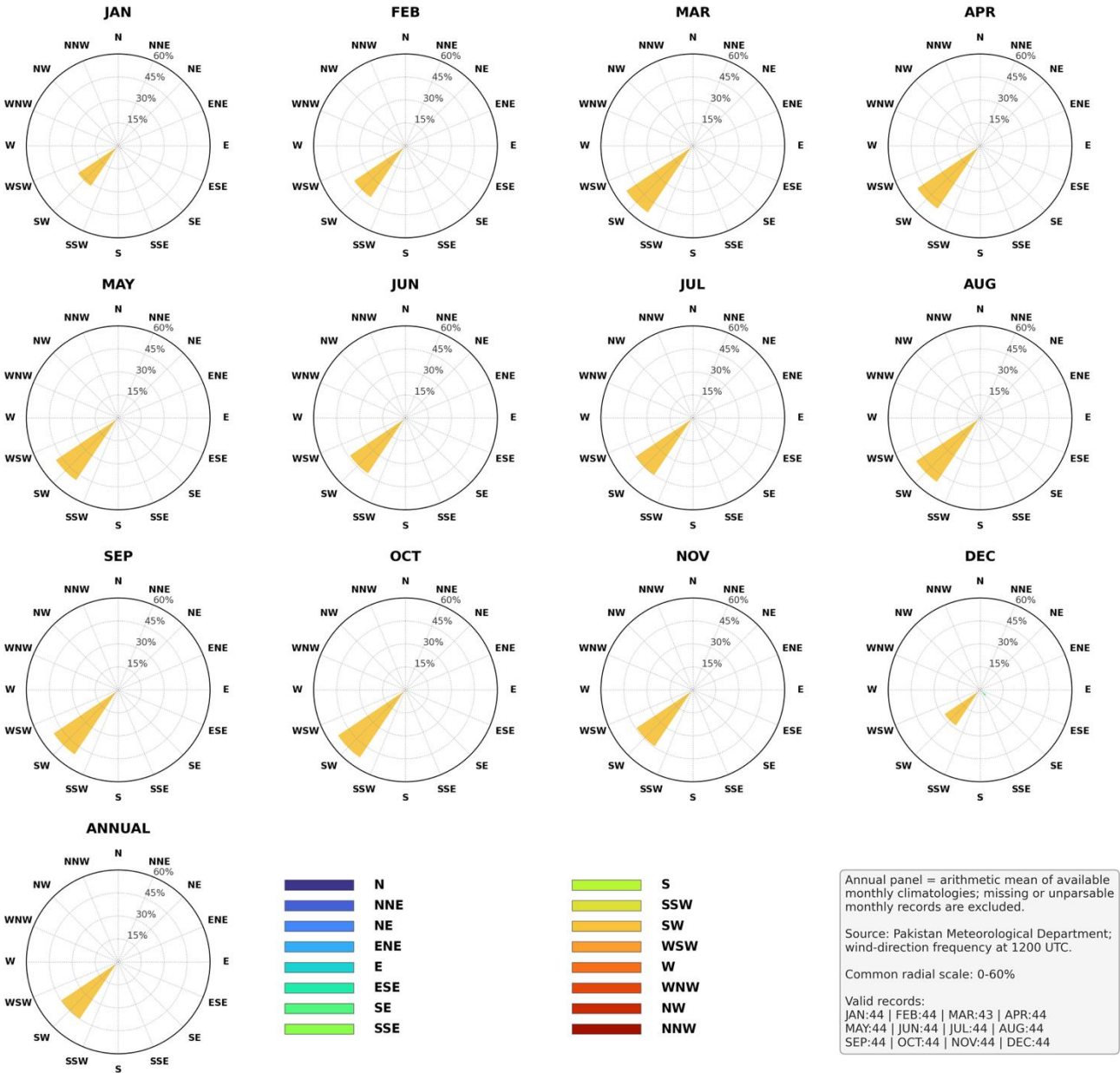
Wind Rose Climatology: KARACHI (AIRPORT)

1945-2000 (56 years) | 1200 UTC | Long-term mean frequency (%)



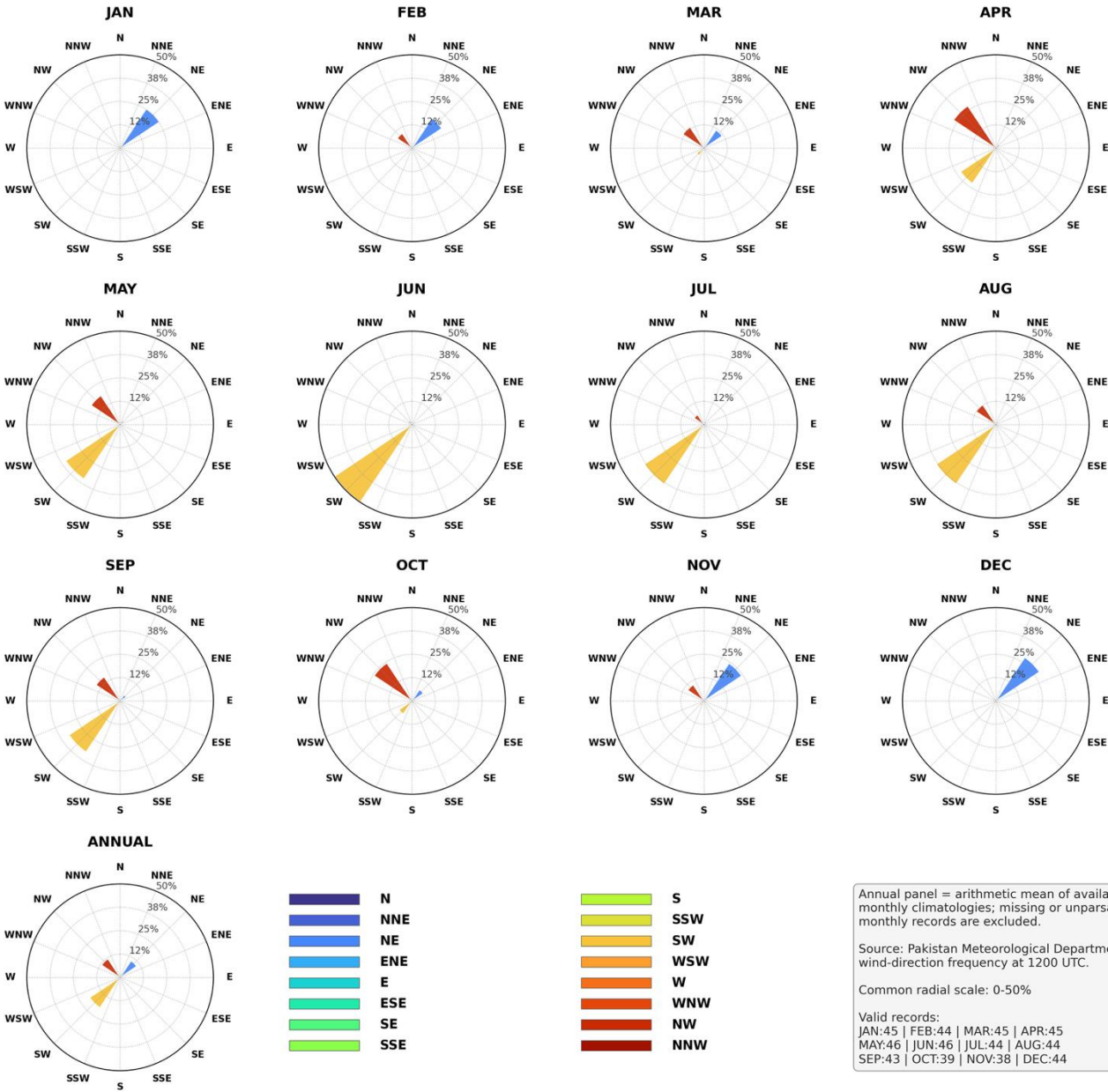
Wind Rose Climatology: KARACHI (MASROOR)

1957-2000 (44 years) | 1200 UTC | Long-term mean frequency (%)



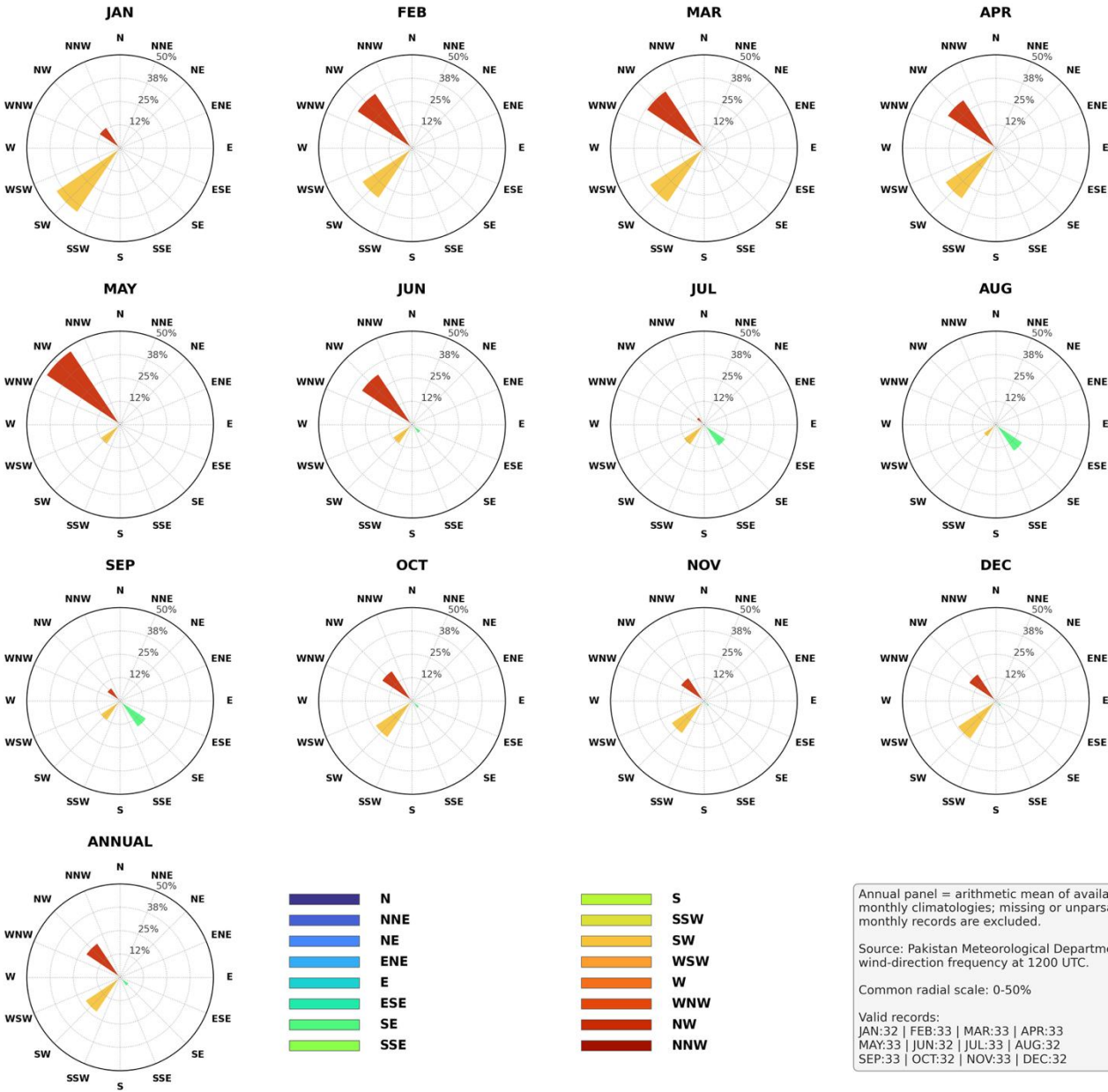
Wind Rose Climatology: KHANPUR (PBO)

1954-2000 (47 years) | 1200 UTC | Long-term mean frequency (%)



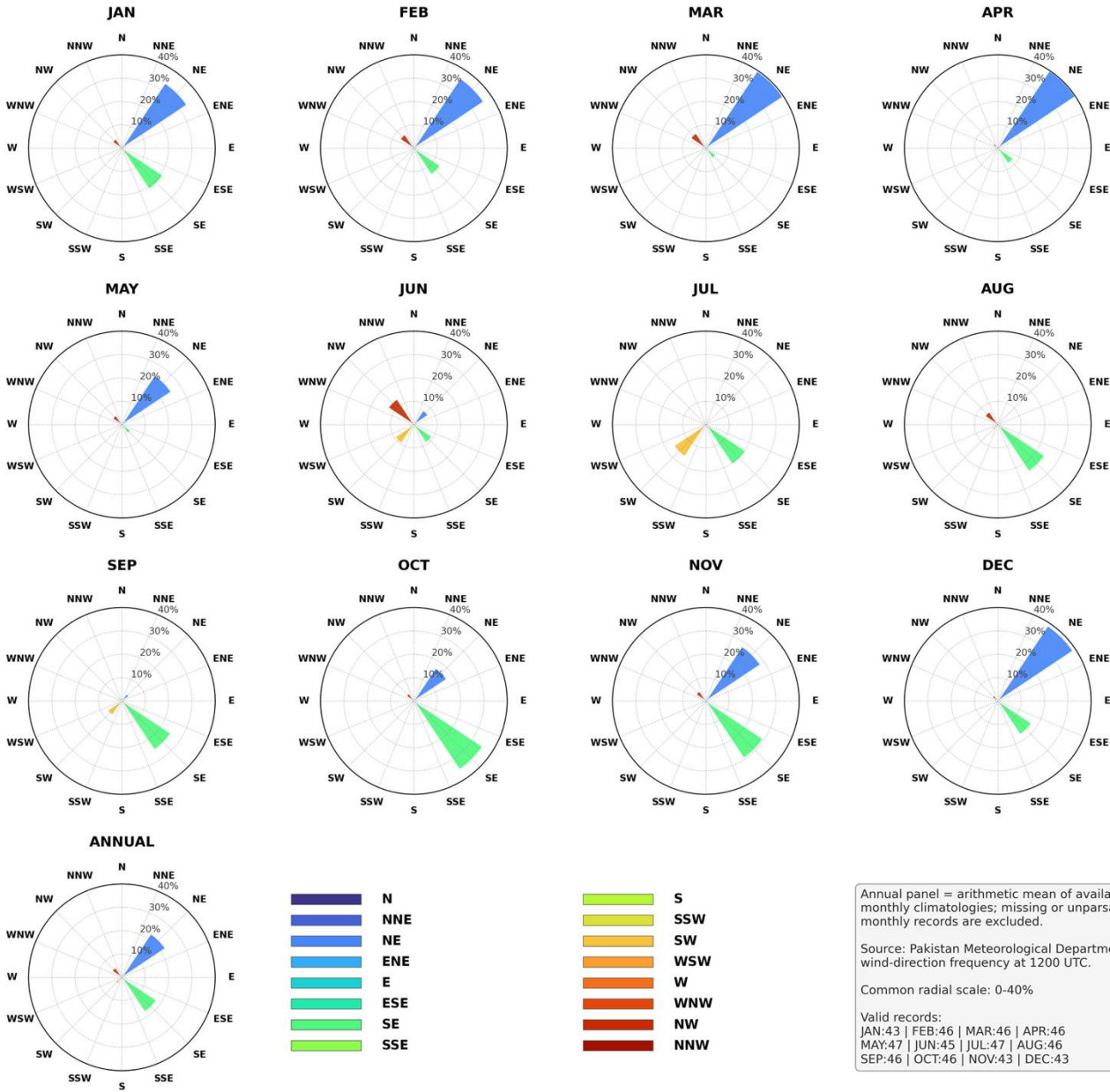
Wind Rose Climatogy: KHUZDAR

1967-2000 (34 years) | 1200 UTC | Long-term mean frequency (%)



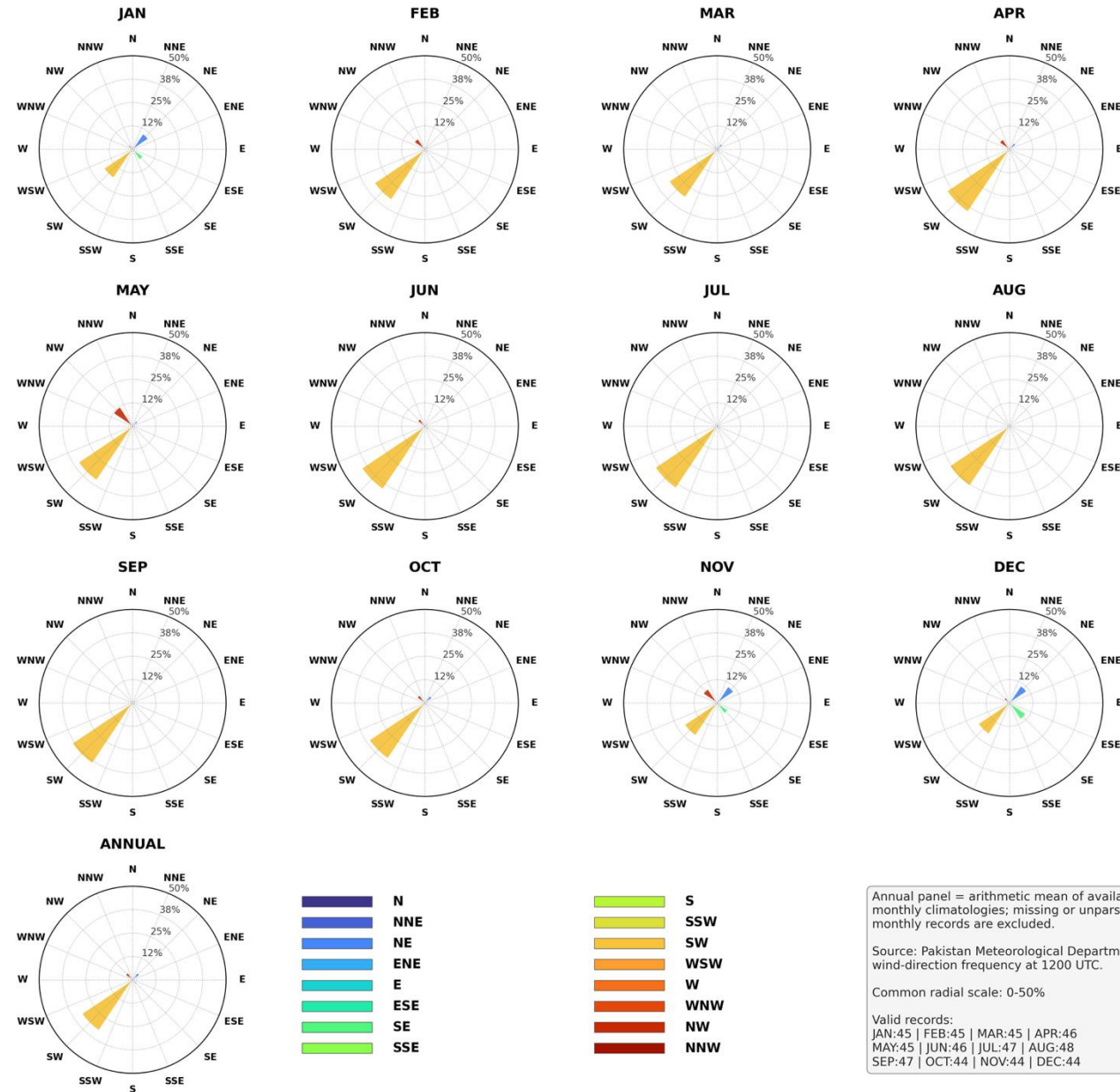
# Wind Rose Climatology: KOHAT

1954-2000 (47 years) | 1200 UTC | Long-term mean frequency (%)



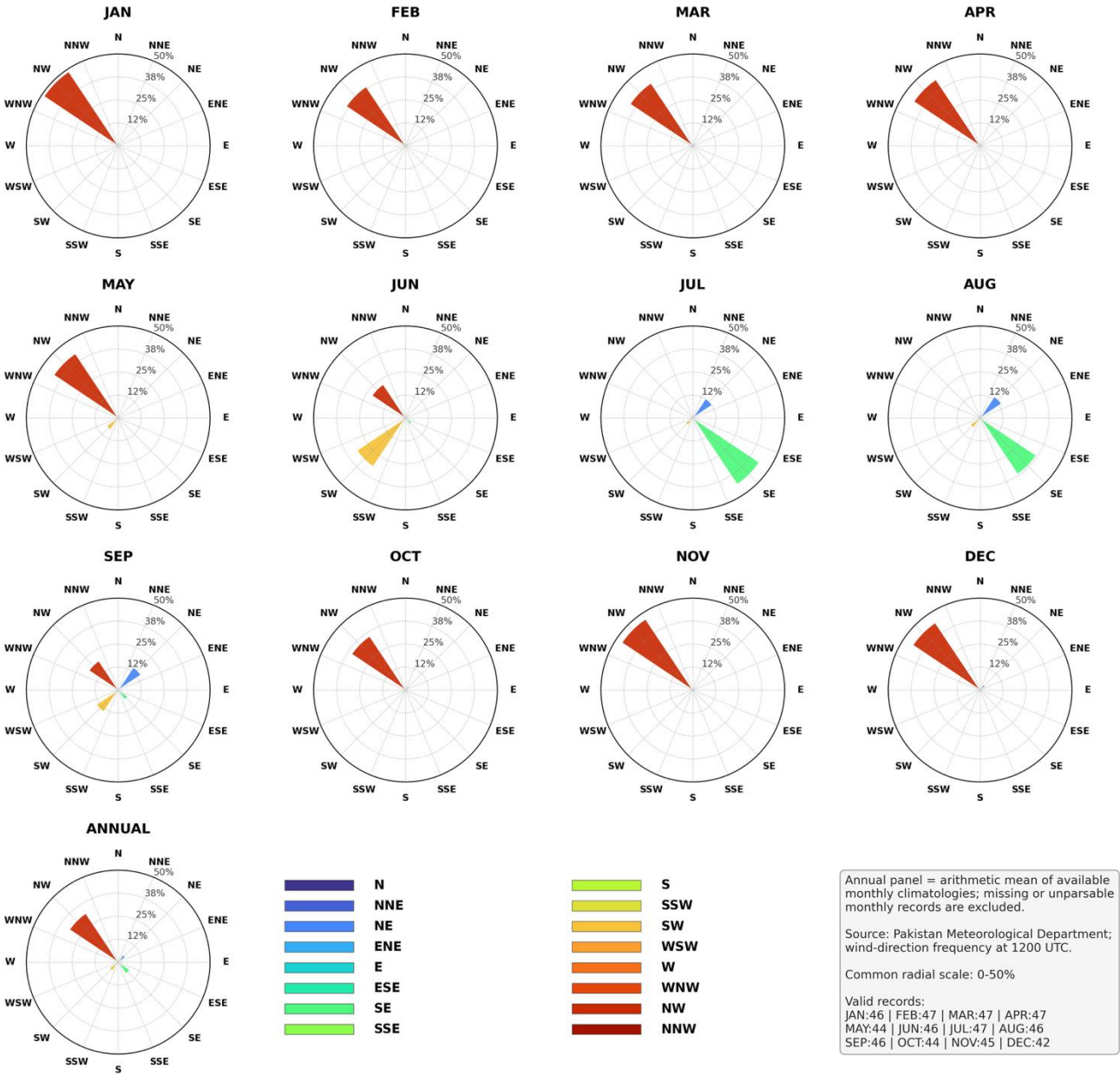
# Wind Rose Climatology: KOTLI

1952-2000 (49 years) | 1200 UTC | Long-term mean frequency (%)



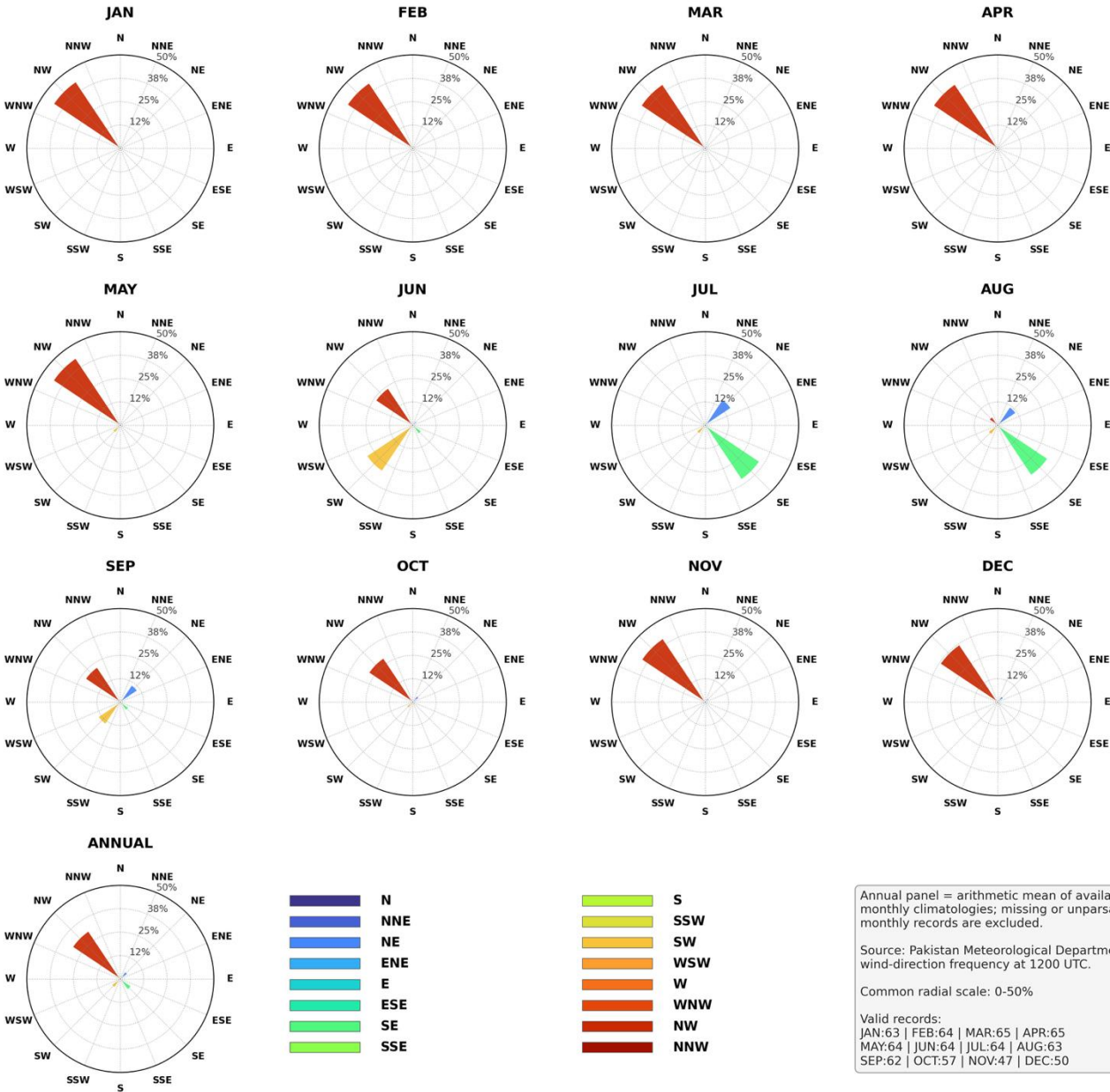
Wind Rose Climatology: LAHORE(AIRPORT)

1953-2000 (48 years) | 1200 UTC | Long-term mean frequency (%)



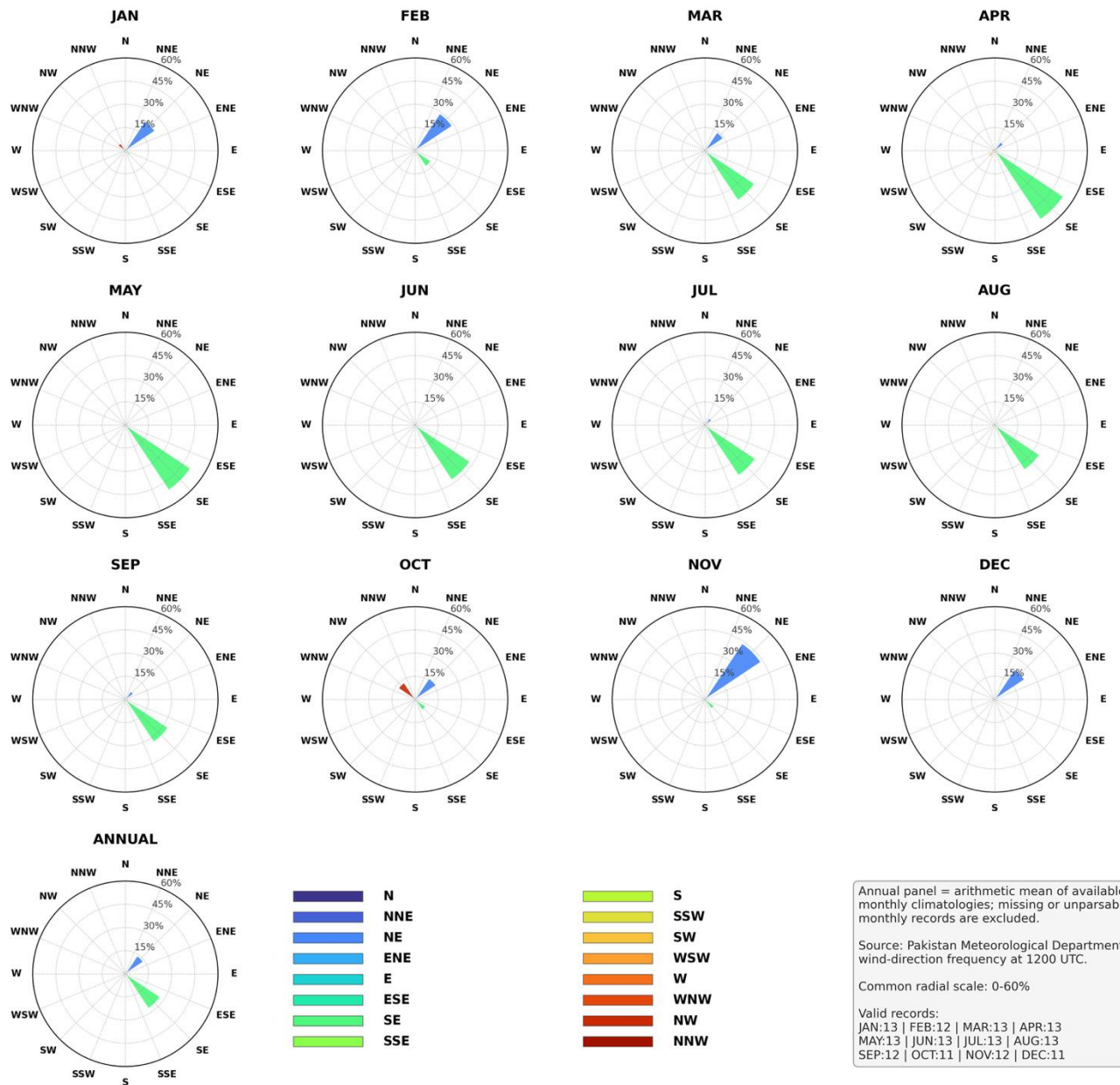
Wind Rose Climatology: LAHORE(PBO)

1933-2000 (68 years) | 1200 UTC | Long-term mean frequency (%)



Wind Rose Climatology: LARKANA

1988-2000 (13 years) | 1200 UTC | Long-term mean frequency (%)



Annual panel = arithmetic mean of available monthly climatologies; missing or unparsable monthly records are excluded.

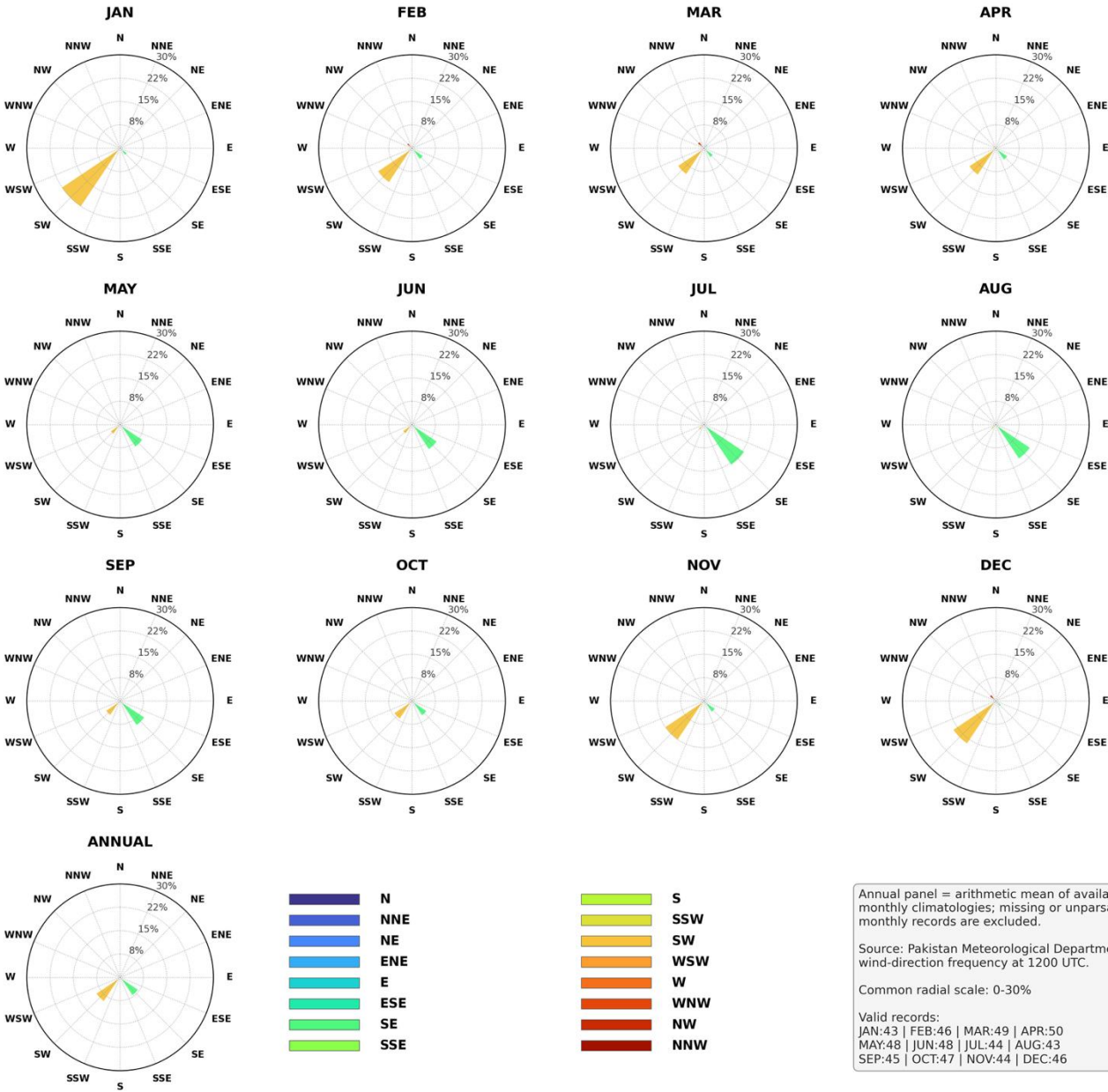
Source: Pakistan Meteorological Department; wind-direction frequency at 1200 UTC.

Common radial scale: 0-60%

Valid records:  
JAN:13 | FEB:12 | MAR:13 | APR:13  
MAY:13 | JUN:13 | JUL:13 | AUG:13  
SEP:12 | OCT:11 | NOV:12 | DEC:11

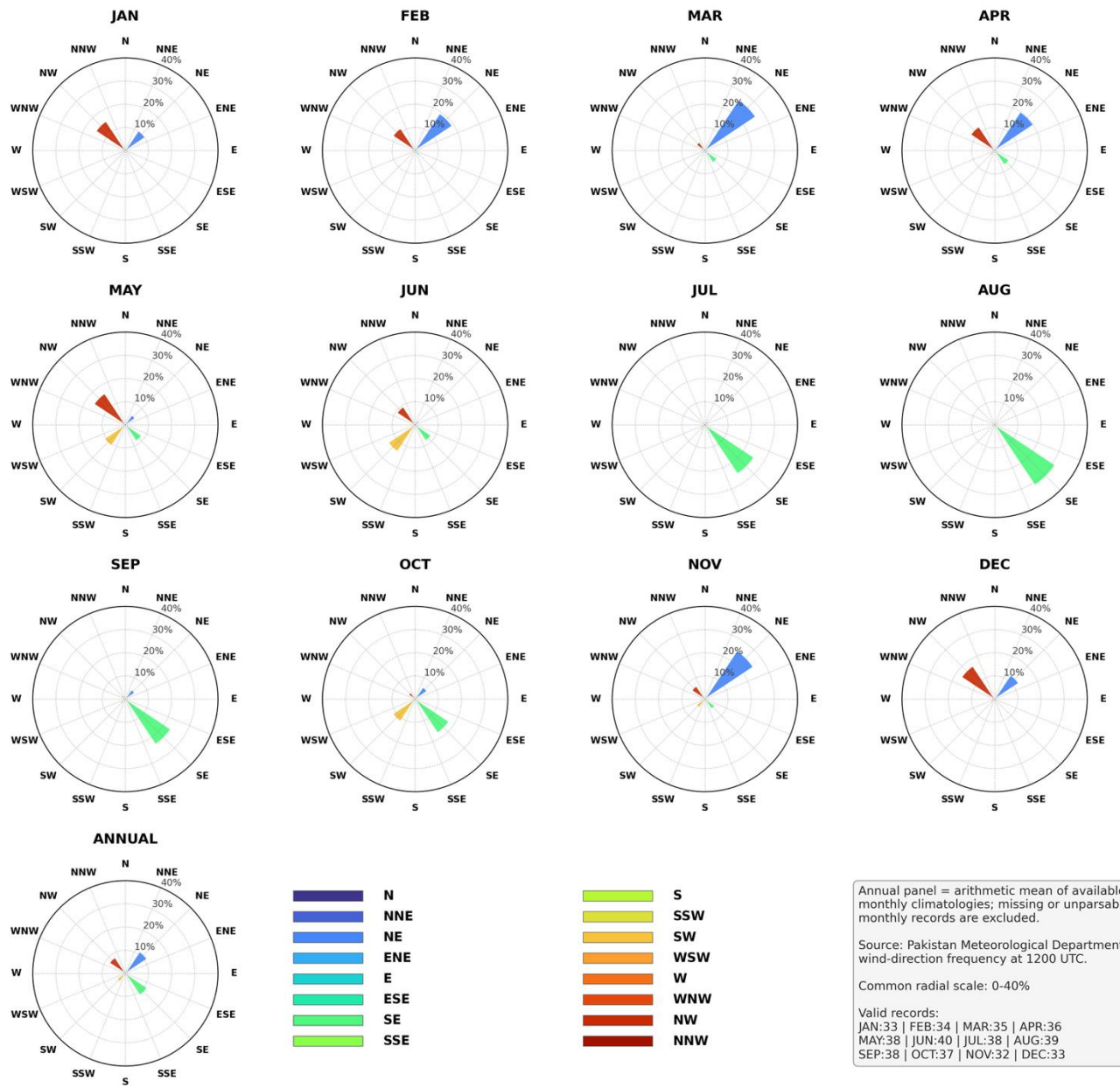
Wind Rose Climatology: LASBELLA

1933-2000 (68 years) | 1200 UTC | Long-term mean frequency (%)



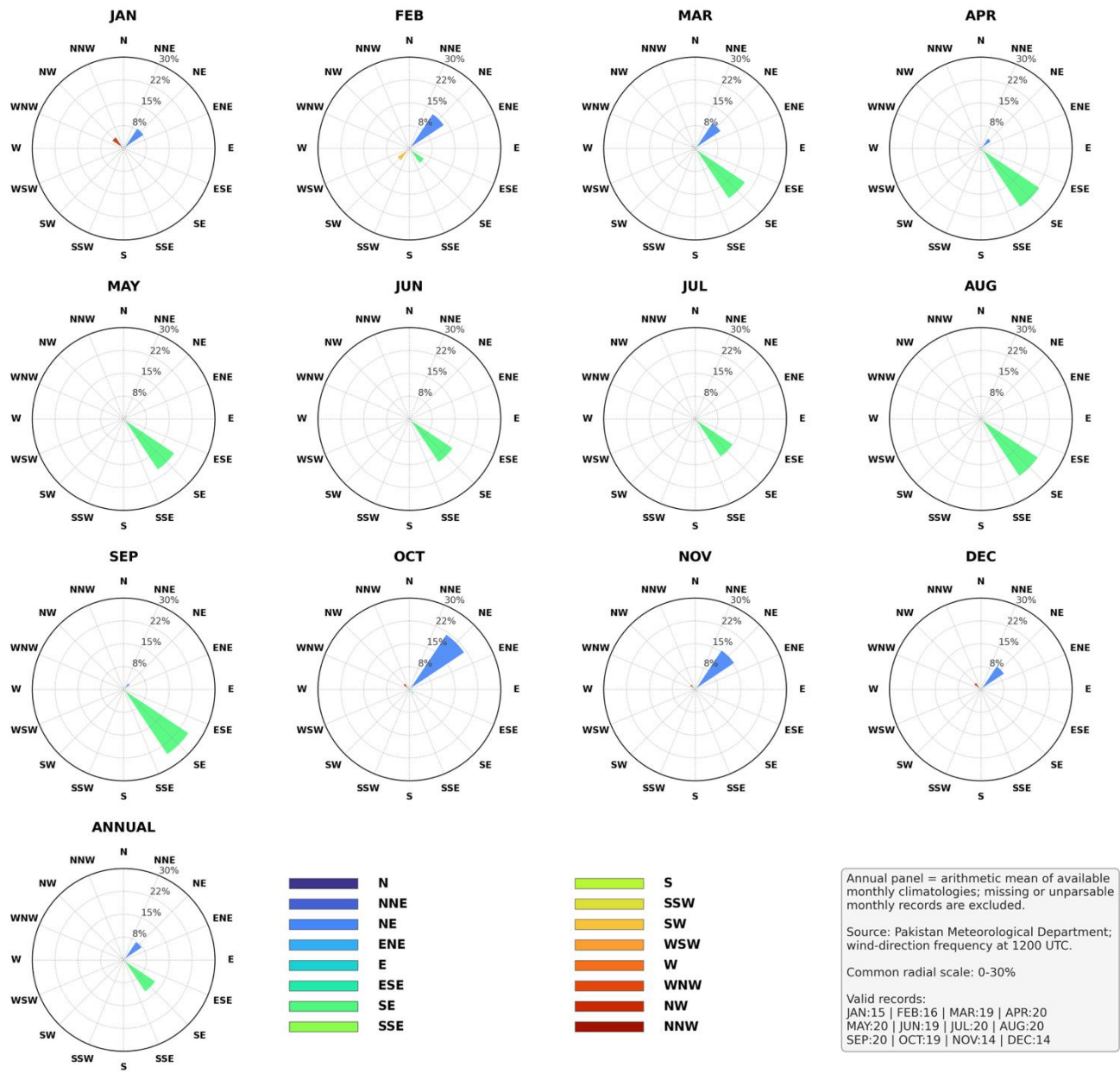
Wind Rose Climatology: MIANWALI

1959-2000 (42 years) | 1200 UTC | Long-term mean frequency (%)



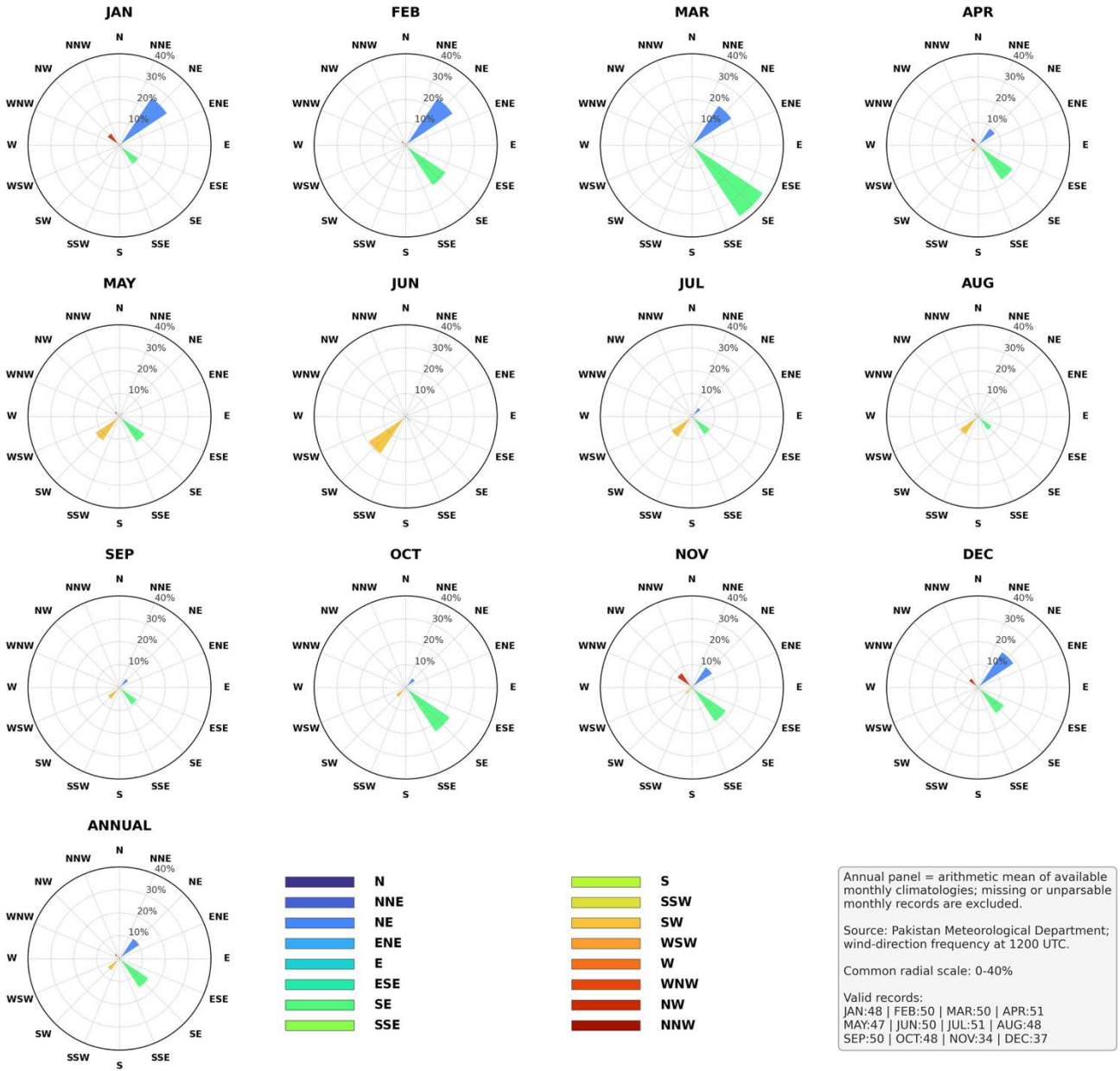
Wind Rose Climatology: MOEN-JO-DARO

1979-2000 (22 years) | 1200 UTC | Long-term mean frequency (%)



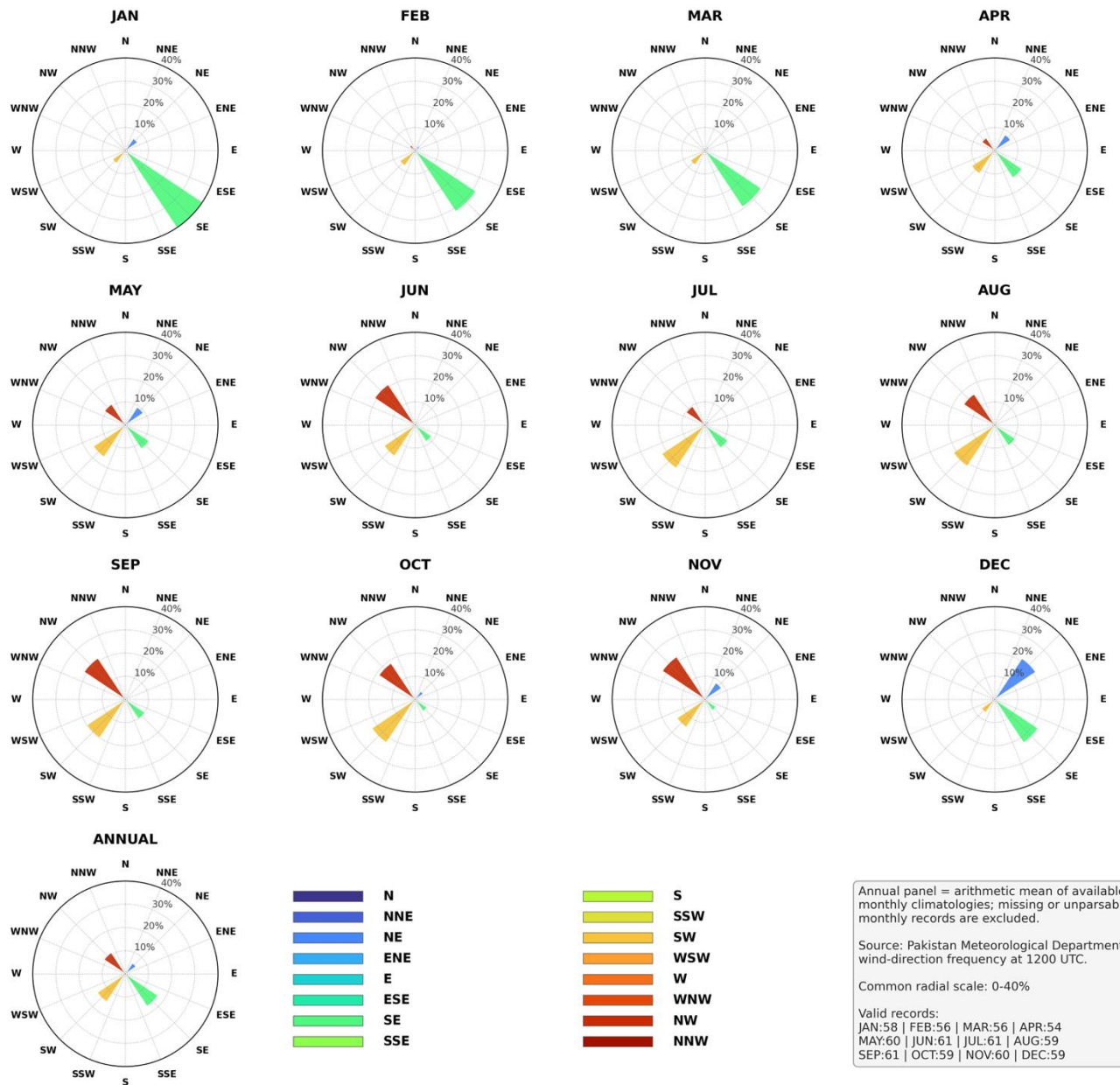
Wind Rose Climatology: MULTAN(PBO)

1950-2000 (51 years) | 1200 UTC | Long-term mean frequency (%)



# Wind Rose Climatology: MURREE

1933-2000 (68 years) | 1200 UTC | Long-term mean frequency (%)



Annual panel = arithmetic mean of available monthly climatologies; missing or unparsable monthly records are excluded.

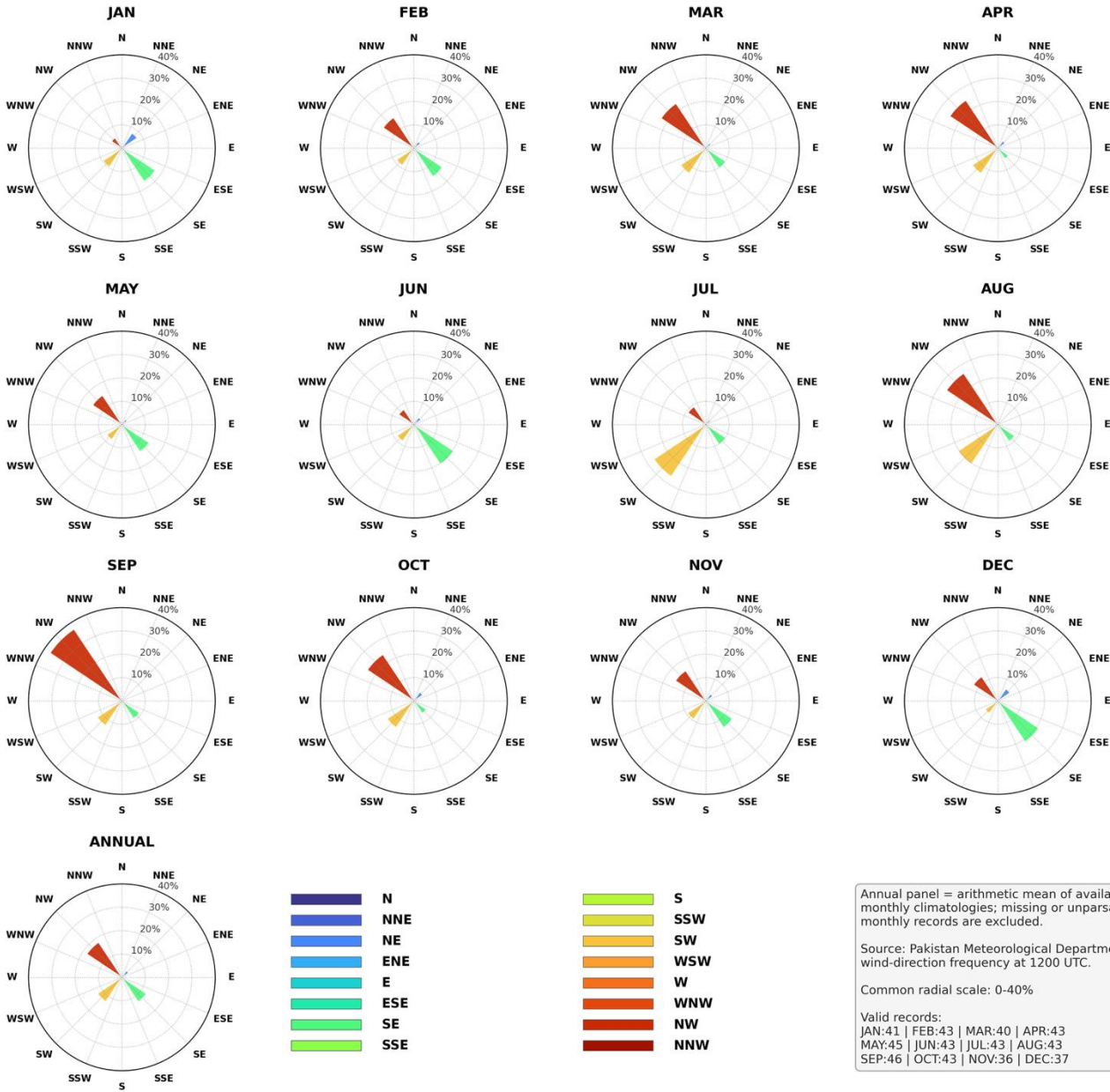
Source: Pakistan Meteorological Department; wind-direction frequency at 1200 UTC.

Common radial scale: 0-40%

Valid records:  
JAN:58 | FEB:56 | MAR:56 | APR:54  
MAY:60 | JUN:61 | JUL:61 | AUG:59  
SEP:61 | OCT:59 | NOV:60 | DEC:59

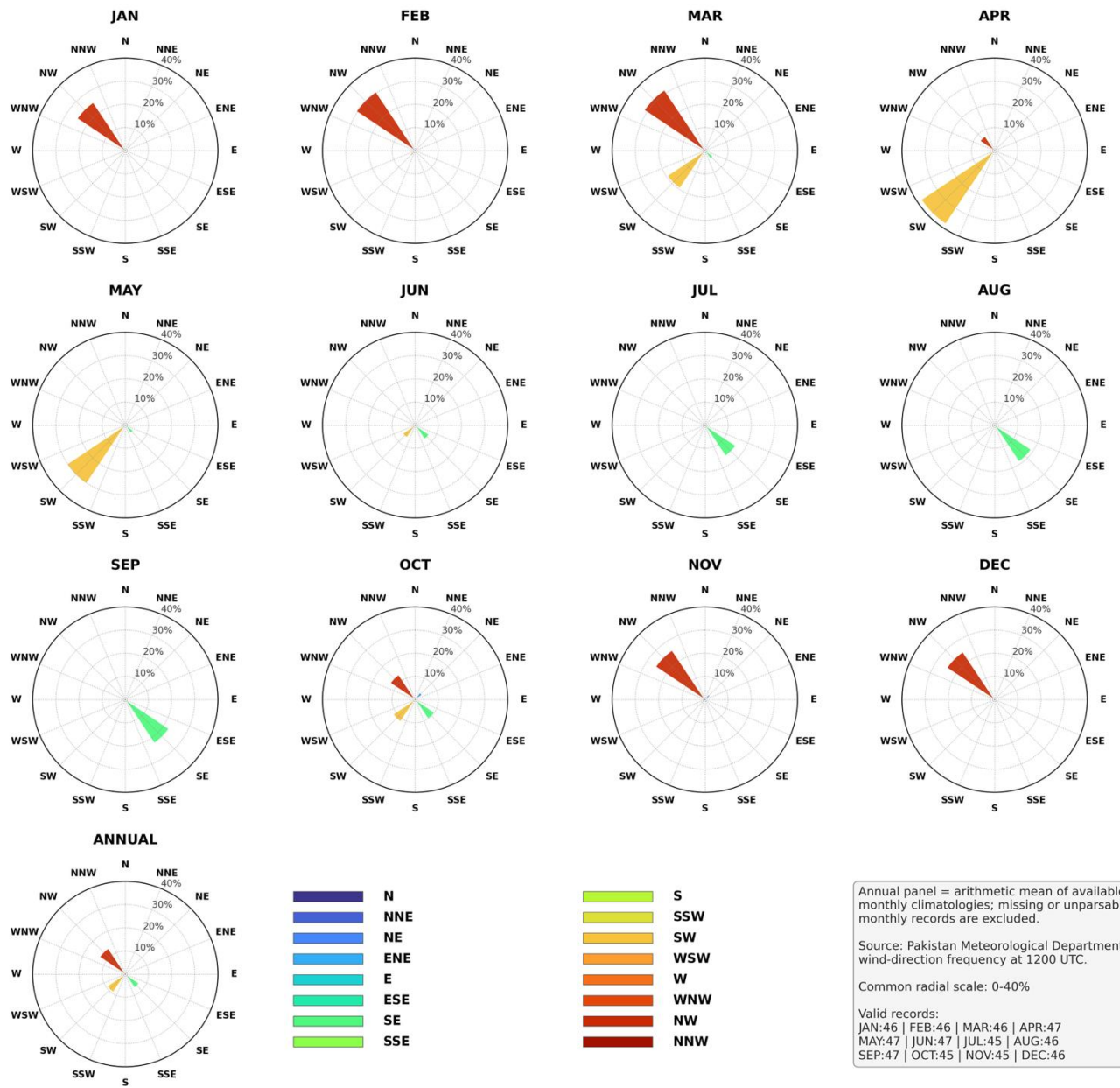
Wind Rose Climatology: MUZAFFARABAD

1955-2000 (46 years) | 1200 UTC | Long-term mean frequency (%)



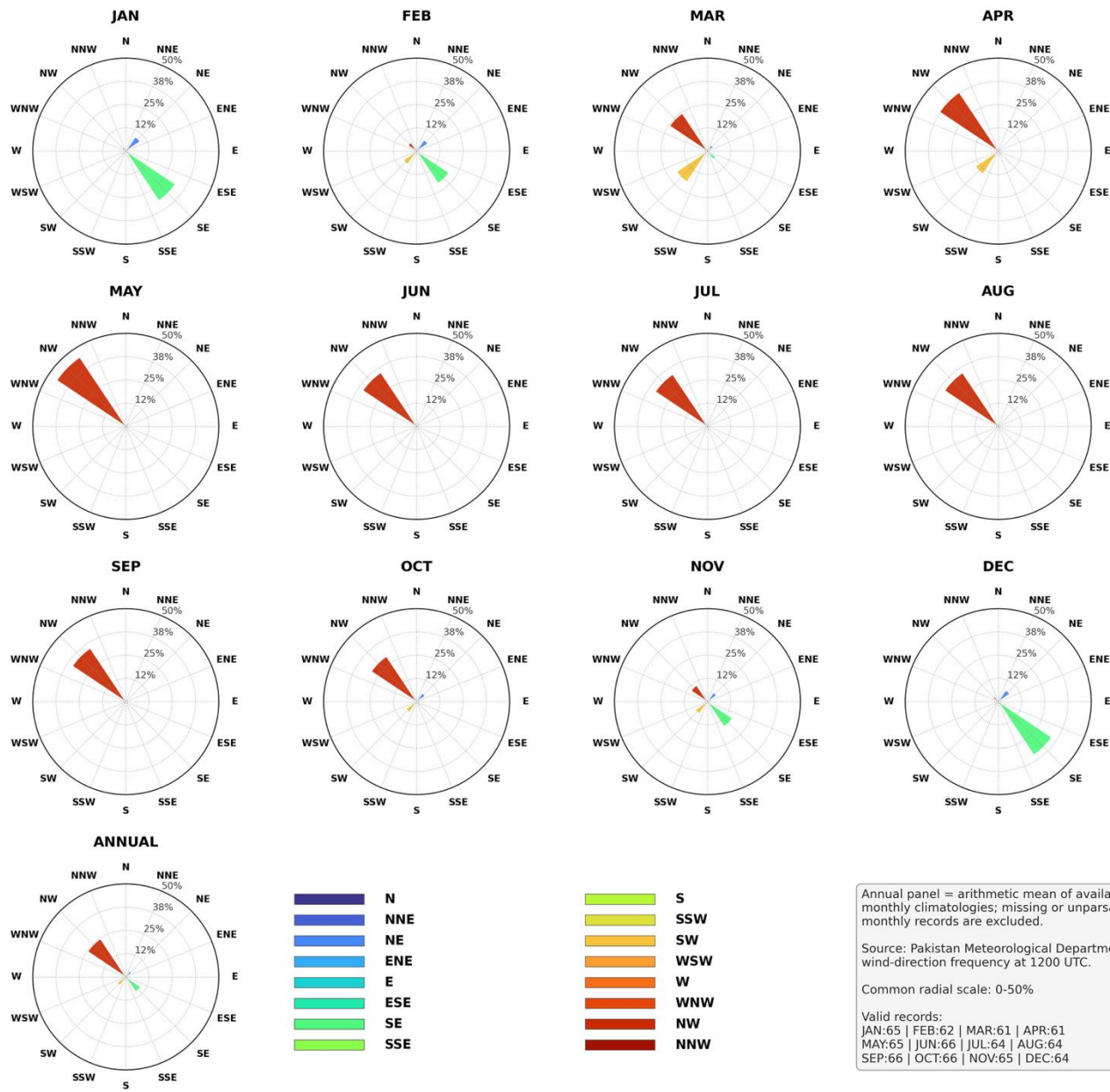
Wind Rose Climatology: NAWABSHAH

1954-2000 (47 years) | 1200 UTC | Long-term mean frequency (%)



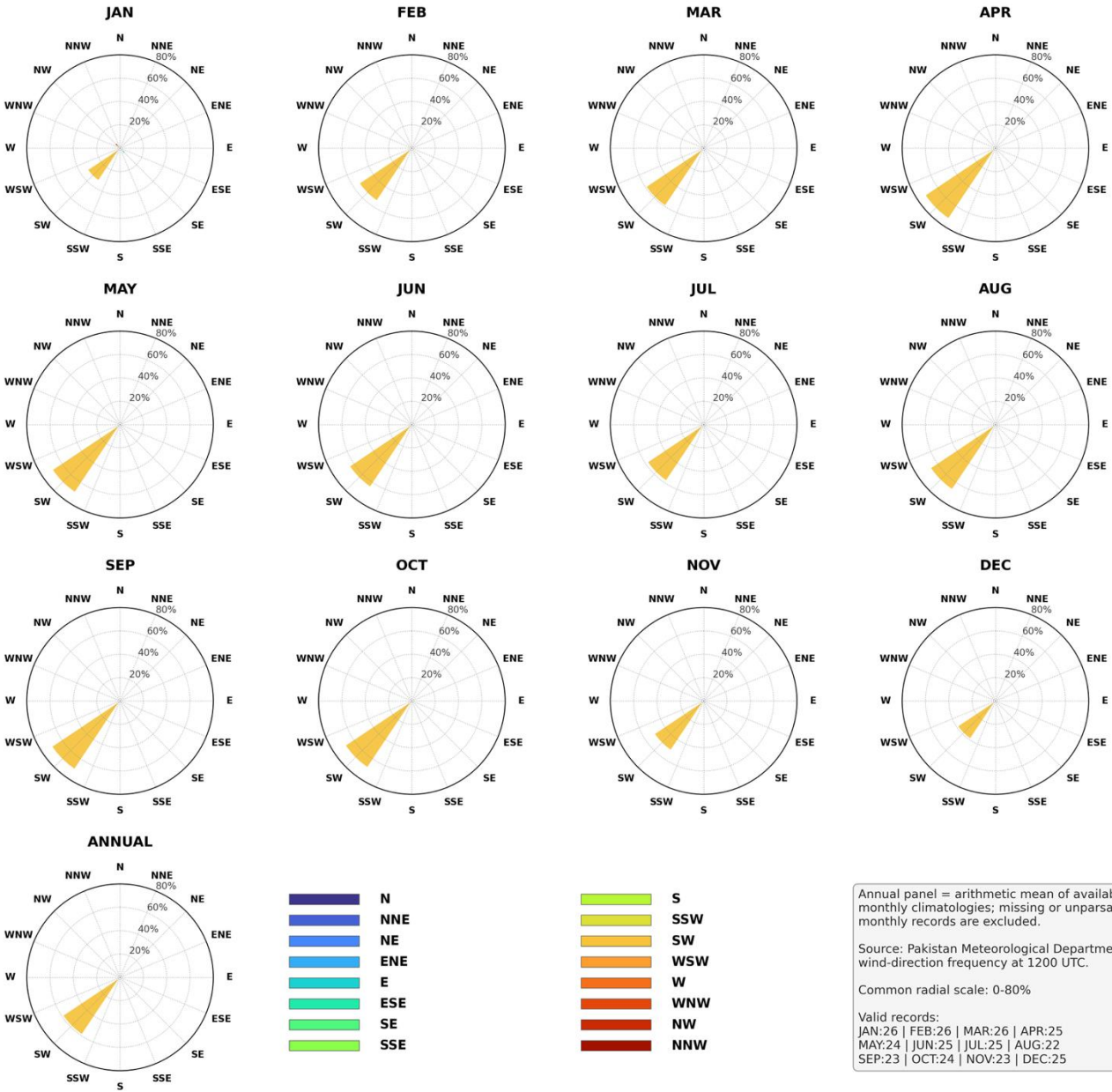
# Wind Rose Climatology: NOKKUNDI

1933-2000 (68 years) | 1200 UTC | Long-term mean frequency (%)



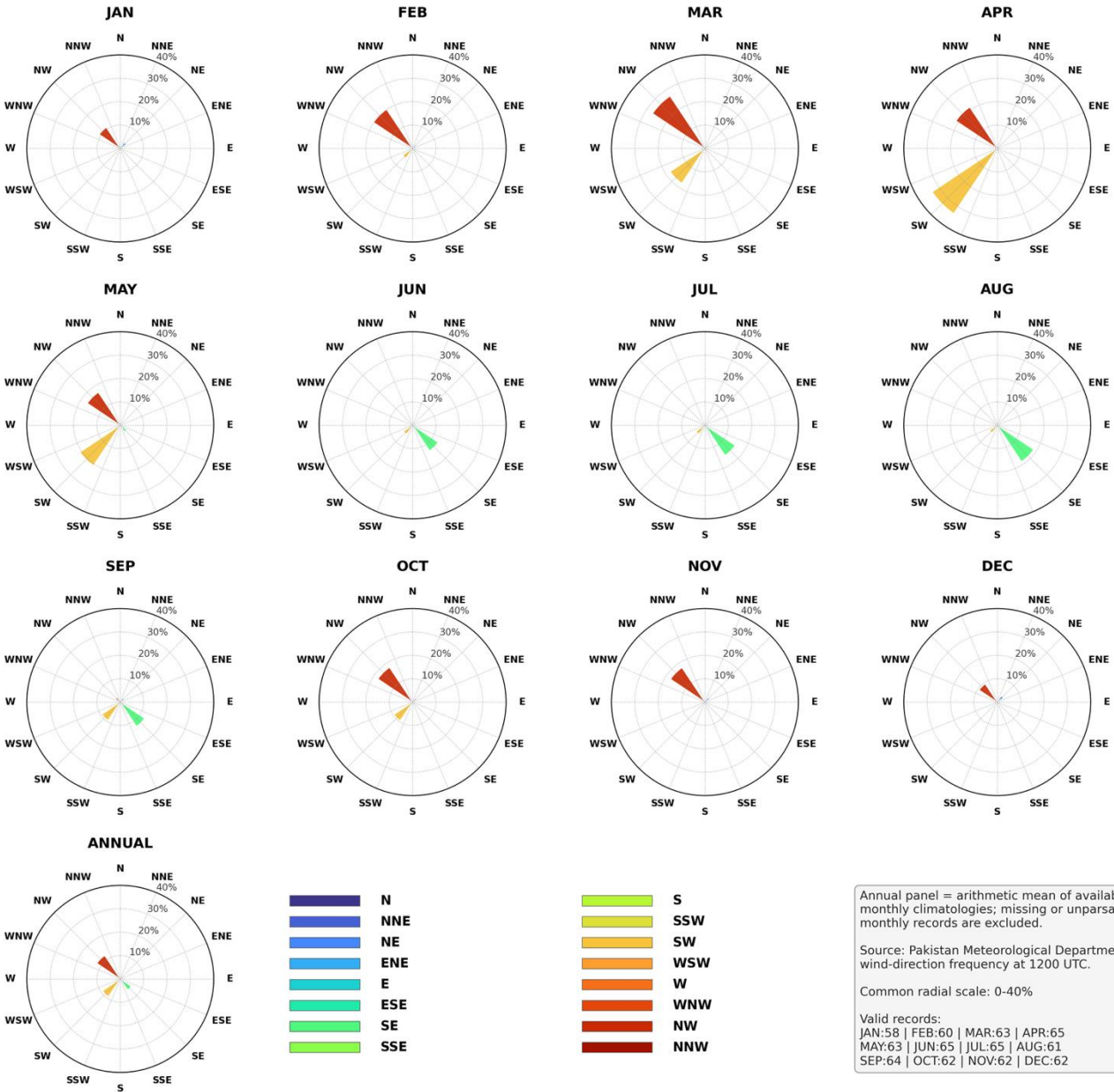
Wind Rose Climatology: ORMARA

1961-2000 (40 years) | 1200 UTC | Long-term mean frequency (%)



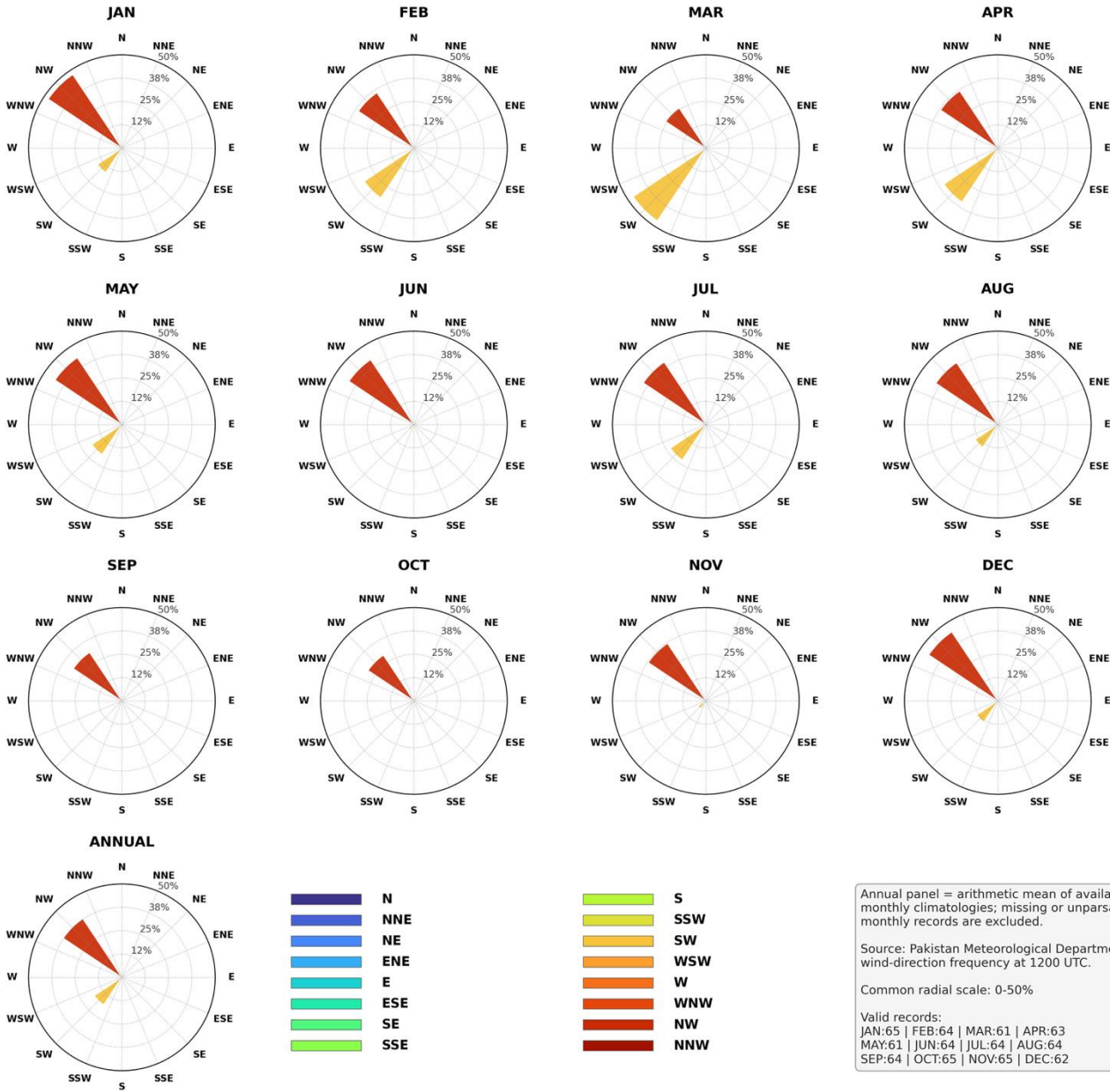
Wind Rose Climatology: PADIDAN

1933-2000 (68 years) | 1200 UTC | Long-term mean frequency (%)



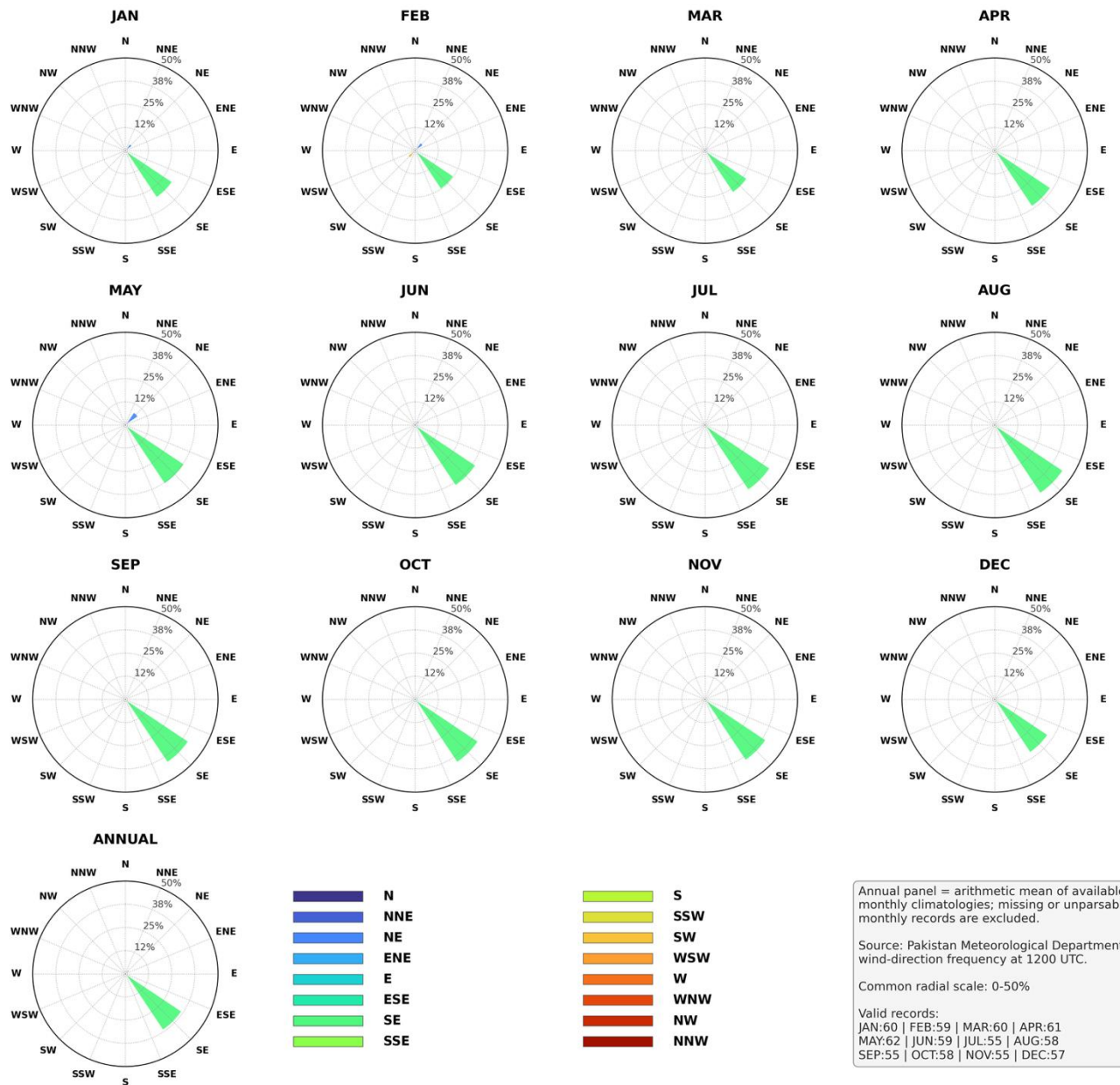
Wind Rose Climatology: PANJGUR

1933-1999 (67 years) | 1200 UTC | Long-term mean frequency (%)



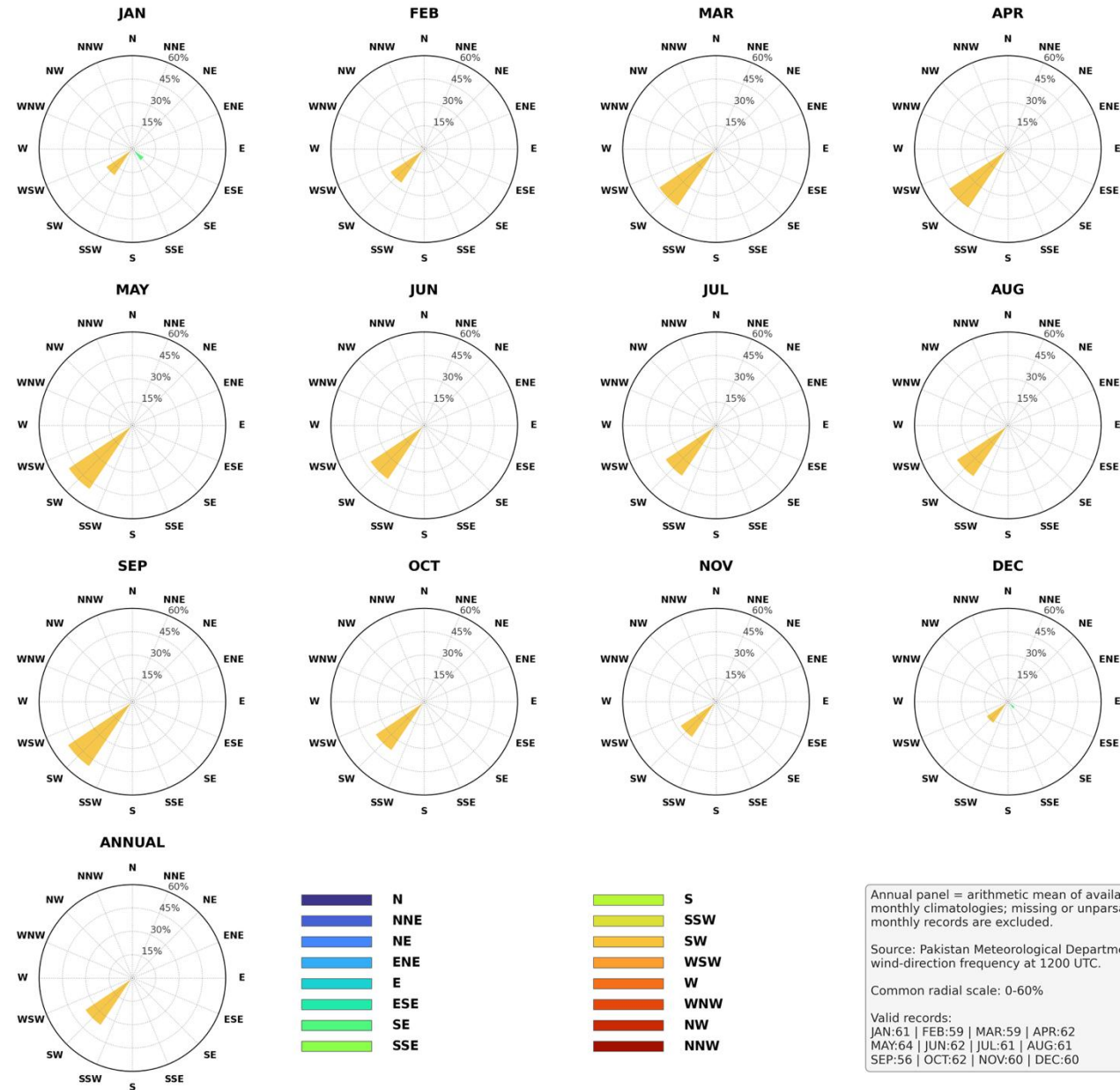
Wind Rose Climatology: PARACHINAR

1933-2000 (68 years) | 1200 UTC | Long-term mean frequency (%)



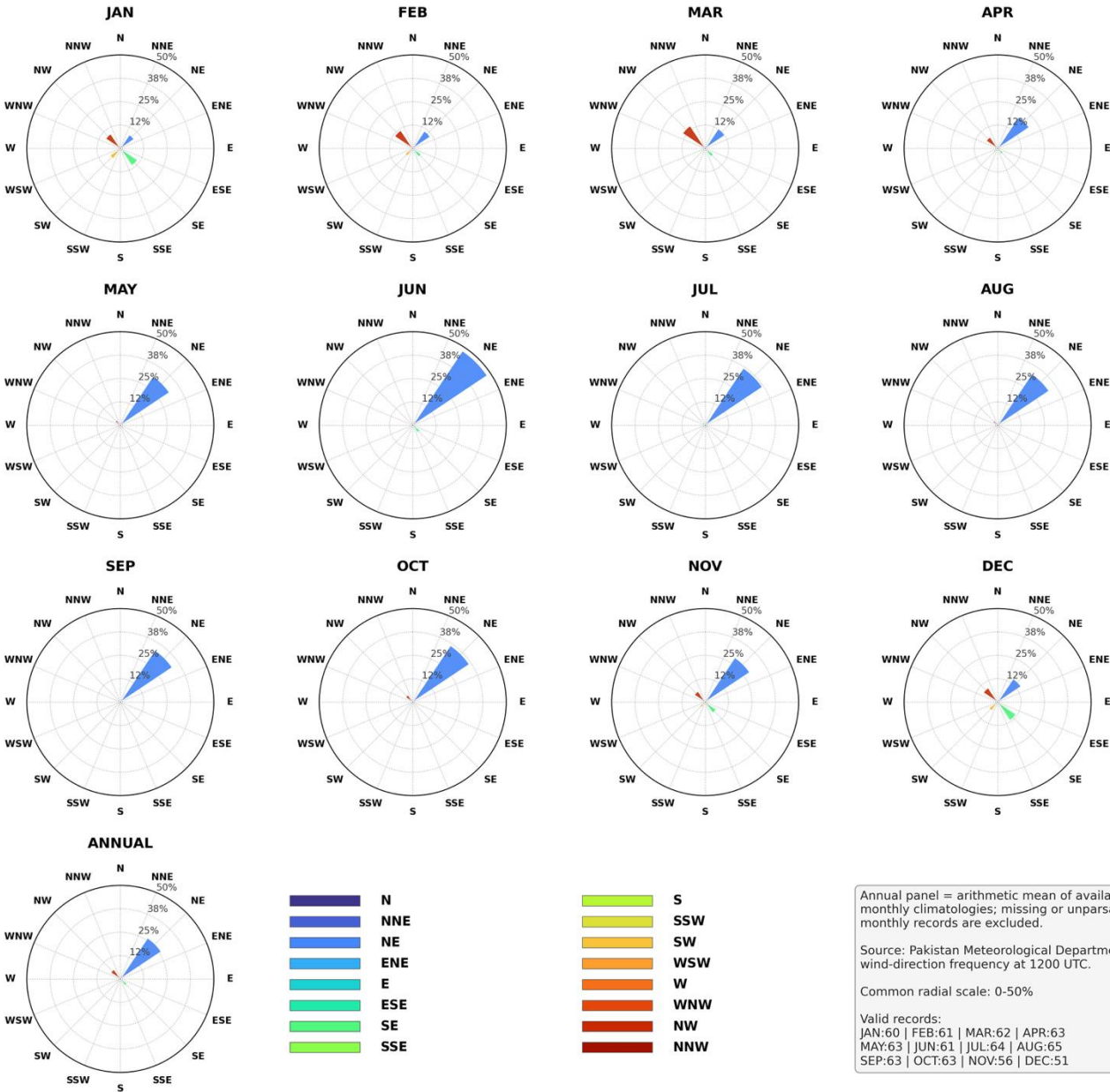
# Wind Rose Climatology: PASNI

1933-2000 (68 years) | 1200 UTC | Long-term mean frequency (%)



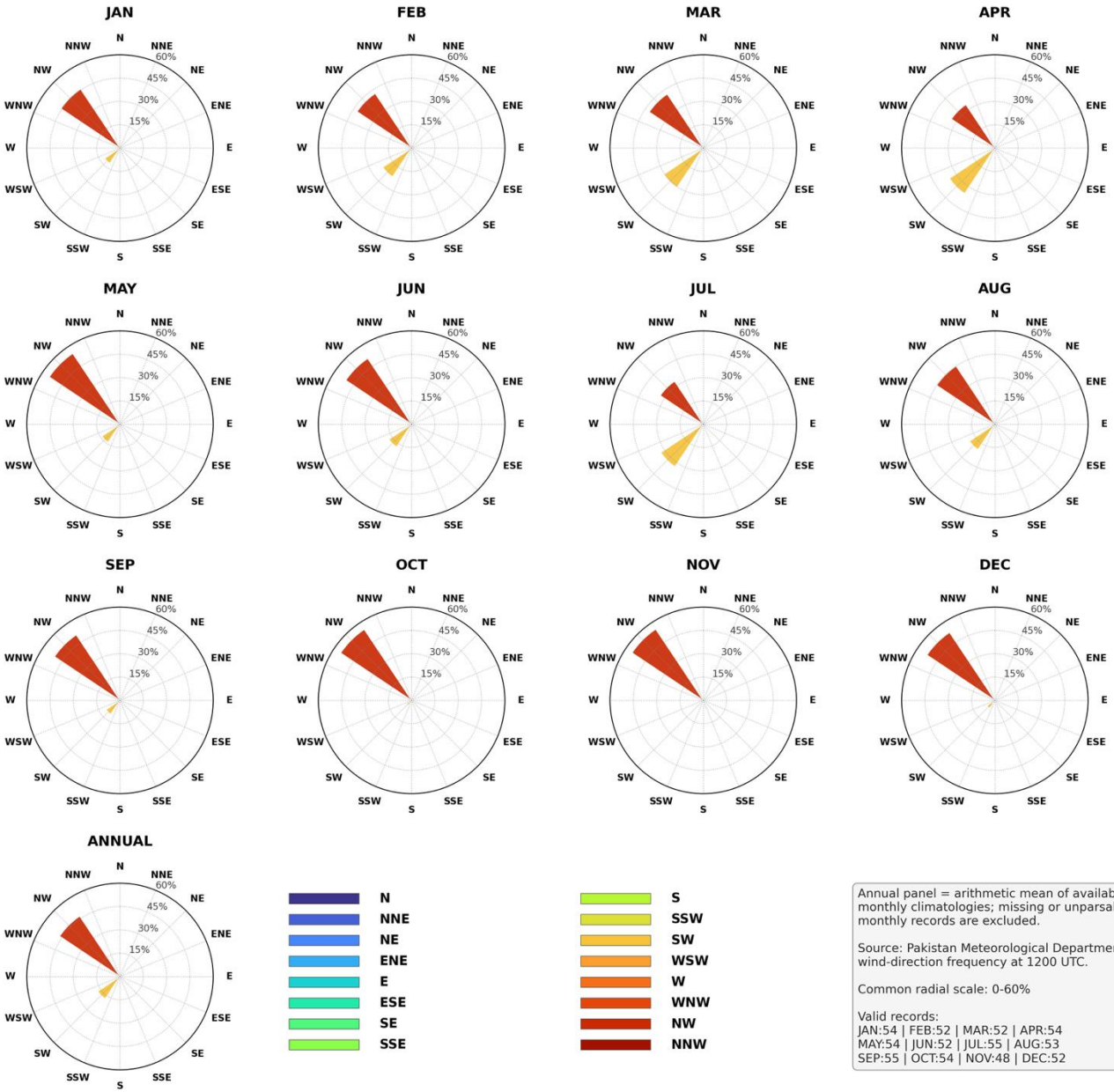
Wind Rose Climatology: PESHAWAR

1933-2000 (68 years) | 1200 UTC | Long-term mean frequency (%)



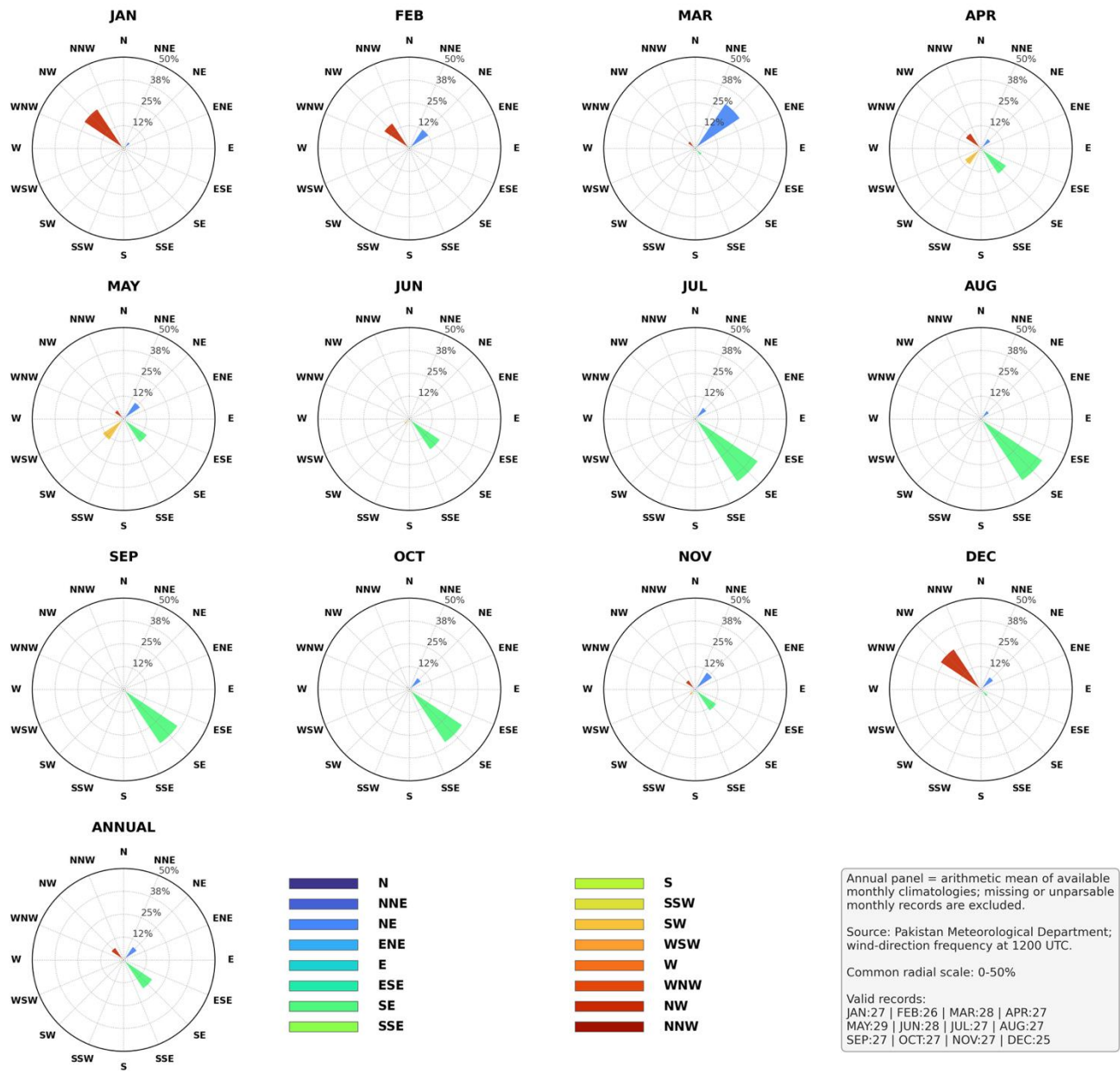
Wind Rose Climatology: QUETTA

1944-2000 (57 years) | 1200 UTC | Long-term mean frequency (%)



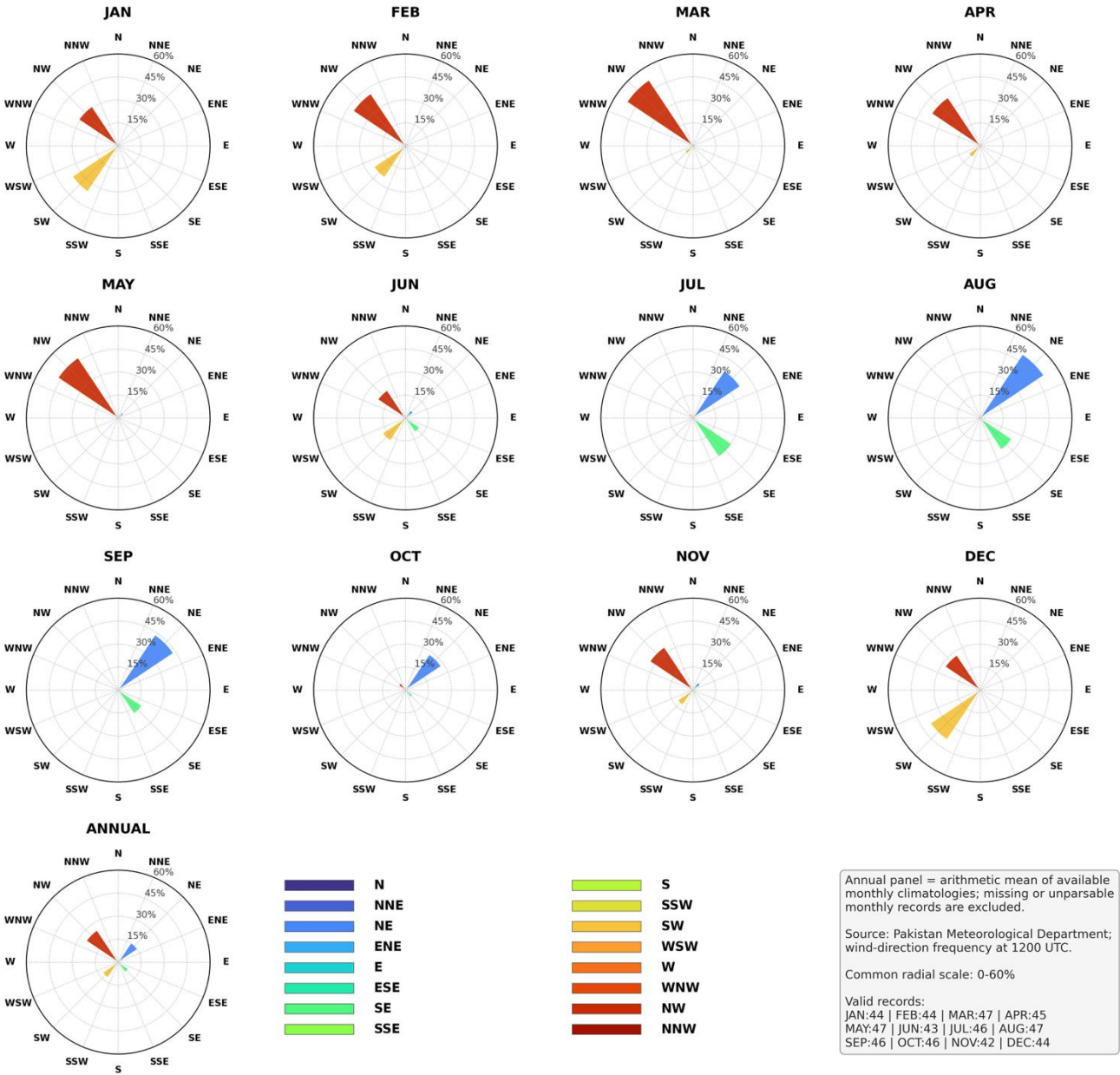
Wind Rose Climatology: RAFIQUE(PAF)

1972-2000 (29 years) | 1200 UTC | Long-term mean frequency (%)



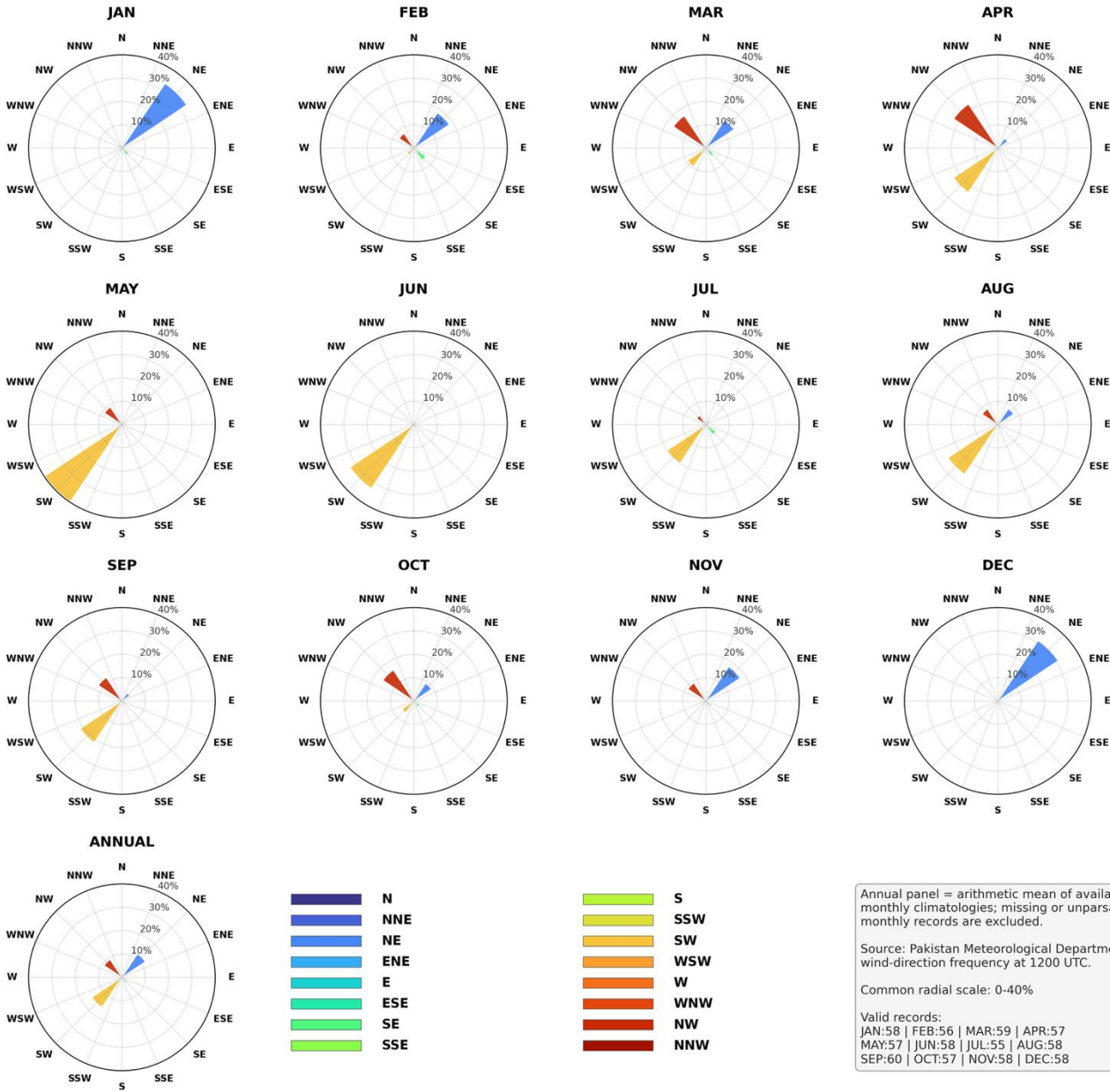
Wind Rose Climatology: RISALPUR

1954-2000 (47 years) | 1200 UTC | Long-term mean frequency (%)



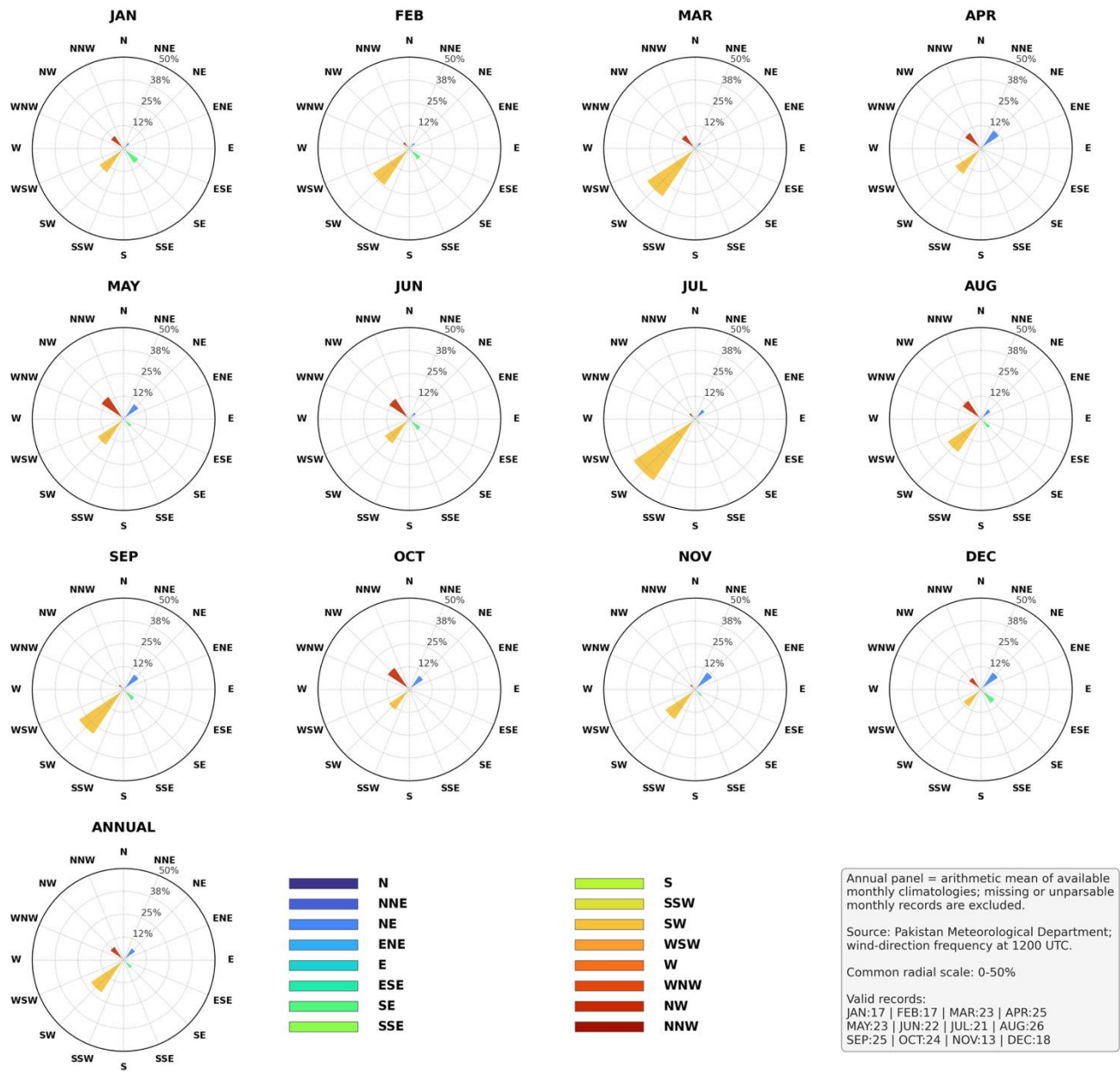
# Wind Rose Climatology: ROHRI

1939-2000 (62 years) | 1200 UTC | Long-term mean frequency (%)



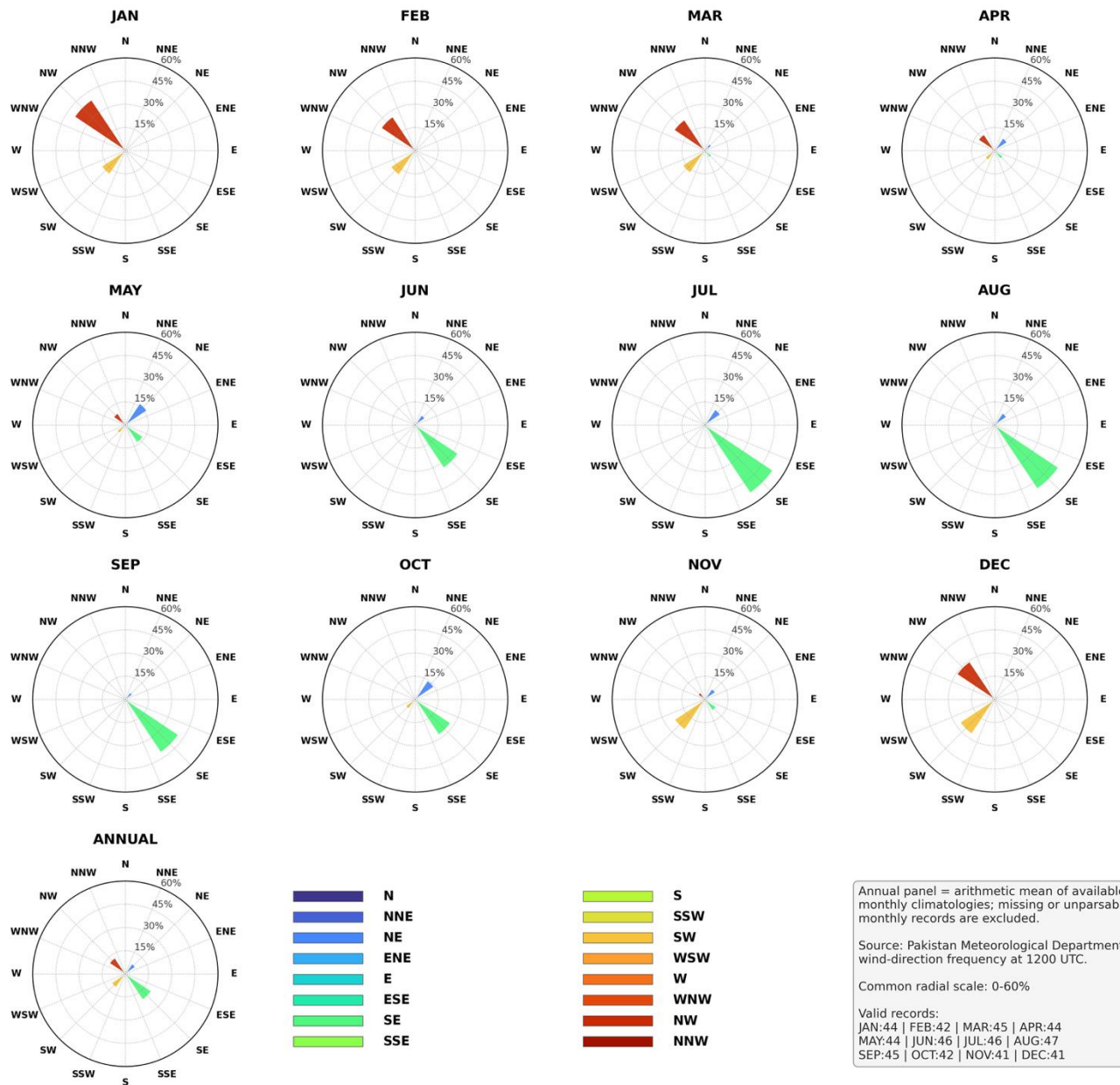
Wind Rose Climatology: SAIDU SHARIF

1974-2000 (27 years) | 1200 UTC | Long-term mean frequency (%)



Wind Rose Climatology: SARGODHA

1947-2000 (54 years) | 1200 UTC | Long-term mean frequency (%)



Annual panel = arithmetic mean of available monthly climatologies; missing or unparsable monthly records are excluded.

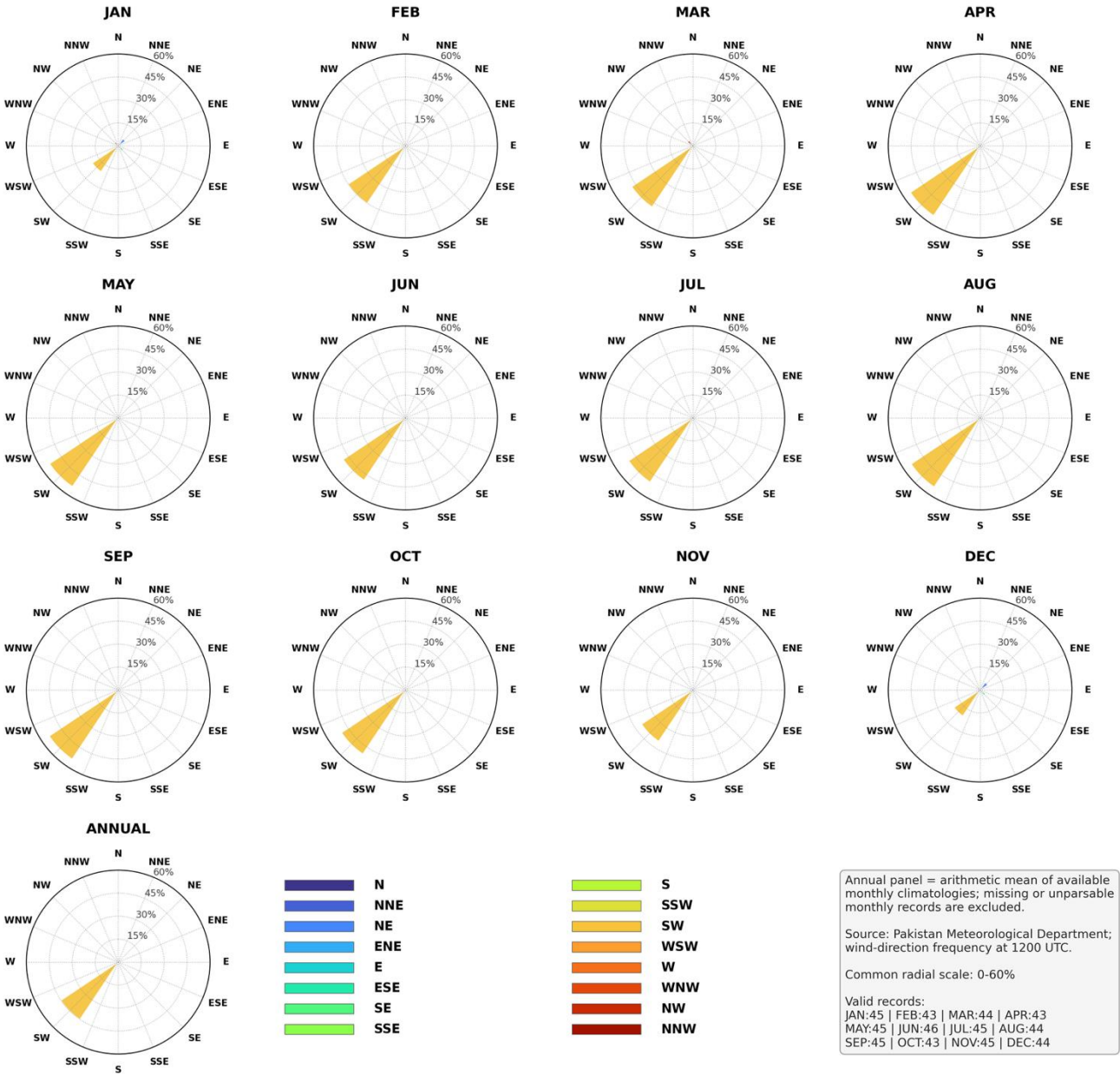
Source: Pakistan Meteorological Department; wind-direction frequency at 1200 UTC.

Common radial scale: 0-60%

Valid records:  
JAN:44 | FEB:42 | MAR:45 | APR:44  
MAY:44 | JUN:46 | JUL:46 | AUG:47  
SEP:45 | OCT:42 | NOV:41 | DEC:41

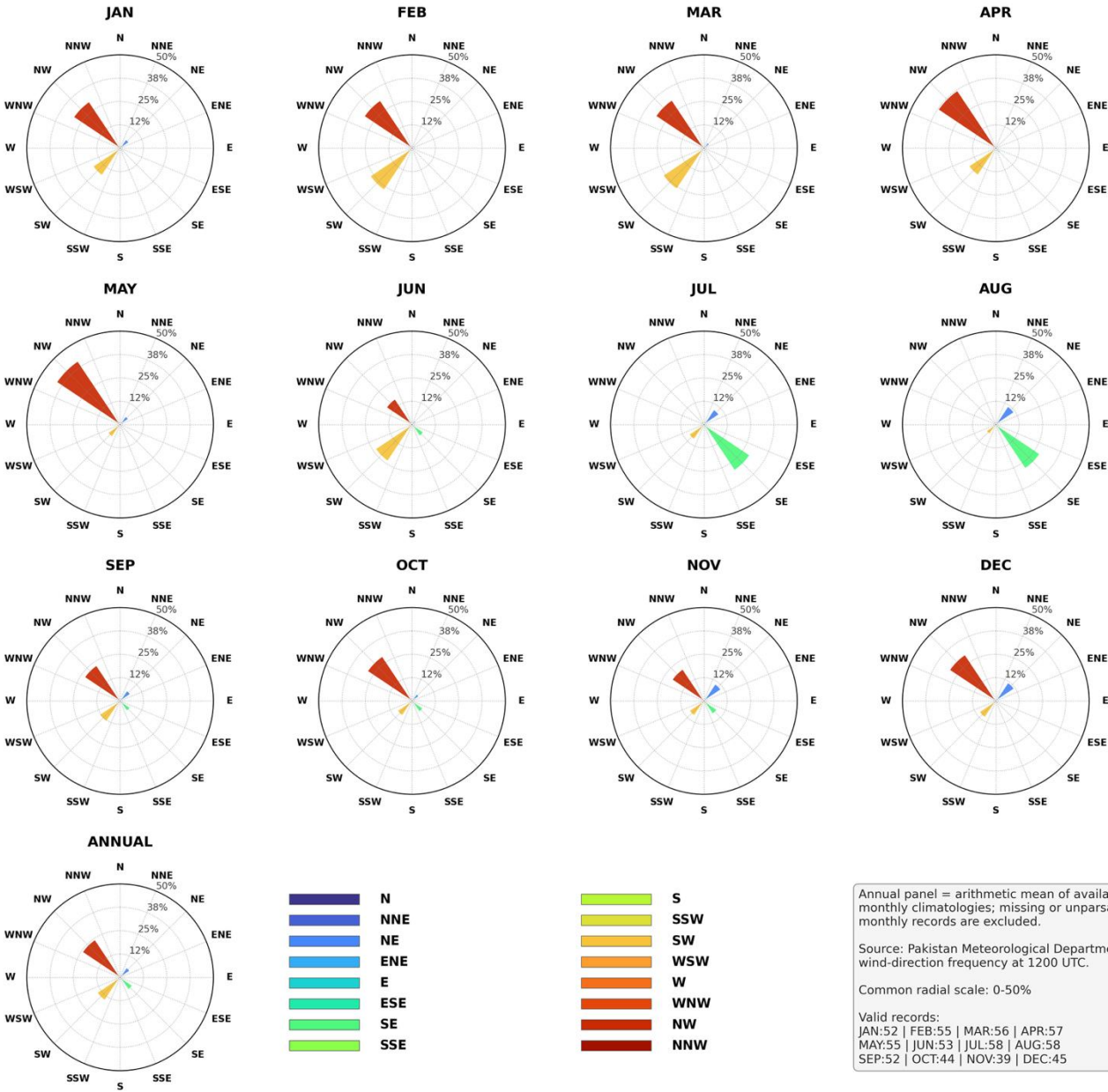
Wind Rose Climatology: SHARA-E-FAISAL

1933-2000 (68 years) | 1200 UTC | Long-term mean frequency (%)



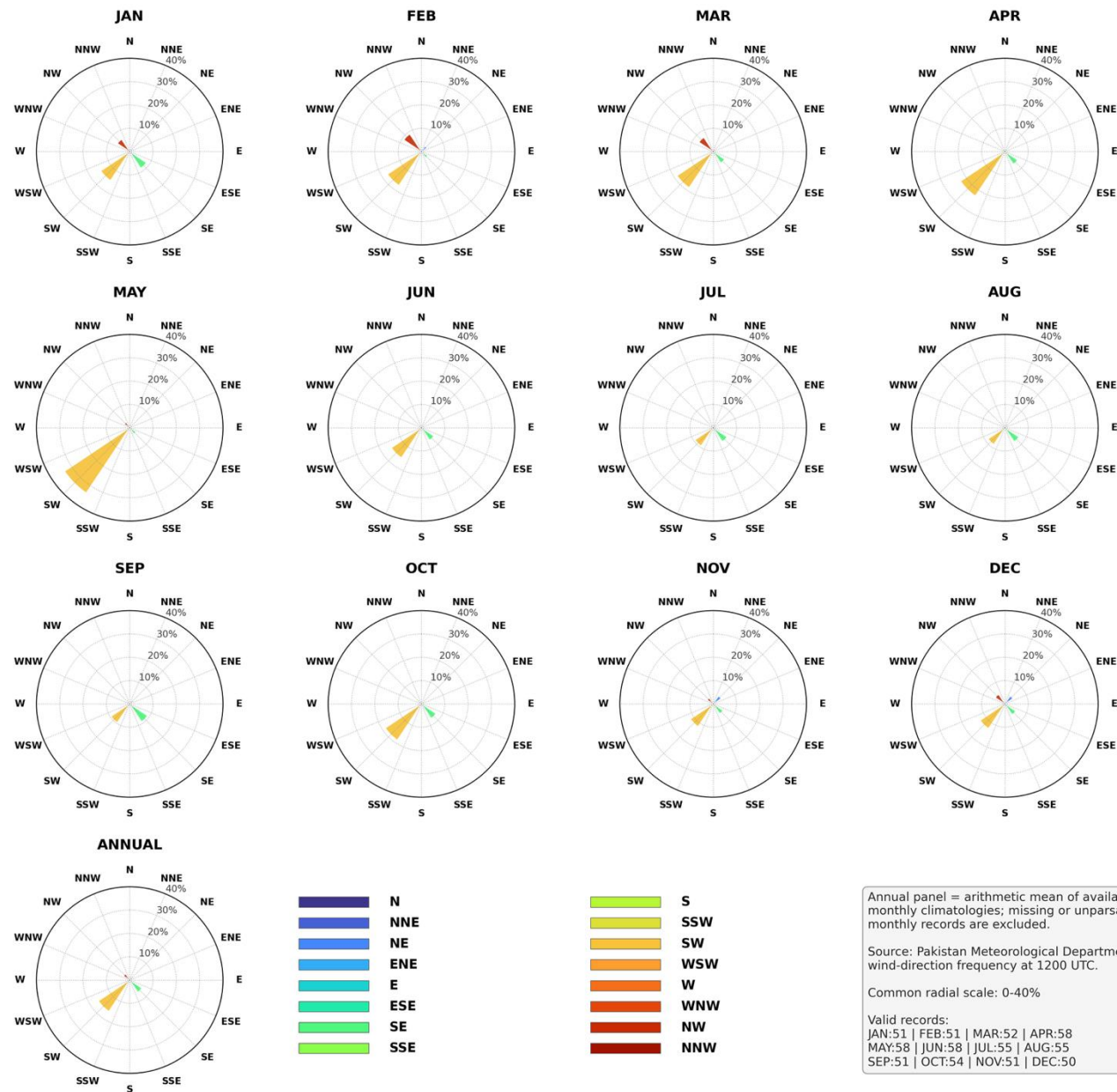
Wind Rose Climatology: SIALKOT

1933-2000 (68 years) | 1200 UTC | Long-term mean frequency (%)



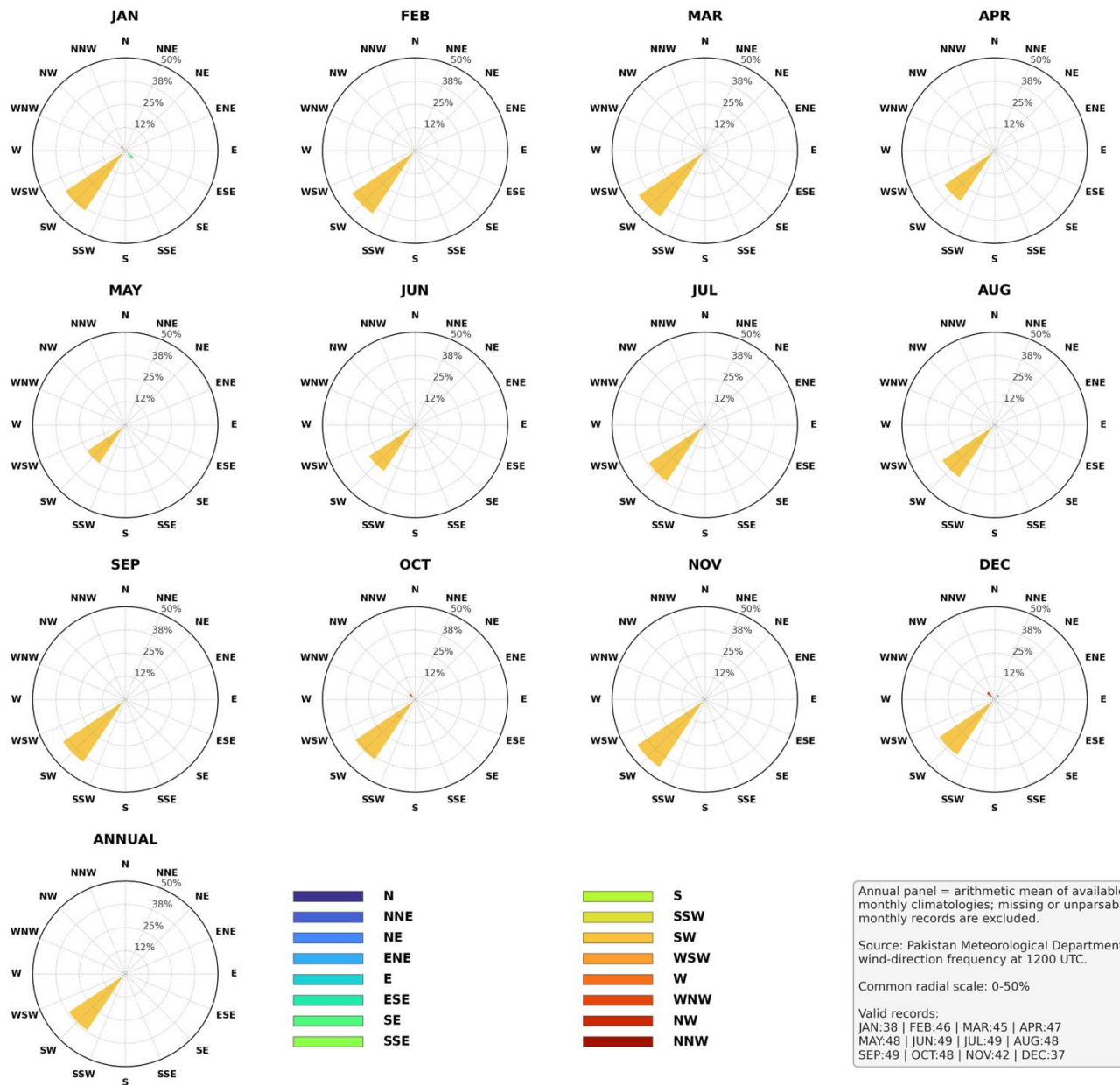
# Wind Rose Climatology: SIBBI

1940-2000 (61 years) | 1200 UTC | Long-term mean frequency (%)



Wind Rose Climatology: SKARDU(PBO)

1952-2000 (49 years) | 1200 UTC | Long-term mean frequency (%)



Annual panel = arithmetic mean of available monthly climatologies; missing or unparsable monthly records are excluded.

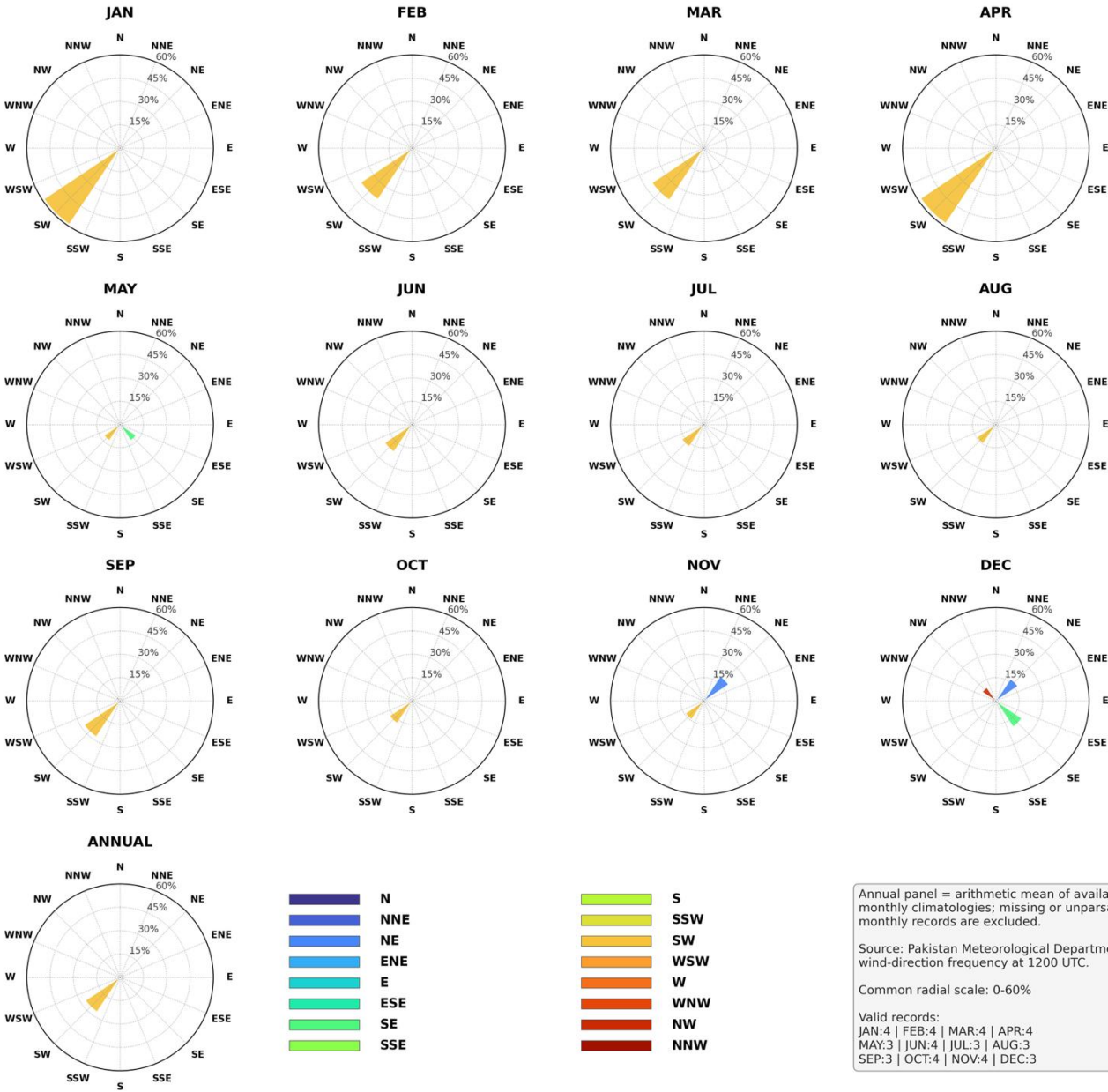
Source: Pakistan Meteorological Department; wind-direction frequency at 1200 UTC.

Common radial scale: 0-50%

Valid records:  
JAN:38 | FEB:46 | MAR:45 | APR:47  
MAY:48 | JUN:49 | JUL:49 | AUG:48  
SEP:49 | OCT:48 | NOV:42 | DEC:37

Wind Rose Climatology: TURBAT

1997-2000 (4 years) | 1200 UTC | Long-term mean frequency (%)



Annual panel = arithmetic mean of available monthly climatologies; missing or unparsable monthly records are excluded.

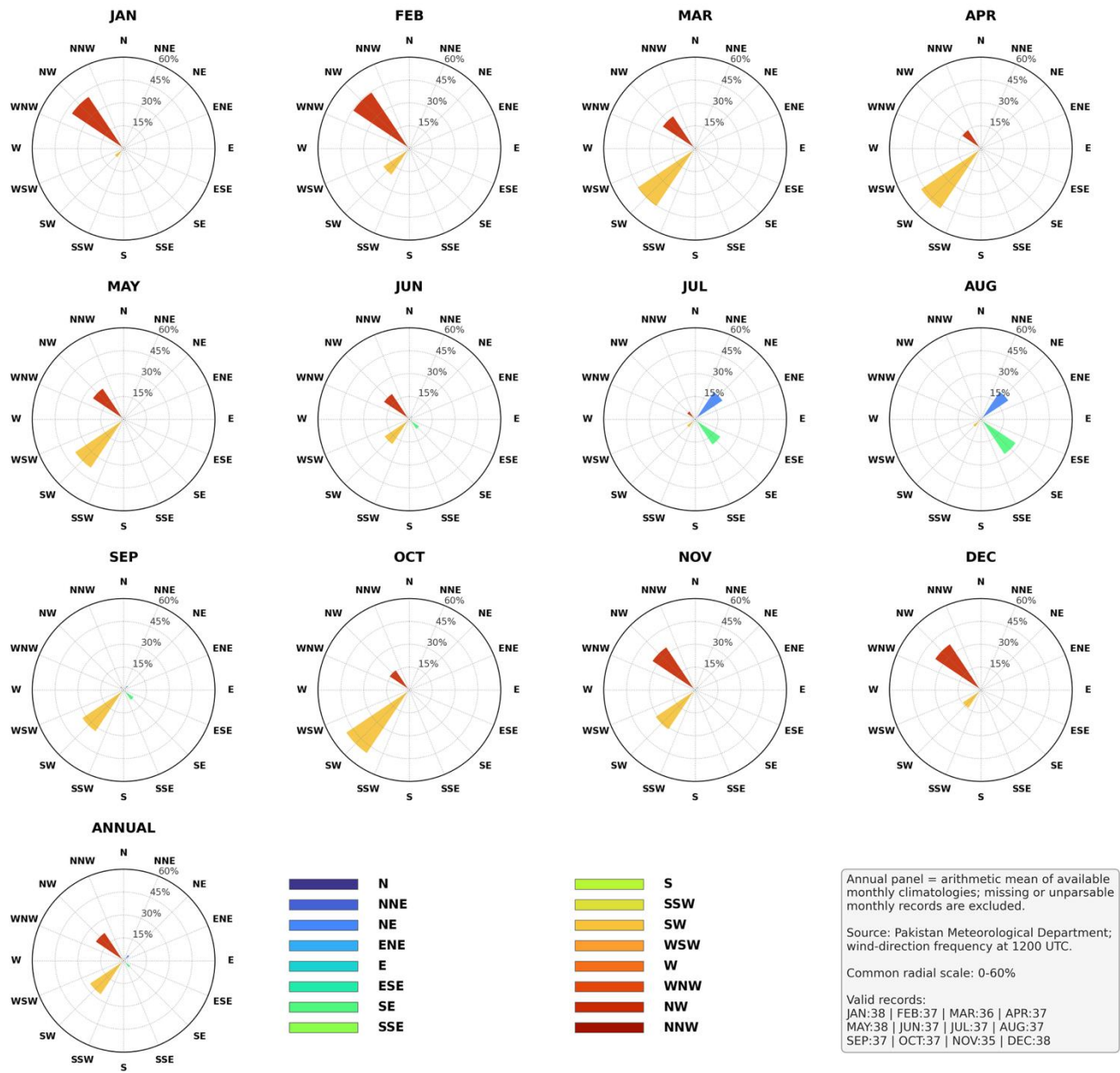
Source: Pakistan Meteorological Department; wind-direction frequency at 1200 UTC.

Common radial scale: 0-60%

Valid records:  
JAN:4 | FEB:4 | MAR:4 | APR:4  
MAY:3 | JUN:4 | JUL:3 | AUG:3  
SEP:3 | OCT:4 | NOV:4 | DEC:3

Wind Rose Climatology: ZHOB(PBO)

1961-2000 (40 years) | 1200 UTC | Long-term mean frequency (%)



Annual panel = arithmetic mean of available monthly climatologies; missing or unparsable monthly records are excluded.

Source: Pakistan Meteorological Department; wind-direction frequency at 1200 UTC.

Common radial scale: 0-60%

Valid records:  
JAN:38 | FEB:37 | MAR:36 | APR:37  
MAY:38 | JUN:37 | JUL:37 | AUG:37  
SEP:37 | OCT:37 | NOV:35 | DEC:38

## **Seasonal and Annual Wind-Direction Regimes in Pakistan**

- **Wind-direction regimes show strong seasonal and geographic variability across Pakistan.**
- **Monthly wind roses reveal changes in prevailing flow directions through the annual cycle.**
- **Annual wind roses provide a concise station-level representation of the long-term directional climate.**
- **Results are based on PMD 1200 UTC observations and station-specific historical periods of record.**
- **These climatologies provide an observational foundation for regional wind-resource assessment, atmospheric-circulation interpretation, and climate-informed planning.**

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**Pakistan Meteorological Department wind-direction observations | 1200 UTC**